CH9　剪力圖與彎矩圖

9-1. 樑的支承及種類

一、梁的支承

樑的支承

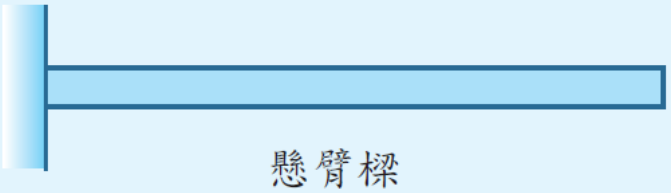
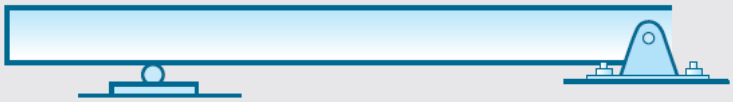
我們把一細長結構件經適當之支承，在其軸向平面施加縱向負荷，該負荷所產生之彎曲力矩使得該結構件以彎曲為主要變形形式，則該結構件稱之為樑(Beam)，常用於樑的支承有三種，如下所示

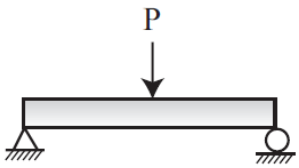
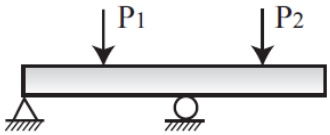
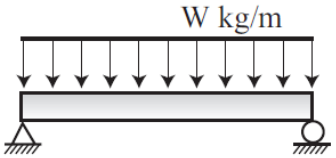
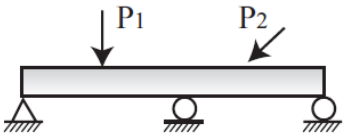
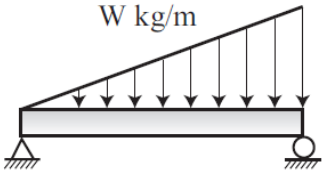
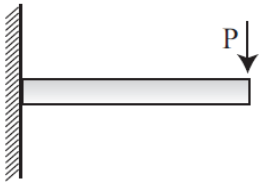
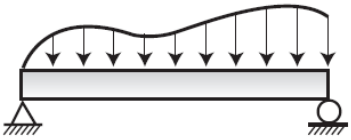
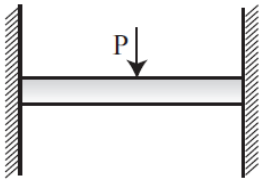
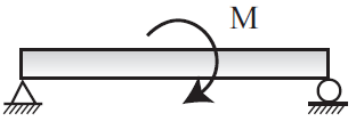
支承	圖示
1. 鉸支承(hinge support)： 用銷(pin)或是鉸鏈(hinge)將樑接支座上，其所受的力可分解成水平及垂直兩方向，可限制樑不作水平及垂直方向的移動，但是可以自由地轉動。	
2. 滾支承(roller support)： 此類支承其所受的力可分解成垂直方向，能限制樑在垂直於支承面的方向移動，但是可沿著支承移動，也可自由地轉動。	
3. 固定支承(fixed support)： 將樑固定於支座上所受的力可分解成水平及垂直兩方向且有承受彎矩，此種支承無法移動也無法轉動。	

二、樑的分類

樑的分類

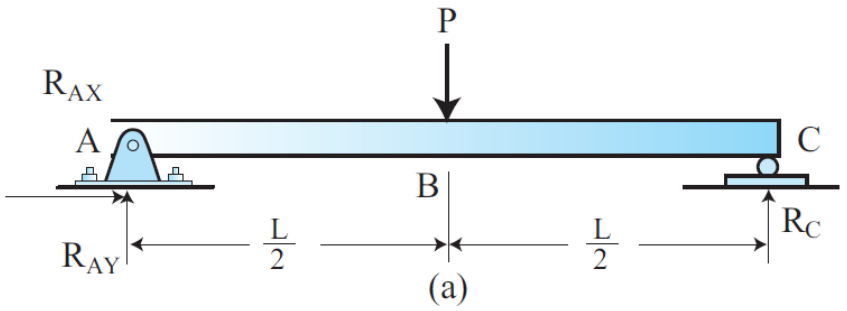
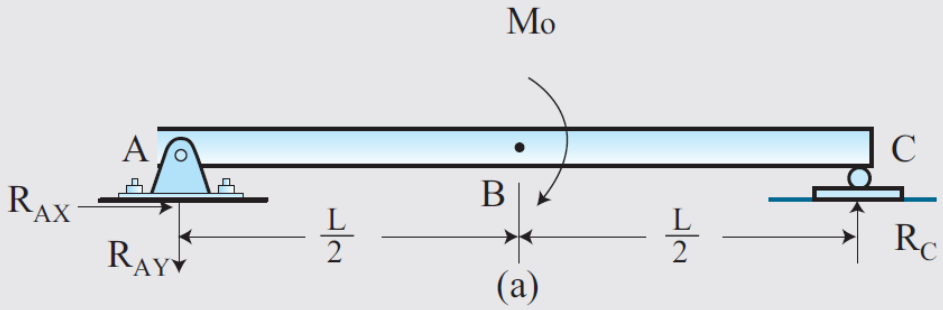
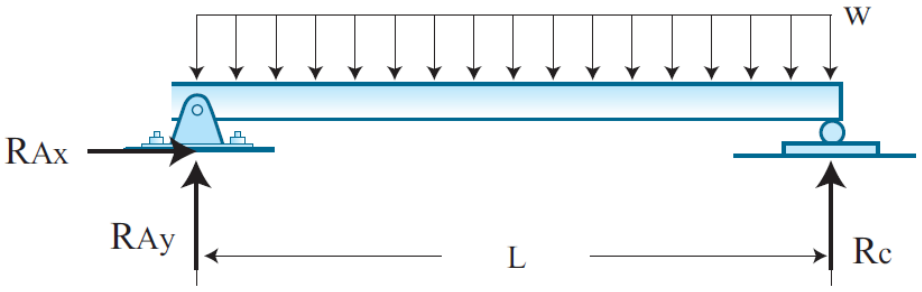
在許多的結構或機件上，均應用到樑，例如橋樑、建築物的橫樑、車輛的車軸及機器的傳動軸等，靜定樑可分為下列三種：

樑的分類	圖示
1.簡支樑 (simply supported beam)	樑的一端為鉸支承，另一端為滾支承。樑在鉸支承的一端不可移動，在滾支承的一端可水平移動。 
2.懸臂樑 (cantilever beam)	樑的一端為固定支承，另一端為自由端。在固定端，樑不能移動。亦不能旋轉， 在自由端兩者均可發生。  懸臂樑
3. 外伸樑	支承的方式如同簡支樑，但是樑的一端或是兩端伸出支承的外側。 

名稱	圖示	名稱	圖示
簡支樑		集中負荷	
外伸樑		均布負荷	
連續樑		均變負荷	
懸臂樑		非均變負荷	
固定樑		彎矩	

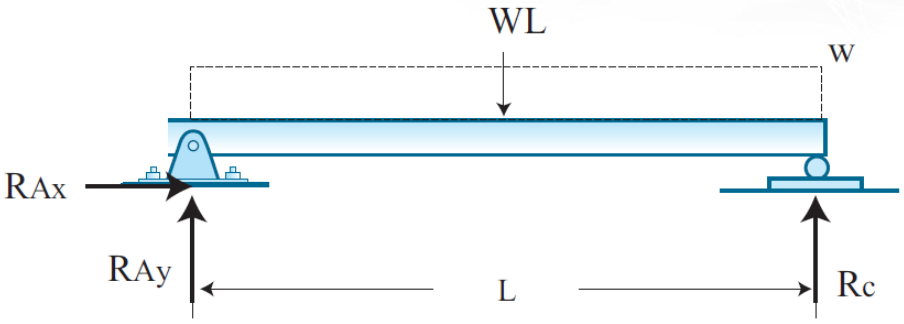
三、樑的負載及支承反力的運算

2. 樑之負荷以及支承反力的運算

	圖示說明
集中負載	 $\sum M_A = 0 \Rightarrow P \times \frac{L}{2} = R_c \times L \Rightarrow R_c = \frac{P}{2}$ $\sum F_Y = 0 \Rightarrow R_{AY} = P - R_c = \frac{P}{2}$
彎矩負載	 $\sum M_A = 0 \Rightarrow M_o = R_c \times L \Rightarrow R_c = \frac{M_o}{L}(\uparrow)$ $\sum F_Y = 0 \Rightarrow R_{AY} = R_c = \frac{M_o}{L}(\downarrow)_V$
均佈負載	

均佈負載

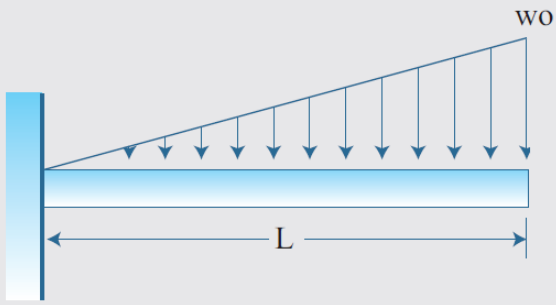
上圖可等效成



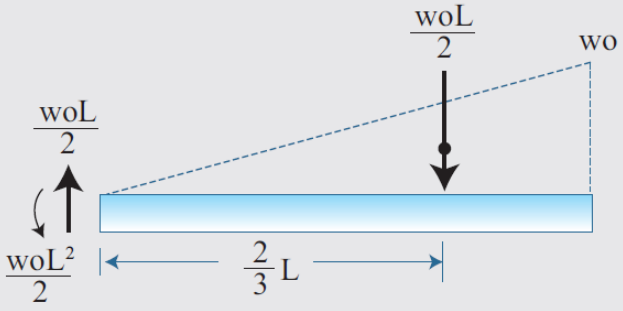
$$\sum M_A = 0 \Rightarrow WL \times \frac{L}{2} = R_c \times L \Rightarrow Rc = \frac{WL}{2}$$

$$\sum F_Y = 0 \Rightarrow R_{AY} = WL - R_c = \frac{WL}{2}$$

4.均變負載



上圖可等效成



四、樑(Beam)之內力

樑之內力

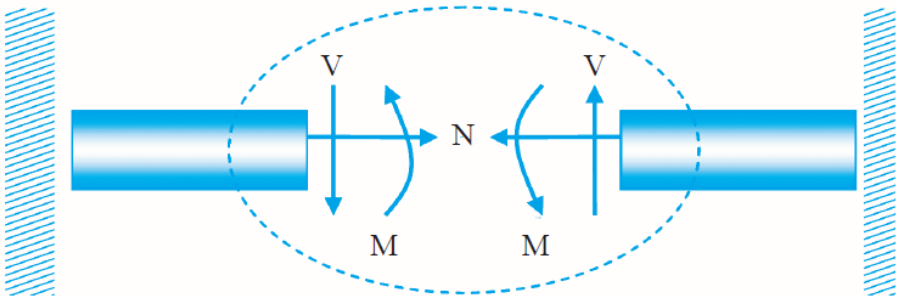
(1)樑之內力(剪力、軸力及彎曲力矩符號)

如下圖所示樑之任一點切開，其斷面所包含之內力有

A.軸力(N)：與樑斷面軸向平行之力。

B.彎矩(M)：使樑產生彎曲或旋轉之力。

C.剪力(V)：與樑斷面相垂直之力。



$\sum F_y=0 \Rightarrow R_A-F_1-V=0$

$V=R_A-F_1 \Rightarrow$ 剪應力 $\tau=\frac{V}{A}$

由$\sum M_C=0 \Rightarrow -R_Ax+F_1(x-a)+M=0$

得$M=R_Ax-F_1(x-a)$

C.如圖12.2所示橫截面左上右下的剪力爲正，反之爲負；使彎曲變形爲凹向上的彎矩爲正，反之爲負。

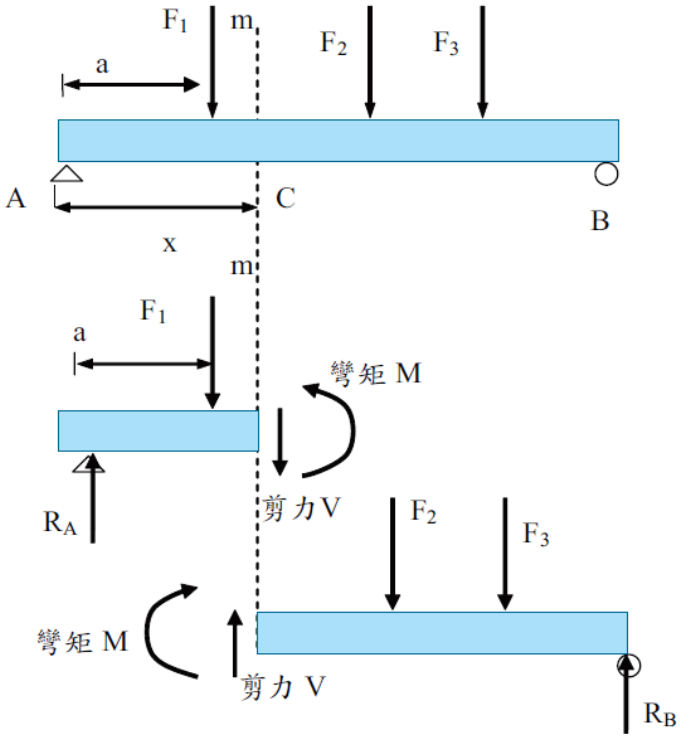


圖12.1　樑截面之剪應力

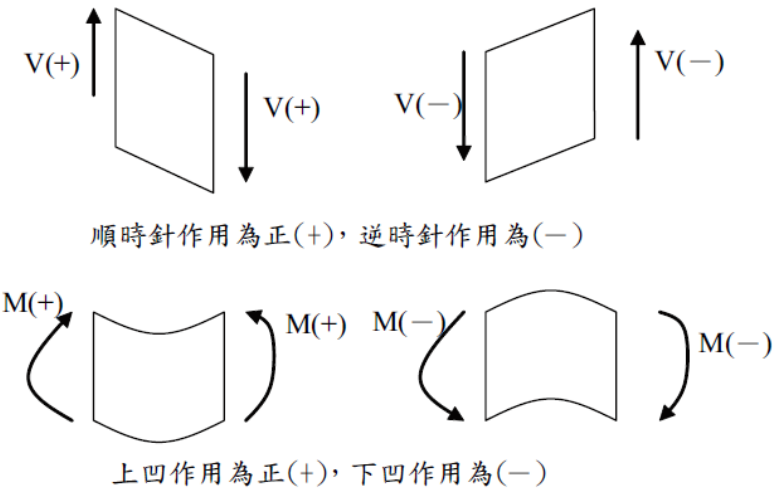


圖12.2　剪力與彎矩之正負

9-2. 剪力圖及彎矩圖

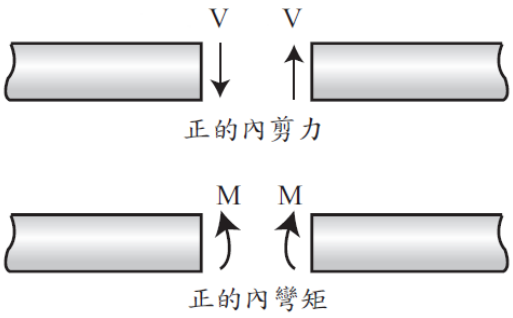
一、剪力圖與彎矩圖

1 樑之內力

(1)樑之內力（剪力、軸力及彎曲力矩符號）

如圖所示樑之任一點切開，其斷面所包含之內力有

A.彎矩(M)：使樑產生彎曲或旋轉之力。



B.剪力(V)：與樑斷面相垂直之力。

剪力(V)	剪力之正負，係對材料之自由體圖取力矩，順時針為“＋”，逆時針為“－”，如圖所示。 V ⊕ V　　V ⊖ V
彎矩(M)	彎矩之正負，當彎矩使材料“凹向上”者為“＋”，使材料“凹向下”者為“－”，如圖所示。 M ⊕ M　　M ⊖ M

2. 剪力與彎矩圖

(1)剪力方程式和彎矩方程式：

A.樑受力時，橫截面上的剪力和彎矩是隨截面的位置不同而變化的，如果沿樑軸線方向選取座標x表示橫截面的位置，則樑的各截面上的剪力和彎矩都可表示爲x的函數，即

V＝V(x)樑的剪力方程，M＝M(x)樑的彎矩方程

B.如果以x爲橫座標軸，以V或M爲縱座標軸，分別繪製V＝V(x)，M＝M(x)的函數曲線，則分別稱爲剪力圖和彎矩圖。

C.從剪力圖與彎矩圖上可以很容易確定樑的受力狀況，且能了解樑受力之最大剪力和最大彎矩位置。

(2)剪力圖(V-dia)與彎矩圖(M-dia)繪製步驟：

A.畫樑的自由體圖，並求樑的支承反力，由左邊畫至右邊。

B.剪力V與彎矩均由零開始，從左向右畫，因為平衡的關係，合力與合力矩在平衡狀態下為零，故最後亦回到零。

C.若是懸臂樑，最好由自由端開始然後推進至全樑。

D.樑受外力無載重部份：

(A)剪力圖為一水平直線⇒彎矩圖為一次線性曲線。

(B)若剪力圖作用力為零⇒彎矩圖為一水平直線。

E.集中負載：剪力圖為垂直跳躍線⇒彎矩圖為一折線轉點。

F.均布負載：剪力圖為一次曲線，為一向下直線，若負載向上時，剪力圖為一上升的直線；彎矩圖為二次拋物線，若剪力V＞0則拋物線開口向上，若剪力V＜0時，拋物線開口向下。

G.一次均變載重：剪力圖為二次拋物線⇒彎矩圖為三次曲線。

H.力矩作用：剪力圖為水平沒有變化⇒彎矩圖為垂直跳躍線，若遇樑上有外加順時針力矩作用時，彎矩圖為向上跳躍；反之逆時針力矩，彎矩圖為向下跳躍。

I. 彎矩值的大小為剪力圖的面積，彎矩圖為剪力圖之高一次函數。

J. 絕對值最大的彎矩總是出現在：

(1)剪力爲零的截面上　　(2)集中力作用處

(3)集中力偶作用處。

K.綜合上述說明，荷重、剪力及彎矩之關係表：

負載 / 圖形	沒有負載	集中負載	均布負載	力矩	均變負載
剪力圖	水平直線	鉛直直線	一次直線	水平直線	N次曲線
彎矩圖	一次直線	轉點折線	二次拋物線	鉛直直線	(N+1)次曲線

【觀念說明】：

荷重 W、剪力 V 及彎矩 M 之關係：

1. 微分關係：

$$\text{均變負載} w = \frac{dV}{dx} \Rightarrow \text{即載重強度等於剪力之變化率。}$$

$$\text{剪力} V = \frac{dM}{dx} \Rightarrow \text{即剪力等於彎矩之變化率。}$$

$V = \frac{dM}{dx} = 0 \Rightarrow$可求出最大彎矩。

2. 積分關係(面積)：

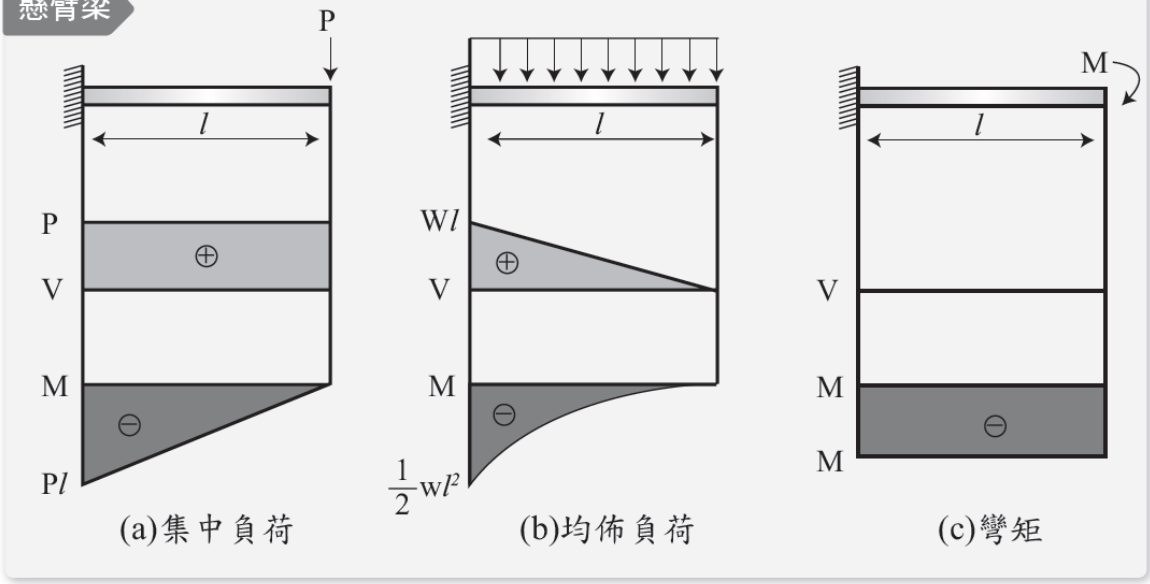
$V_{B/A} = \int w dx \Rightarrow$即 A 至 B 兩點之剪力差值等於 A 至 B 兩點之向上載重圖面積。

$M_{B/A} = \int V dx \Rightarrow$即 A 至 B 兩點之彎矩差值等於 A 至 B 兩點之剪力圖面積。

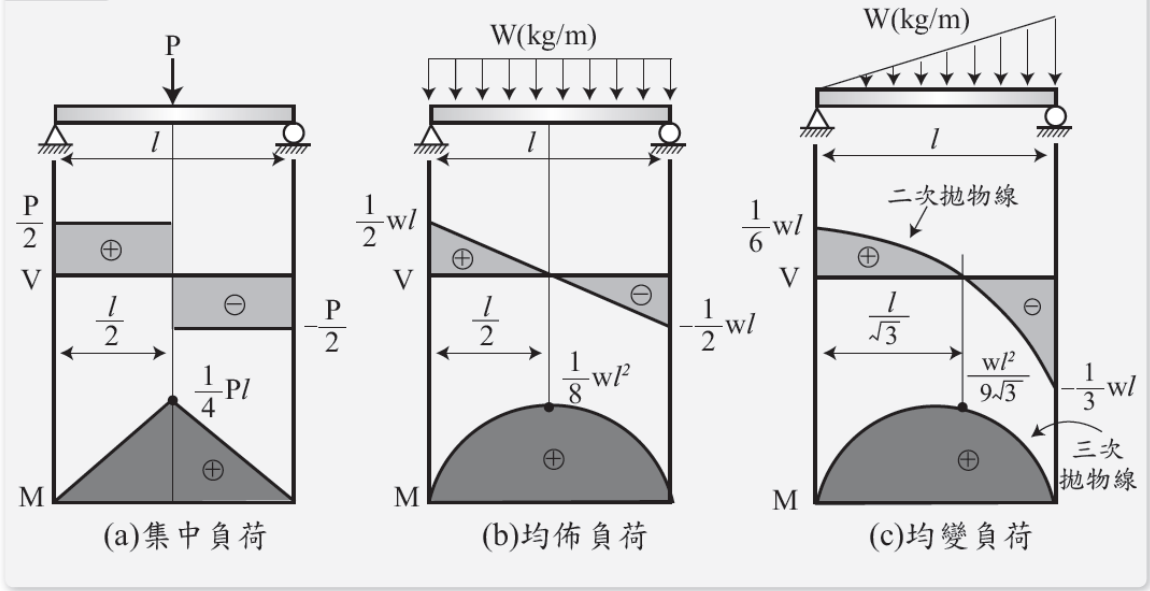
3. 危險斷面（臨界斷面）：
   當梁之斷面尺寸相同時，產生彎矩之絕對值為最大時之截面；亦為應力最大之截面（若梁之斷面尺寸不相同時）。
4. 剪力等於零（V=0）處未必是危險截面，必須就彎矩圖之局部極大值加以比較，其中彎矩值之絕對值為最大時之截面才是危險截面。
5. 懸臂梁之危險斷面一定在固定端。
6. 簡支梁之危險斷面在剪力圖之剪力由正變負處或由負變正處。

3. 基本常見的剪力圖與彎矩圖

懸臂梁

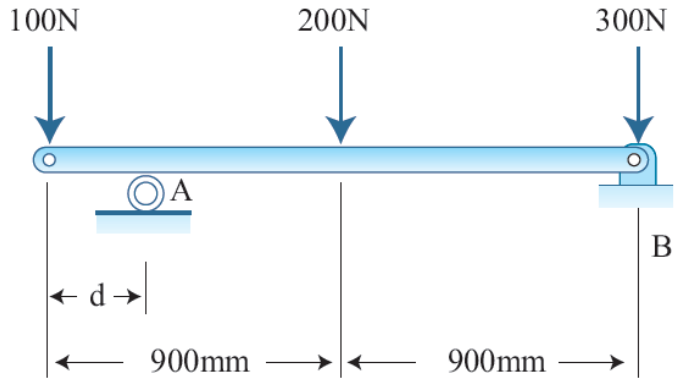


簡支梁



範例 1

如下圖所示，(1)若A的支承反力為360N試求B的支承反作用力及d的距離。
(2)若B的支承反力為360N試求A的支承反作用力及d的距離。



【解析】

$$\sum F_y = 0 \ : \ A + B - (100 + 200 + 300)N = 0 \Rightarrow A + B = 600N$$

(1) 若A的支承反力為360N，B=240N

$$\overset{+}{\nearrow} M_B = 0 \ : \ (100N)(1.8) - A(1.8 - d) + (200N)(0.9) = 0$$

$$d = \frac{1.8A - 360}{A} \Rightarrow d = \frac{1.8(360) - 360}{360} = 0.800m$$

(2) 若B的支承反力為360N，A=240N

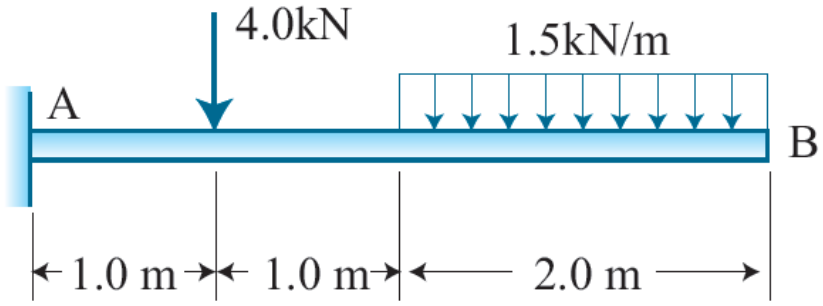
$$\overset{+}{\nearrow} M_A = 0 \ :$$

$$(100N)(d) - (200N)(0.9 - d) - (300N)(1.8 - d) + B(1.8 - d) = 0$$

$$d = \frac{720 - 1.8B}{600 - B} \Rightarrow d = \frac{720 - 1.8(360)}{600 - 360} = 0.300m$$

範例 2

試求支承的反作用力。

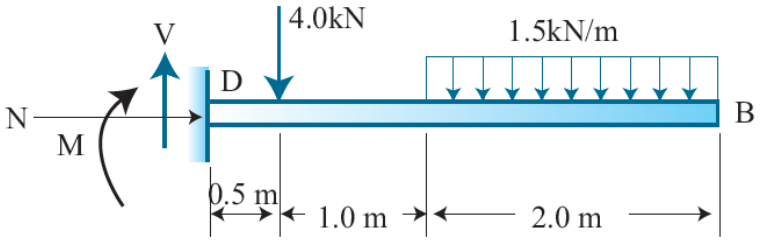


【解析】

$$\sum F = 0 \ : \ V = 4.0\,kN + (1.5\,kN/m)(2.0m) = 4.0kN + 3.0kN = 7.0kN$$

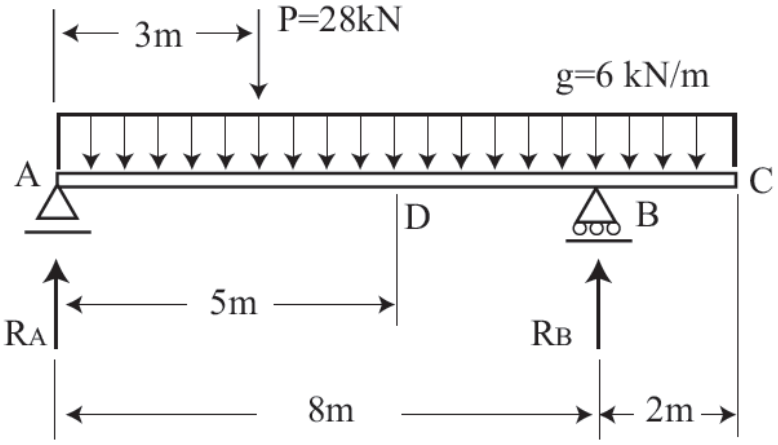
$$\sum M_D = 0 \ : \ M = -(4.0\,kN)(0.5m) - (1.5\,kN/m)(2.0m)(2.5m)$$

$$= -2.0\,kN\cdot m - 7.5\,kN\cdot m = -9.5\,kN\cdot m$$



範例 3

如圖示之一過懸樑(a simple beam with an overhang)**簡支撐**(simply supported)**於**A、B **兩點，承受一均勻分佈負荷**(a uniformly distributed load)q = 6 kN/m **及一集中負荷**(a concentrated load)P = 28 kN。**試求圖中**D **點之剪力**(shear force)**及彎距**(bending moment)。(94年機械關務三等)

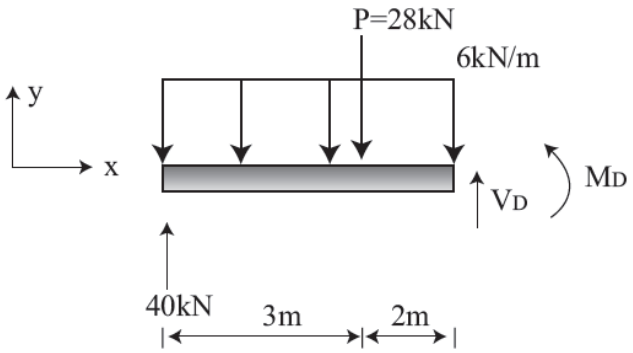


【解析】

(1) 先求支承反力：

$$\Sigma M_A = 0 \Rightarrow R_B = \frac{1}{8} \times [28 \times 3 + 6 \times 10 \times 5] = 48\text{kN}(\uparrow)$$

$$\Sigma M_B = 0 \Rightarrow R_A = 40\text{kN}(\uparrow)$$



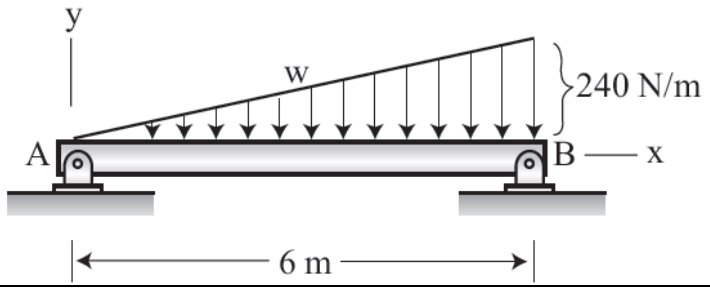
(2) 取AD自由體圖：

$$\Sigma F_y = 0 \Rightarrow V_D = 28 + 6 \times 5 - 40 = 18\text{kN}(\uparrow)$$

$$\Sigma M_D = 0 \Rightarrow M_D = -28 \times 2 - 6 \times 5 \times 2.5 + 40 \times 5 = 69\text{kN} \cdot \text{m}(\circlearrowleft)$$

範例 4

一簡支樑受分佈力w = 40x N/m 如圖所示，梁內受最大力矩之處離A 端多少m？

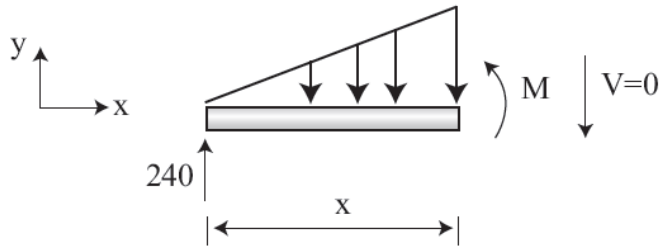


【解析】

最大力矩發生在剪力V = 0之處，先求支承反力

$$R_A = \frac{1}{6}[6\times 240\times \frac{1}{2}\times 6\times \frac{1}{3}] = 240(N)$$ 取A至V = 0自由體圖

$$\Sigma F_y = 0$$

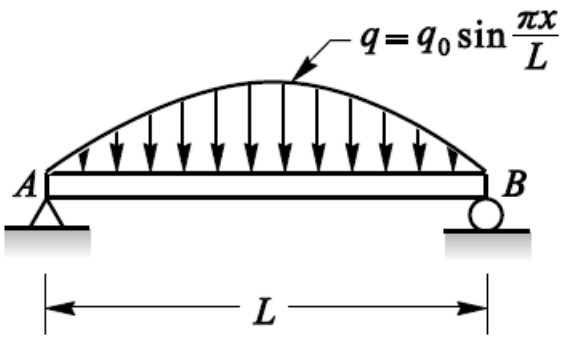


$$\Rightarrow 240 = (\frac{x}{6})^2 \times 240\times 6\times \frac{1}{2}$$

$$\Rightarrow x = 3.46(m)$$

範例 5

試求圖中簡支梁之最大彎矩。



【解析】

$$\frac{dV}{dx} = -q = -q_0 \sin\frac{\pi x}{L} \quad , \quad V = q_0 \frac{L}{\pi}\cos\frac{\pi x}{L} + C_1$$

$$\frac{dM}{dx} = V = q_0 \frac{L}{\pi}\cos\frac{\pi x}{L} \quad , \quad M = q_0 \frac{L^2}{\pi^2}\sin\frac{\pi x}{L} + C_1 x + C_2$$

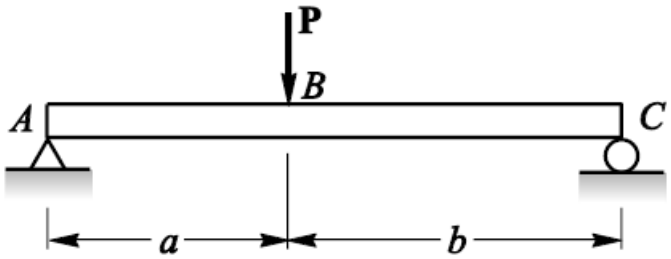
邊界條件(1)：$x = 0$，$M = 0$，得 $C_2 = 0$

邊界條件(2)：$x = L$，$M = 0$，得 $C_1 = 0$

故得 $V = q_0 \frac{L}{\pi}\cos\frac{\pi x}{L}$ ， $M = q_0 \frac{L^2}{\pi^2}\sin\frac{\pi x}{L}$ ， $M_{\max} = q_0 \frac{L^2}{\pi^2}$

範例 6

圖中簡支梁在 B 點承受一集中負荷 P，試求梁之剪力及彎矩方程式，並繪剪力圖及彎矩圖。

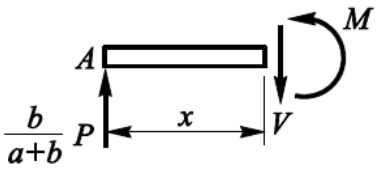


【解析】

(一)取自由體圖先求支承反力

$$\text{支承反力：}\ R_A=\frac{bP}{a+b}\ ,\ R_B=\frac{aP}{a+b}$$

(二)取 AB 段自由體圖

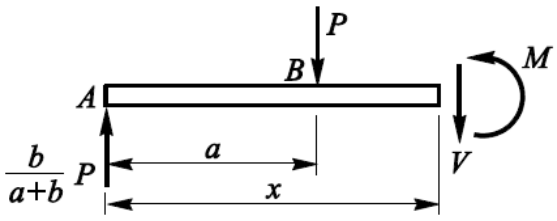


$0\le x\le a\ (AB\ 段)$

$$\Sigma F_y=0\ ,\ V=\frac{bP}{a+b}$$

$$\Sigma M=0\ ,\ M=\frac{bP}{a+b}x$$

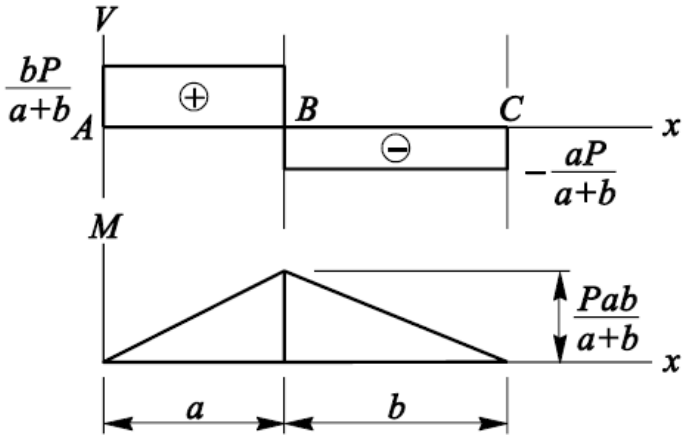
(三)取 BC 段自由體圖



$$a \le x \le (a+b)$$

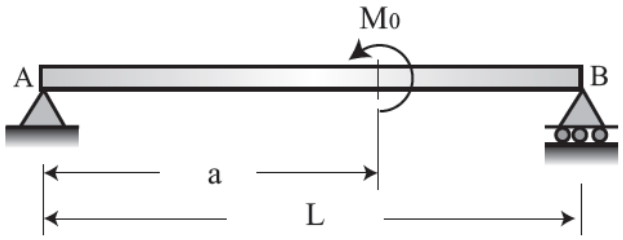
$$\Sigma F_y = 0 \ ， \ V = \frac{bP}{a+b} - P = -\frac{aP}{a+b}$$

$$\Sigma M = 0 \ ， \ M = \frac{bP}{a+b}x - P(x-a) = Pa\left(1 - \frac{x}{a+b}\right)$$

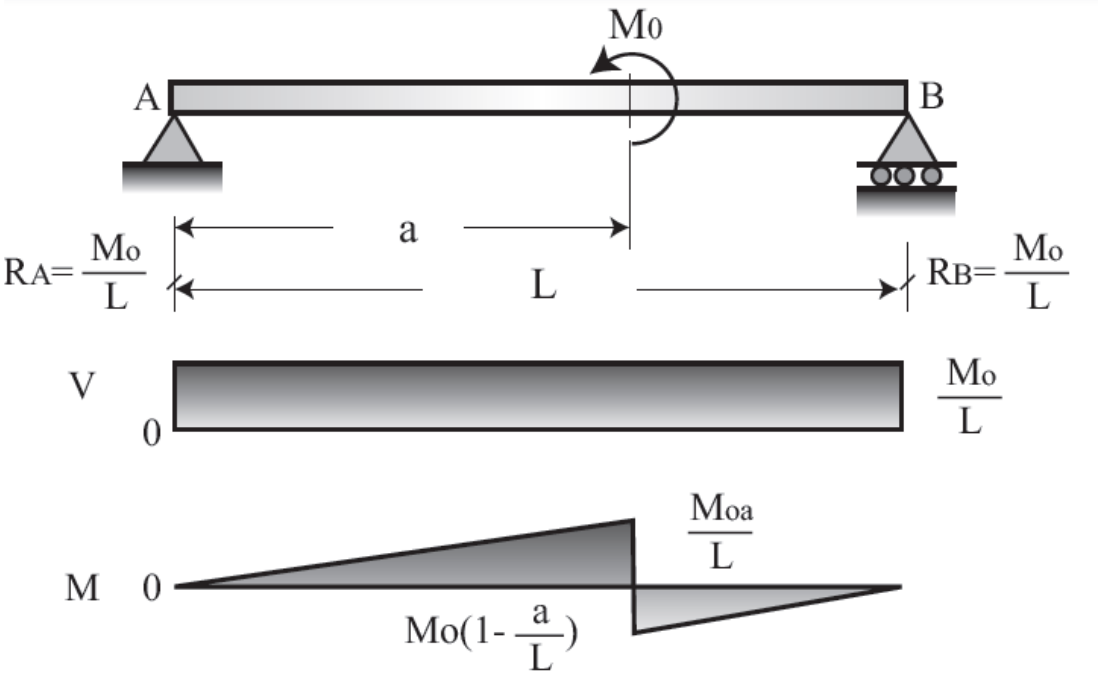


範例 7

試繪出如圖所示之剪應力和力矩圖
(shear and moment diagram)。

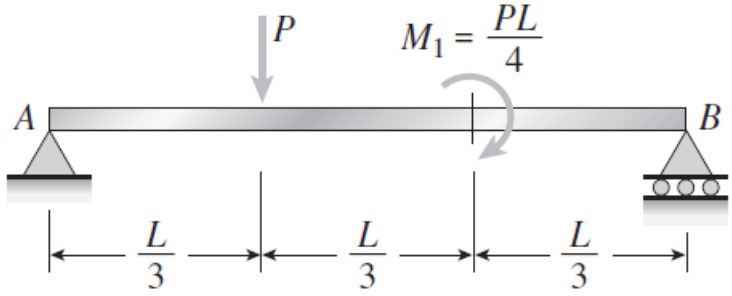


【解析】



範例 8

The simple beam AB shown in the figure is subjected to a concentrated load P and a clockwise couple $M_1 = \frac{PL}{4}$ acting at the third points. Draw the shear-force and bending-moment diagrams for this beam.



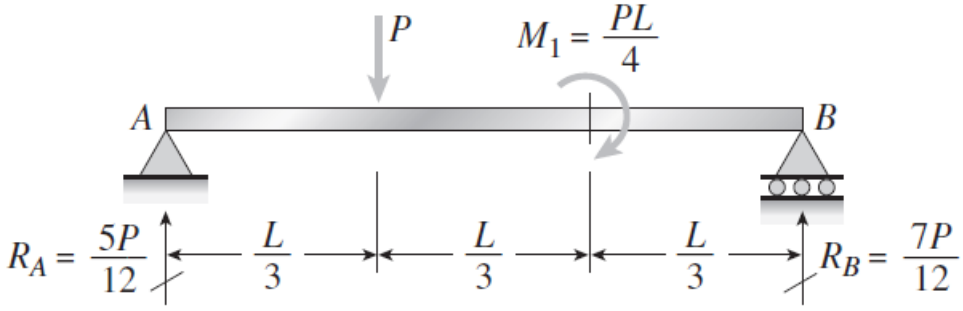
(96 台大工科) (100 台大工科) (100 中央土木)

【中文翻譯】

如圖之簡樑，受集中負載 P 及力偶 $M_1 = \frac{PL}{4}$ 。繪此樑之剪力及彎矩圖。

【解析】

(二) 先求支承反力



由靜力平衡式計算支承反力

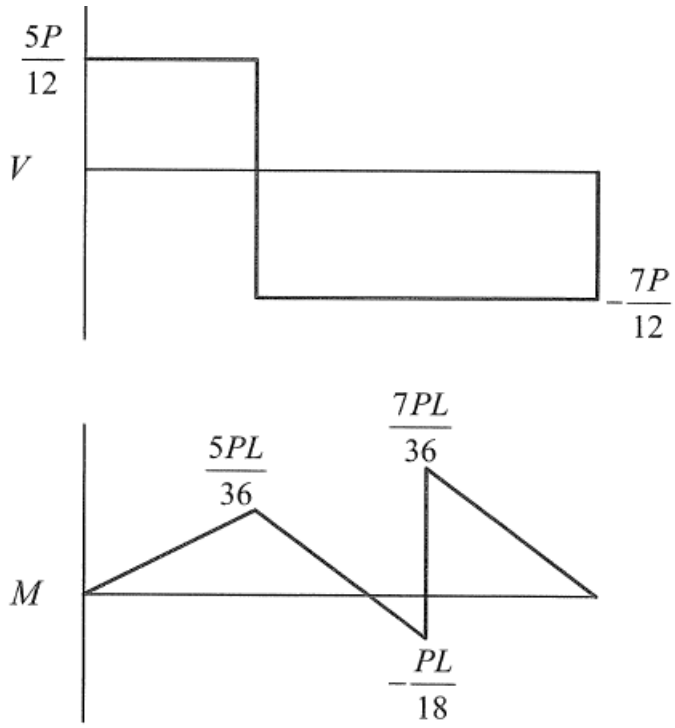
$\sum M_A = 0$

$\Rightarrow P \times \frac{L}{3} + \frac{PL}{4} = R_B \times L \Rightarrow R_B = \frac{7P}{12}(\uparrow)$

$\sum F_y = 0$

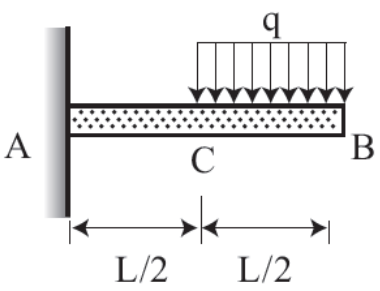
$\Rightarrow R_A + R_B = P \Rightarrow R_A = \frac{5P}{12}(\uparrow)$

(三) 剪力圖與彎矩圖


5P/12
V
−7P/12
7PL/36
5PL/36
M
−PL/18

範例 9

如圖所示之樑長L，斷面積為b×h 之矩形，材料之楊氏模數為E，在CB 段上受一均勻分布載重q。樑重不計， 試求圖中樑上之剪力圖、彎矩圖。



【機械高考】

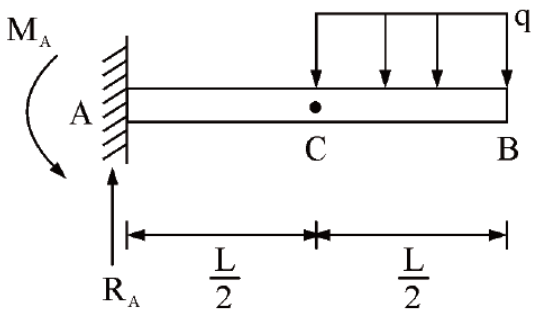
【解析】

(1) 計算支承反力：

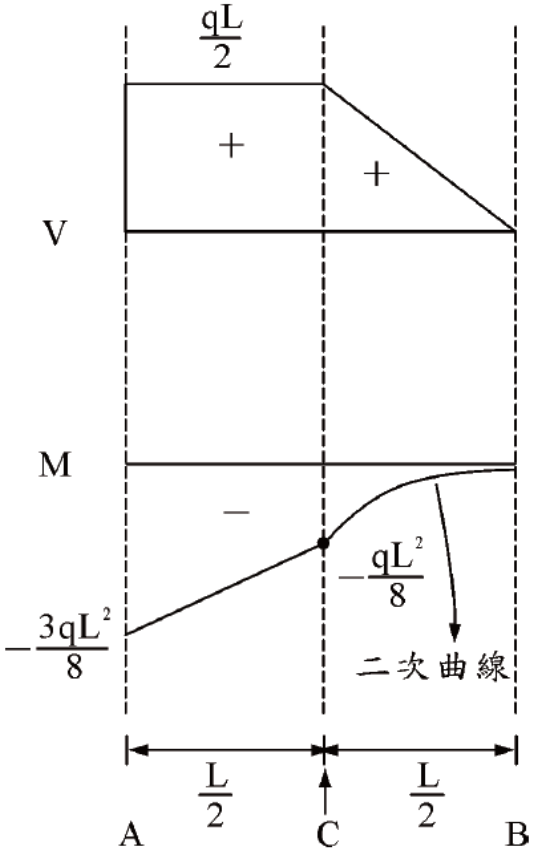
$$\Sigma F_y = 0 \Rightarrow R_A = \frac{qL}{2}$$

$$\Sigma M_A = 0$$

$$\Rightarrow M_A = \frac{qL}{2} \times \left(\frac{L}{2} + \frac{1}{2} \times \frac{L}{2}\right) = \frac{3qL^2}{8} (\circlearrowleft)$$

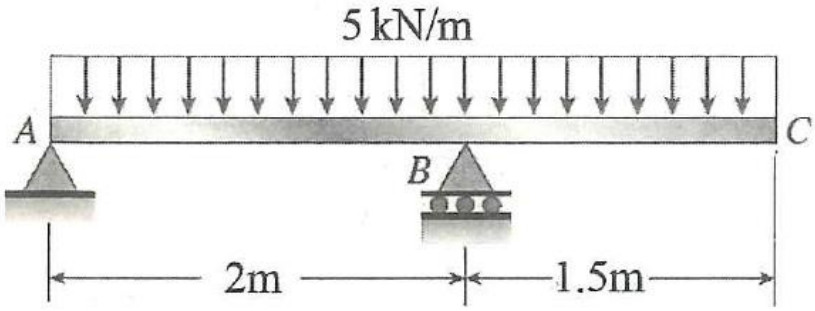


(2) 畫剪力彎矩圖：



範例 10

The uniformly loaded beam ABC has simple supports at A and B and an overhang BC. Draw the shear-force and bending-moment diagrams for this beam. (20%)



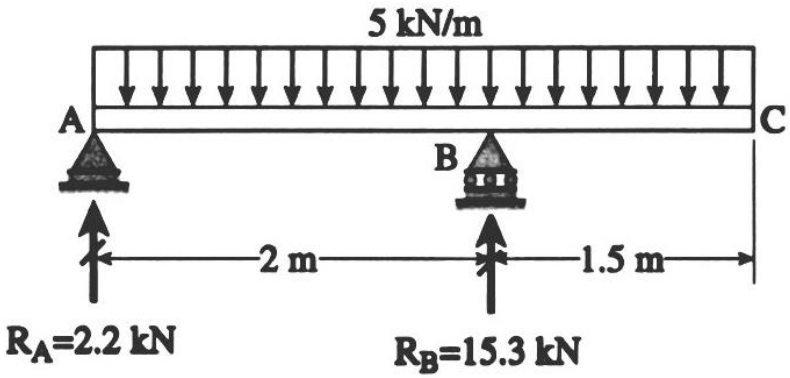
(103 台大工科) (106 地特三等)

【中文翻譯】

一均勻分布負載強度作用在樑 ABC 上，繪出此樑的剪力與彎矩圖。

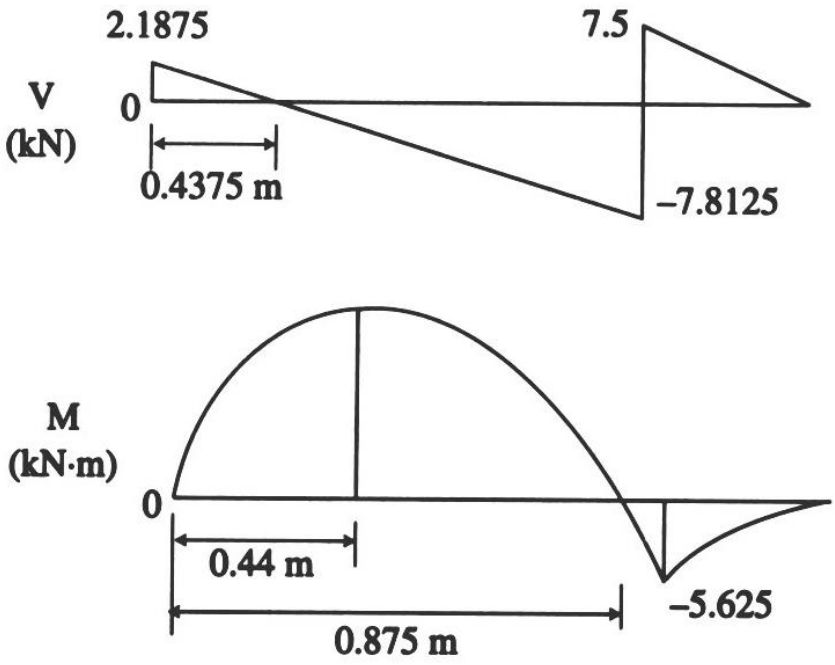
【解析】

(一)取整體自由體圖



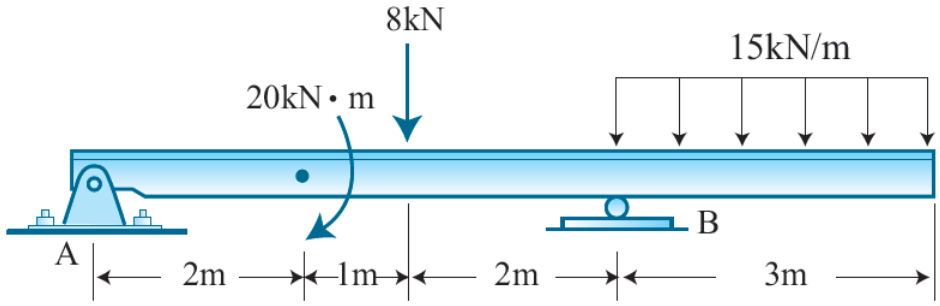
$$\Sigma M_A = 0 \qquad 17.5 \times 1.75 = R_B \times 2 \qquad R_B = 15.3125\ \text{kN}$$

$$\Sigma F_y = 0 \qquad R_A + R_B = 17.5 \qquad R_A = 2.1875\ \text{kN}$$


2.1875
7.5
V
(kN)
0
0.4375 m
–7.8125
M
(kN·m)
0
0.44 m
–5.625
0.875 m

範例 11

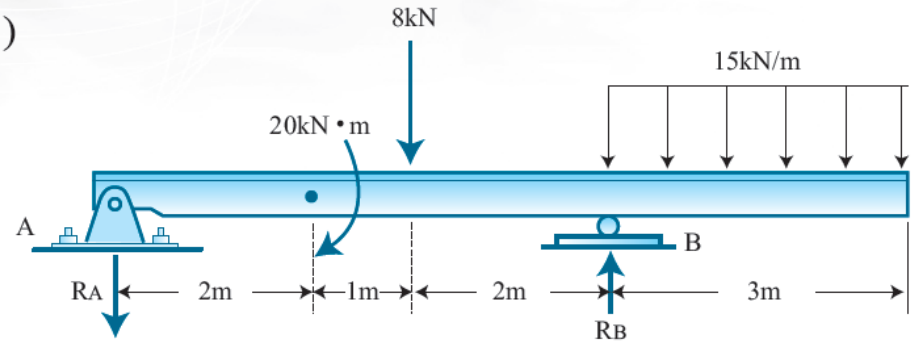
如圖所示，試繪出樑(Beam)之剪力圖與彎矩圖，並標示最大剪力與最大彎矩為何？



[機械高考]

【解析】

(1)

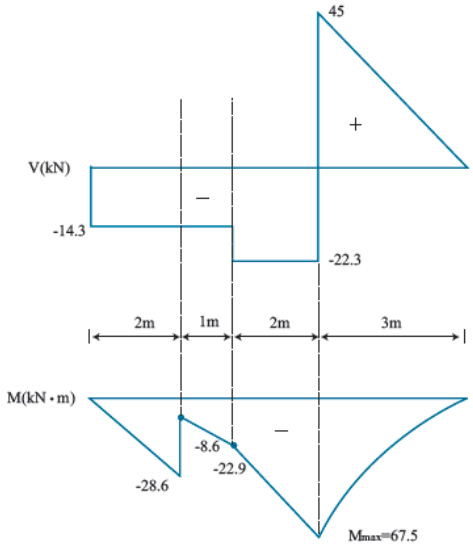


先求支承反力

$$\sum M_B = 0 \Rightarrow 5R_A = 20\times10^3 + 15\times10^3 \times 3\times1.5 - 8\times10^3 \times 2$$

$$R_A = 14300\text{N} = 14.3\text{kN}(\downarrow) \quad \sum F_y = 0 \Rightarrow R_B = 67300\text{N} = 67.3\text{kN}(\uparrow)$$

(2)畫剪力圖與彎矩圖：

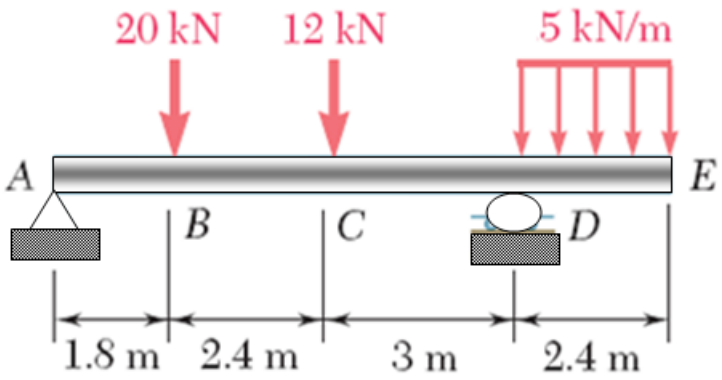


(3)$|M_{max}|$值為剪力圖正面積之值或負面積和：

$$M_{max} = \frac{1}{2}\times45\times3 = 67.5\text{kN}\cdot\text{m}$$

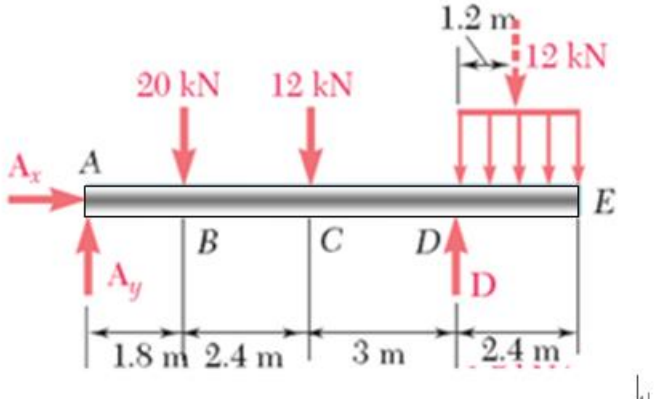
範例12

4. (10%) For the beam and loading shown, determine the magnitude of the maximum bending moment in the beam.



(99年成大工科)

【解析】



$\sum M_A = 0:$

$D(7.2\text{ m}) - (20\text{ kN})(1.8\text{ m}) - (12\text{ kN})(4.2\text{ m})$

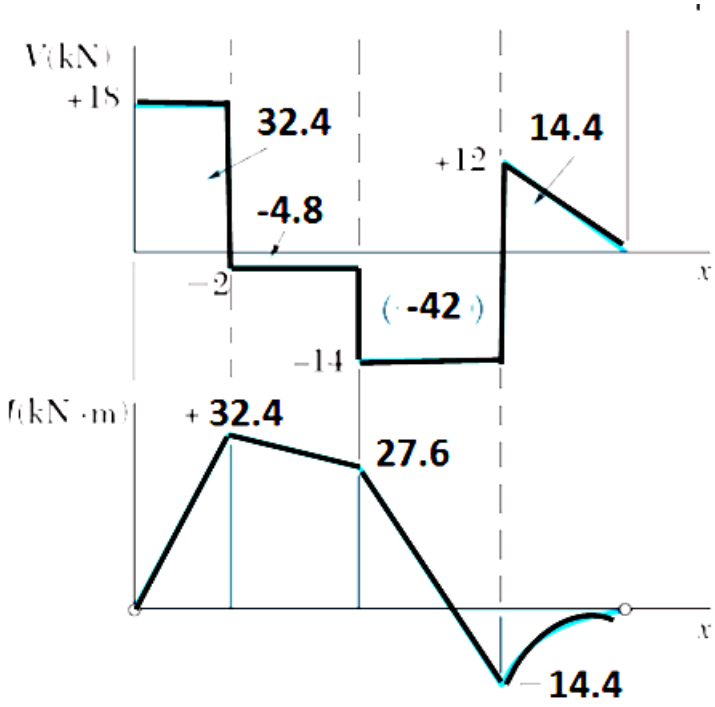
$\quad - (12\text{ kN})(8.4\text{ m}) = 0$

$\boxed{D = 26\text{ kN}}$

$\sum F_y = 0:$

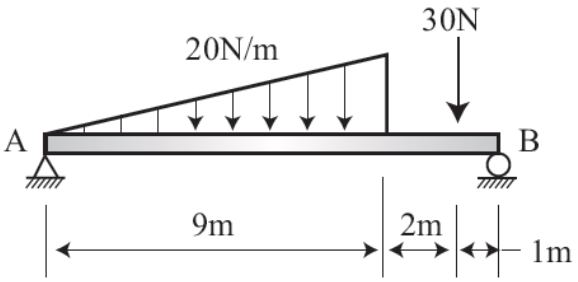
$A_y - 20\text{ kN} - 12\text{ kN} + 26\text{ kN} - 12\text{ kN} = 0$

$\boxed{A_y = 18\text{ kN}}$


V(kN)
+18
32.4
-4.8
+12
14.4
x
−2
(-42)
−14
M(kN·m)
+ 32.4
27.6
x
− 14.4

範例 13

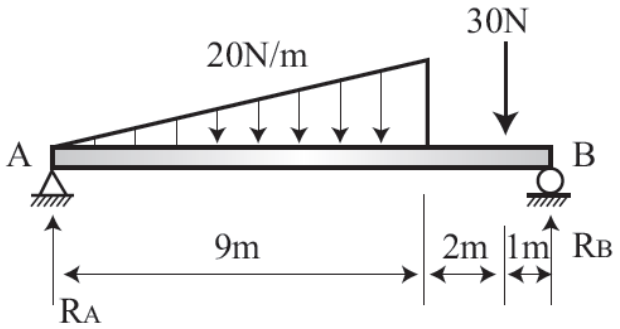
試求圖中樑上之剪力圖、彎矩圖，並求出在A 與B 處的反作用力大小。(93 年機械普考)



(96 年交大應用力學)

【解析】

(1) 如圖所示：



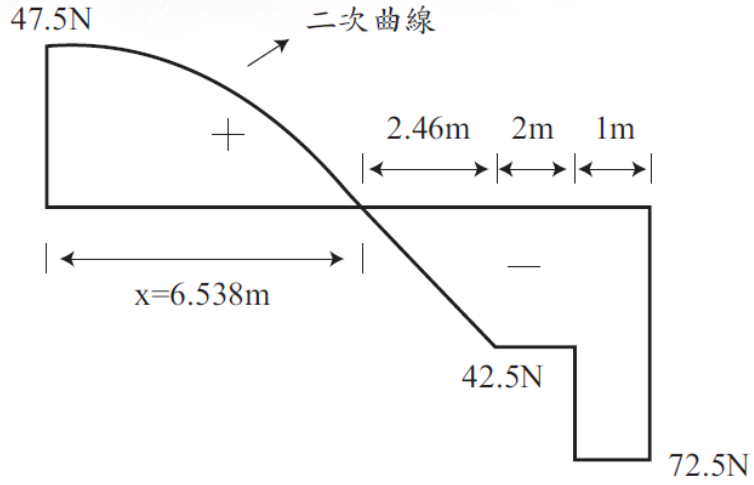
$$\sum M_A = 0$$

$$\Rightarrow 20\times 9\times \frac{1}{2}\times(9\times\frac{2}{3}) + 30\times(9+2) = 12\ R_B$$

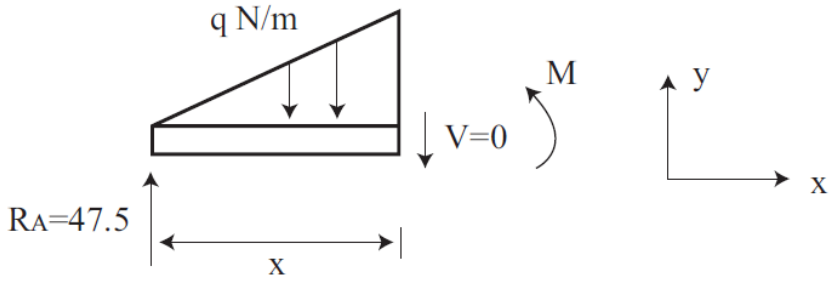
$$R_B = 72.5(N)\uparrow$$

$$\sum M_B = 0 \Rightarrow 12\,R_A = 20\times 9\times \frac{1}{2}\times(3+9\times\frac{1}{3}) + 30\times 1 \Rightarrow R_A = 47.5(N)\uparrow$$

(2) 剪力圖：



取左半端剪力為零之自由體圖



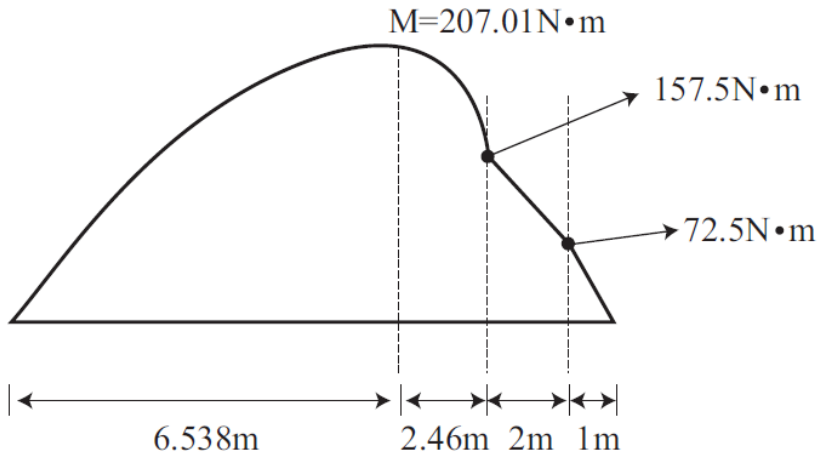
$$\sum F_y = 0 \Rightarrow 20 \times 9 \times \frac{1}{2} \times \left(\frac{x}{9}\right)^2 = 47.5$$

三角形面積

$$x = 6.538\text{m} \quad q = 20 \times \frac{6.538}{9} = 14.529\text{N/m}$$

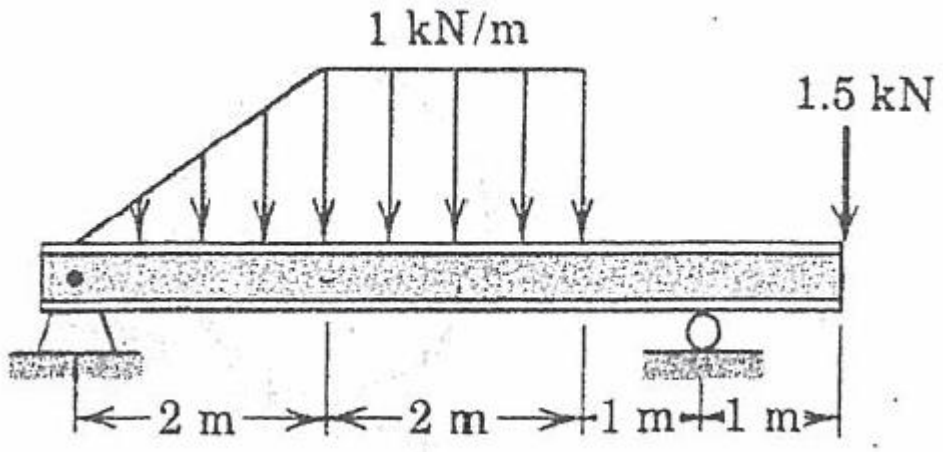
$$\sum M_A = 0 \Rightarrow M = 14.529 \times 6.538 \times \frac{1}{2} \times 6.538 \times \frac{2}{3} = 207.01\text{N} \cdot \text{m}$$

(3) 畫彎矩圖：



範例 14

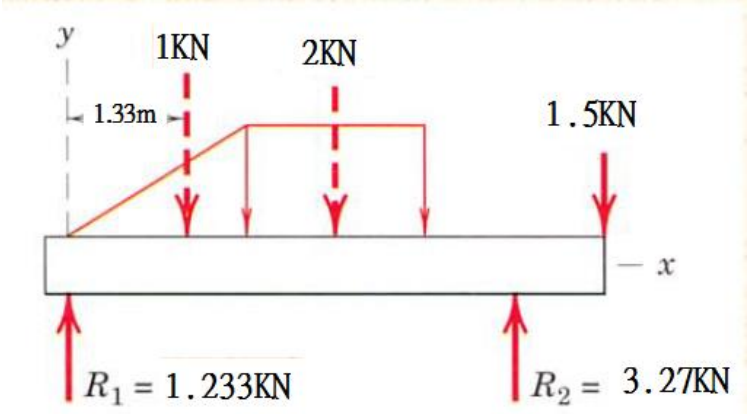
Draw the shear-force and bending-moment diagrams for the loaded beam and determine the maximum moment ***M*** and its x from the left end. (15%)



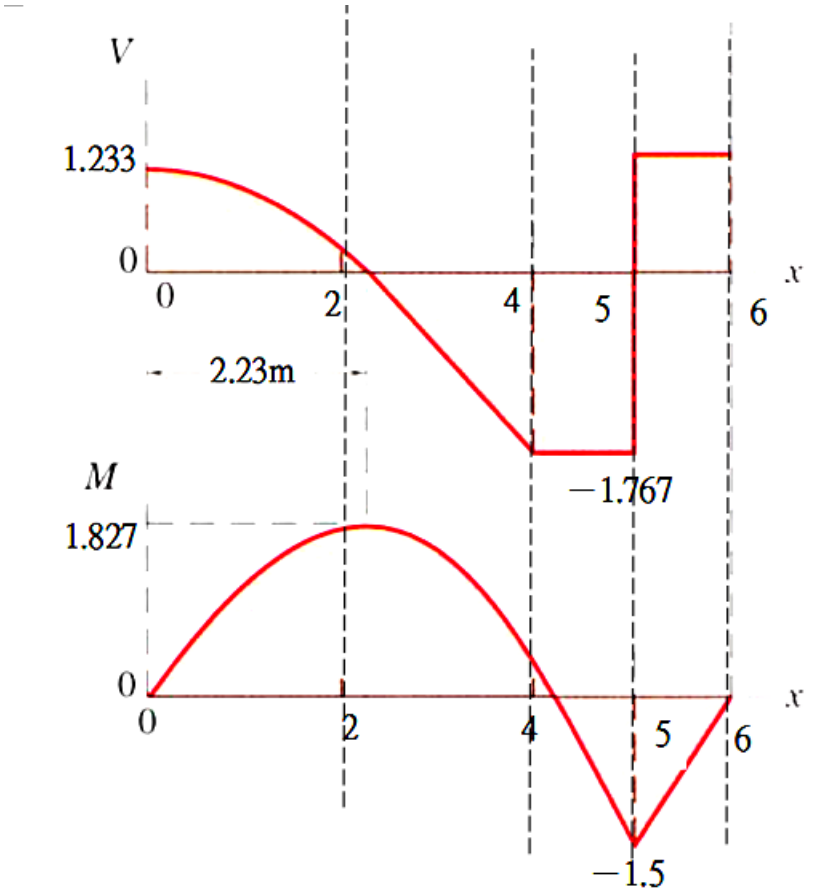
(100 年台大醫工)

【解析】

(一)先求支承反力



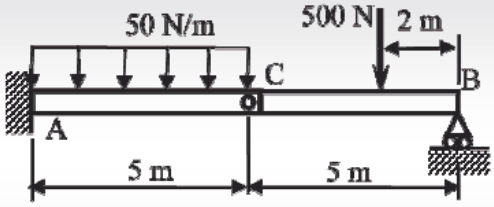
(二)畫力矩灣矩圖



最大彎矩距左端2.23m

範例 15

如圖所示之連續樑 ACB，左側為固定端、右側為簡支端、中間 C 為銷接（pinned）且無摩擦。



(1) 試求連接銷 C 處之反力（reaction）。

(2) 繪出整體 ACB 樑之剪力圖（shear diagram）。

(3) 繪出整體 ACB 樑之彎矩圖（bending moment diagram）。

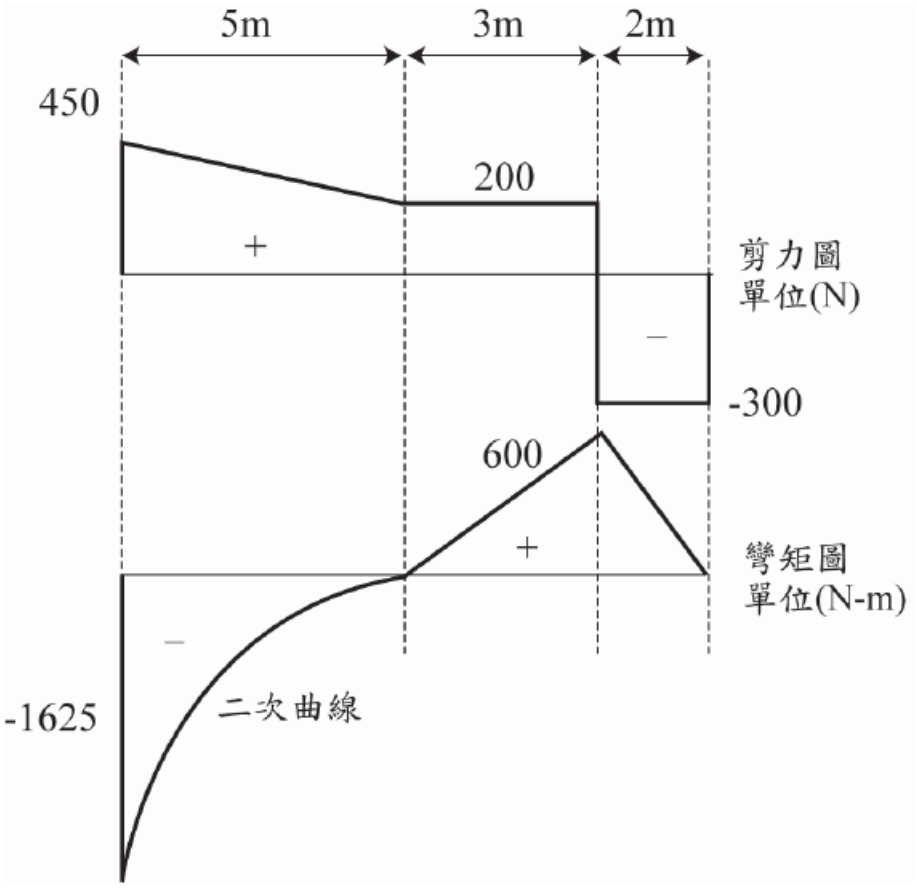
(100 年地特三等)

【解析】

解 (1) 取 BC 自由體圖

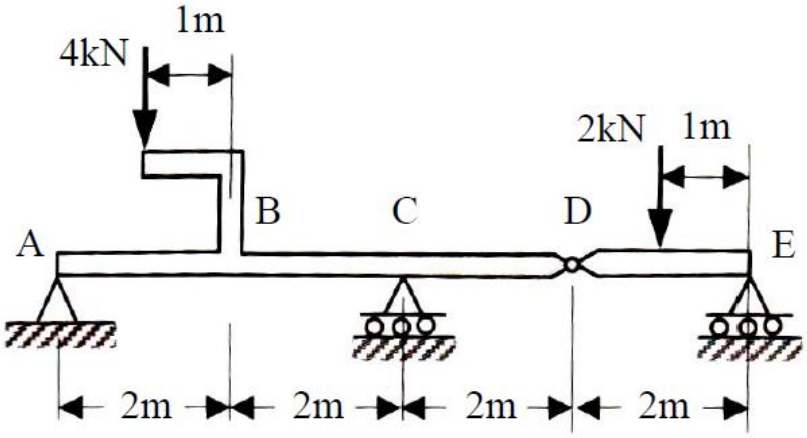
$M_B = 0 \Rightarrow R_c = 200(N)$

(2)



範例 16

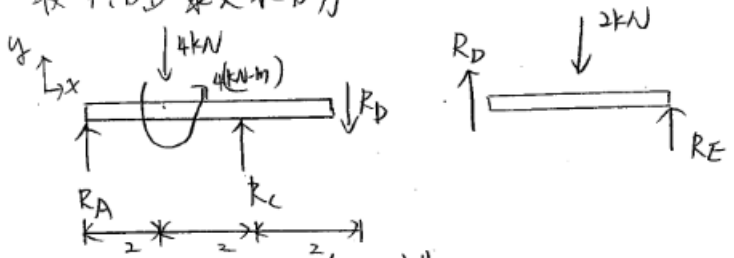
請畫出下圖所示組合樑 ABCDE 之剪力圖與彎矩圖。（20 分）



(101 機械高考) (102 台大工科) (106 台科營建)

【解析】

取 F.B.D 求支承力



由 DE F.B.D 為對稱結構

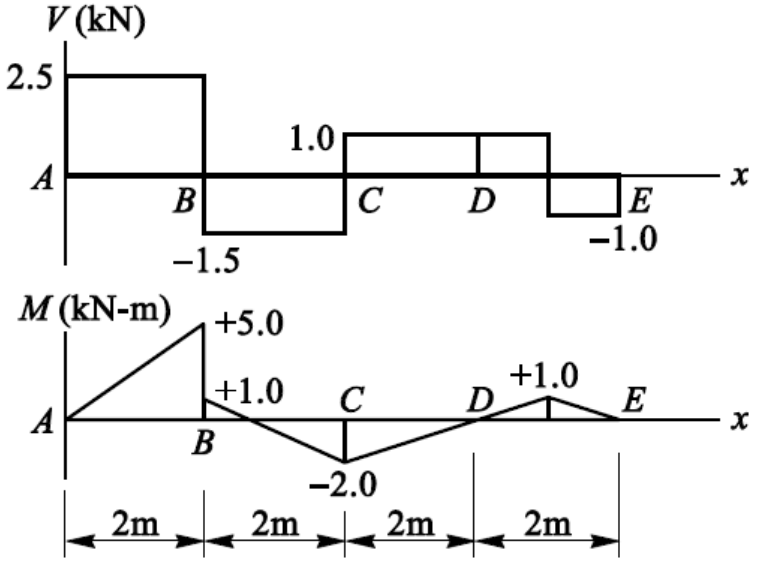
故 $R_E = R_D = 1(kN)$

由 $\Sigma M_A = 0$

$4R_C = 1\times6 + 4\times2 - 4 \Rightarrow R_C = 2.5(kN)$

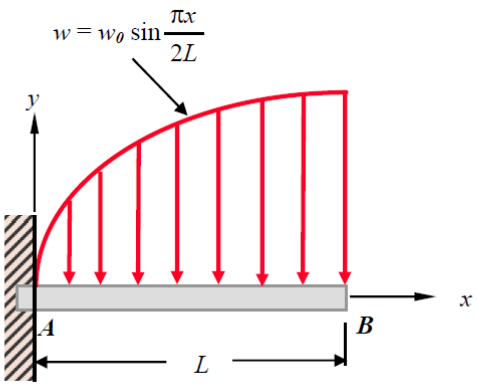
由 $\Sigma F_y = 0$

$R_A = 4 - 2.5 + 1 = 2.5(kN)$


V (kN)
2.5
1.0
A
B
C
D
E
x
−1.5
−1.0
M (kN-m)
+5.0
+1.0
+1.0
A
B
C
D
E
x
−2.0
2m
2m
2m
2m

範例 17

無重量懸臂樑 ***AB*** 受分佈力作用如下圖。試繪懸臂樑 ***AB*** 之自由體圖（Free Body Diagram），剪力分佈圖（Shear force diagram）及彎矩分佈圖（Bending moment diagram）。並試求最大彎矩之大小及其發生位置。（20 分）



(105 機械高考)

【解析】

$$(1)\ \frac{dV}{dx} = -w \Rightarrow V(x) = \omega_0\left(\frac{2L}{\pi}\right)\cos\left(\frac{\pi x}{2L}\right) + C_1 \qquad \text{(a)}$$

$$\frac{dM}{dx} = V \Rightarrow M(x) = \omega_0\left(\frac{2L}{\pi}\right)^2 \sin\left(\frac{\pi x}{2L}\right) + C_1 x + C_2 \qquad \text{(b)}$$

考慮邊界條件

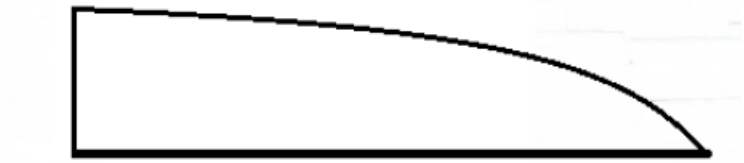
$V(L) = 0 \quad M(L) = 0$ 帶入(a)(b)兩式得 $C_1 = 0$ 和 $C_2 = -\omega_0\left(\frac{2L}{\pi}\right)^2$

解出剪力與彎矩函數

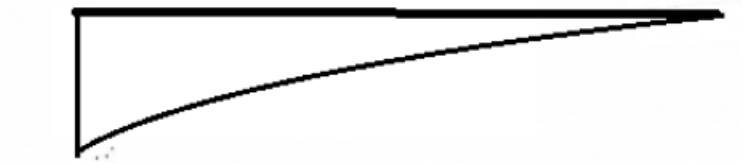
$$V(x) = \omega_0\left(\frac{2L}{\pi}\right)\cos\left(\frac{\pi x}{2L}\right) \qquad \text{(c)}$$

$$M(x) = \omega_0\left(\frac{2L}{\pi}\right)^2\left[\sin\left(\frac{\pi x}{2L}\right) - 1\right] \text{(d)}$$

SFD



BMD



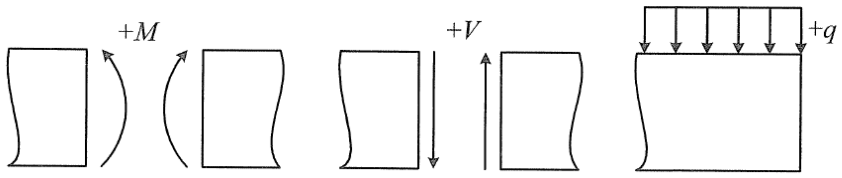
(2)當 $x=0$ 可分別求出梁內最大剪力與彎矩

$$V(x=0)=\omega_0(\frac{2L}{\pi})=V_{\max}$$

$$M(x=0)=-\omega_0(\frac{2L}{\pi})^2=M_{\max}$$

範例 18

如果定義梁的外力 q，剪力 V，與彎矩 M 的正向如圖所示，請指出哪一個關係是正確的。



(A) $\int_a^b q\,dx = V(b) - V(a)$　　(B) $\frac{dM}{dx} = -V$　　(C) $\frac{dV}{dx} = q$

(D) $\int_a^b V\,dx = M(b) - M(a)$　　(E) $\frac{d^2M}{dx^2} = q$

(台大機械)

【解析】

$$\frac{dV}{dx} = -q \Rightarrow \int_a^b q\,dx = V(a) - V(b)$$

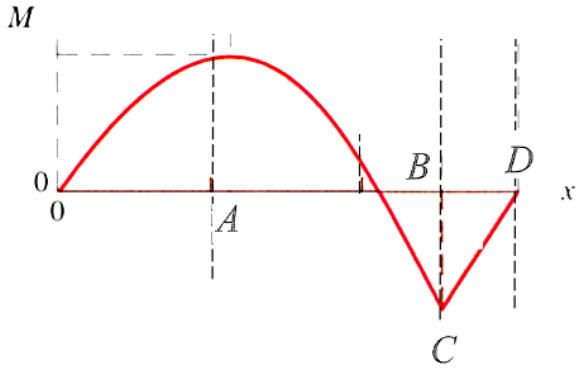
$$\frac{dM}{dx} = V \Rightarrow \int_a^b V\,dx = M(b) - M(a)$$

故選(D)

範例 19

參考下列彎矩圖，下列敘述何者正確?

(A)D 點為自由端不受力 (B) B 點位移最大 (C)C 點受集中力作用 (D) A 點位移最大



(台大機械)

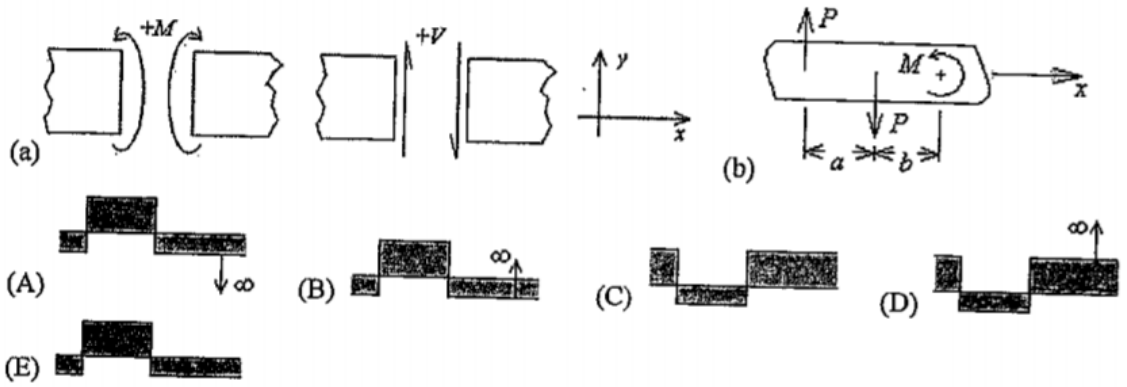
【解析】

C點受集中力剪力圖有不連續面

故選(C)

範例 20

如果定義樑的剪力 V 與彎矩 M 的正向如圖(a)所示。附圖(b)爲樑的一段，其中 $M = Pa$ ，請選出在這一段中合理的剪力圖。



(台大機械)

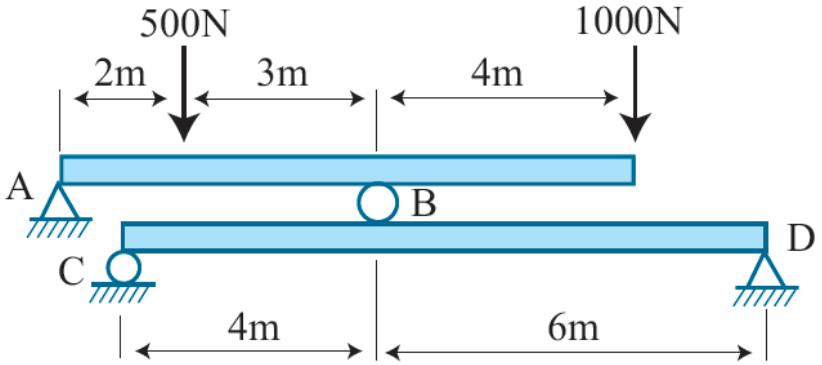
【解析】

故選(C)

精選試題演練—計算題型(樑之內力)

例 1

試求D支承的反作用力。



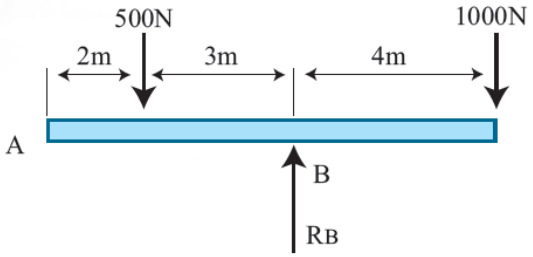
【解析】

(1)作AB桿之自由體圖

$[\sum M_A = 0]$

$500\times 2+1000\times 9=R_B\times 5$

$\therefore R_B=2000(N\uparrow)$

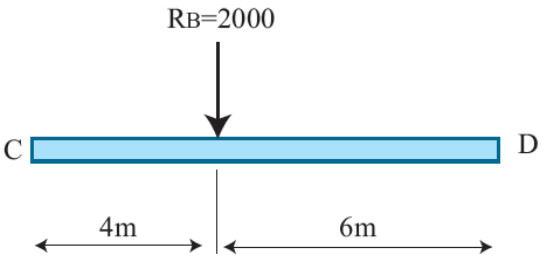


(2)作CD桿之自由體圖

$[\sum M_C = 0]$

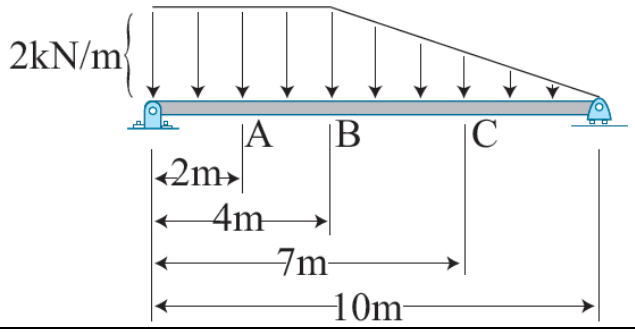
$2000\times 4=R_D\times 10$

$\therefore R_D=800(N\uparrow)$



例 2

圖中A點所受之力矩應為：



【解析】

先求左端支承反力

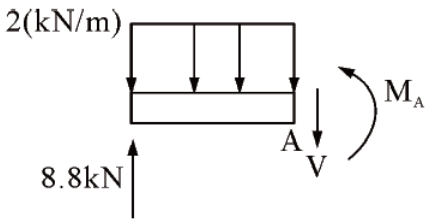
$\Sigma M=0$

$$R_{左}=\frac{1}{10}\left[2\times4\times(6+2)+\frac{1}{2}\times2\times6\times6\times\frac{2}{3}\right]=8.8\ (kN)$$

取A點以左之自由體圖如右

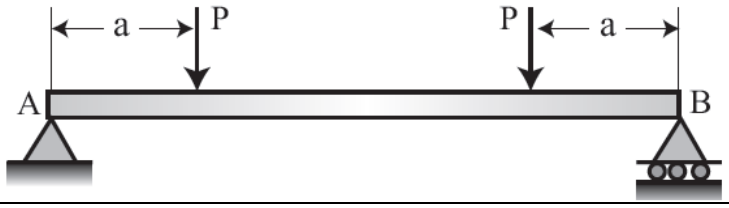
$\Sigma M_A=0$

$\Rightarrow M_A=8.8\times2-2\times2\times1=13.6\ (kN\cdot m)$

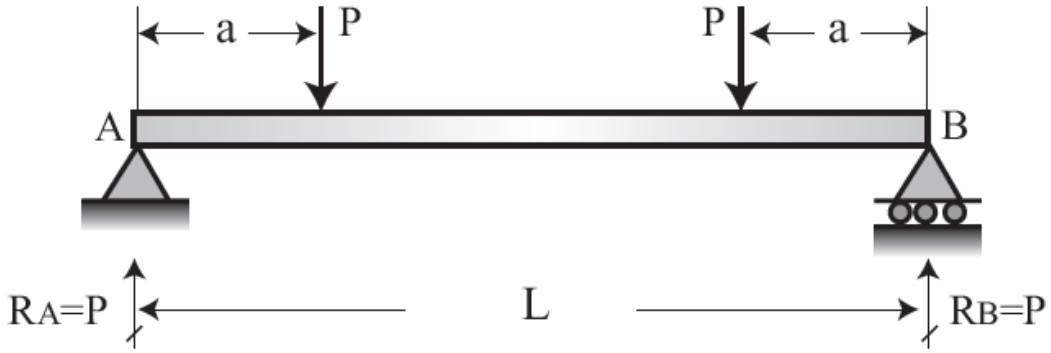


例 3

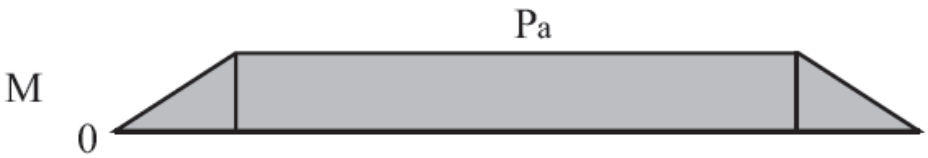
試繪出如圖所示之剪應力和力矩圖(shear and moment diagram)。



【解析】

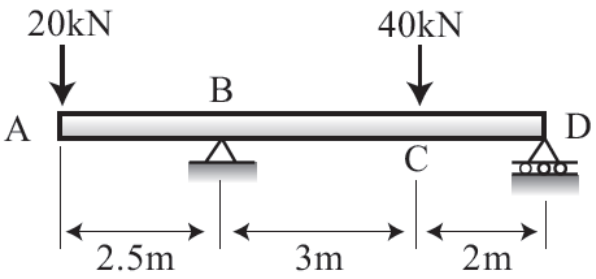




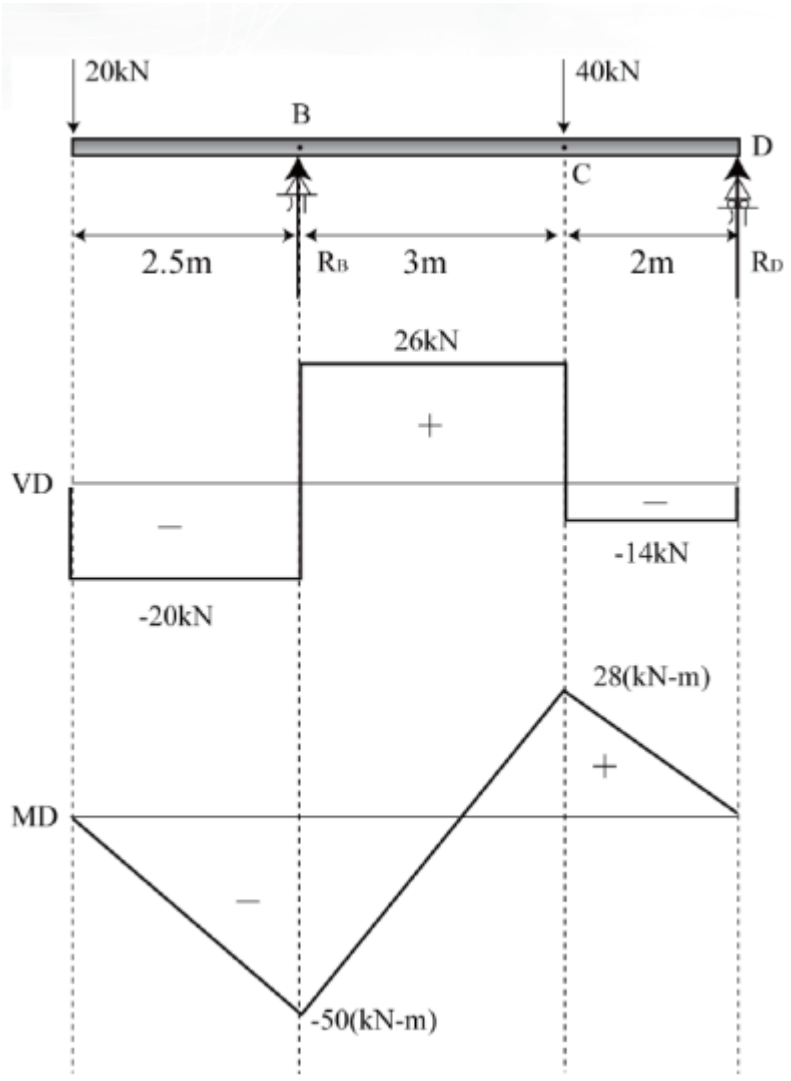


例 4

如圖所示之延伸樑，已知其截面積為均勻：(1)試繪剪力圖及彎曲力矩圖；(2)請找出發生最大彎曲應力及最大剪應力的截面位置，並說明其理由。



【解析】



(1) 先求R_B、R_D後再繪製剪力圖(V_D)及彎矩圖(M_D)

$\sum M_B = 0 \quad \Rightarrow 20\times 2.5 + R_D \quad 5 = 40\times 3 \quad \Rightarrow R_D = 14\,(N)$

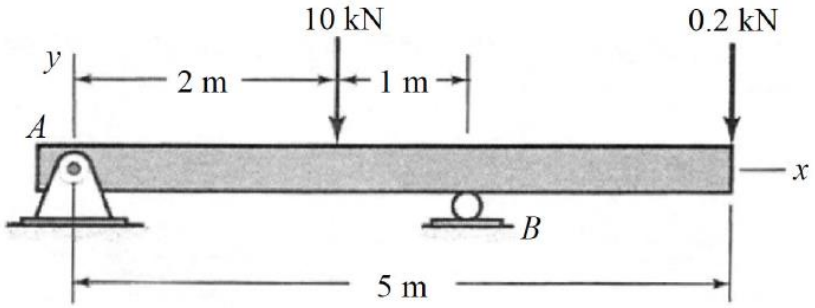
$\sum F_g = 0 \quad \Rightarrow R_B = 46\,(N)$

$M_B = -20\times 2.5 = -50$ (kN -m)　$M_C = -50 + 26\times 3 = 28$ (kN-m)

(2) 由於最大彎距M_B發生於B處，因此發生最大彎曲應力為B處，上下邊緣處最大剪力R_B發生於B處，因此發生最大剪應力的截面位置為B處截面中心處。

例 5

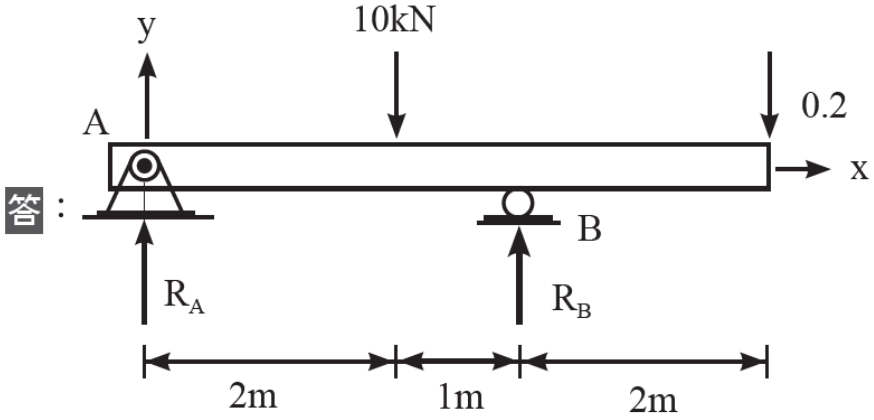
如圖表示一個等截面的橫樑（prismatic beam）及其受力狀態，A為銷接（pin）支持，B為滾柱（roller）支持。求解A與B處的反作用力，並繪製剪力圖（shear diagram）。



【106 普考】

【解析】

答：

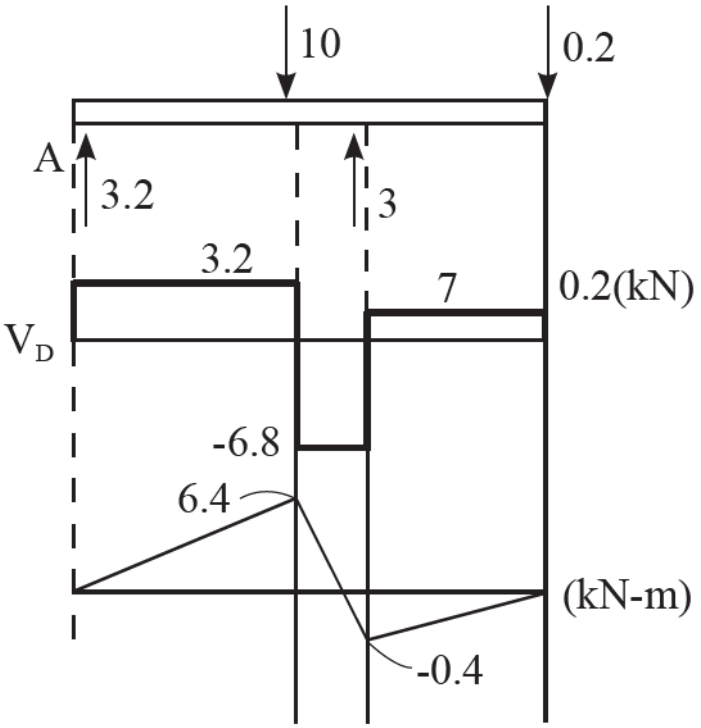


$$\circlearrowleft \sum M_A = 0$$

$$0.2\times 5 + 10\times 2 - R_B \times 3 = 0 \Rightarrow R_B = 7$$

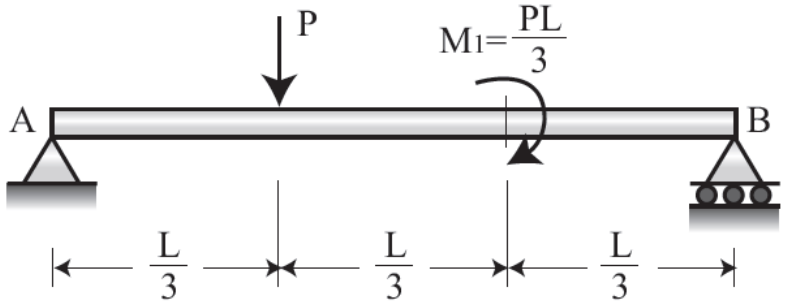
$$+\uparrow \sum F_y = 0$$

$$R_A - 10 - 0.2 + 7 = 0 \Rightarrow R_A = 3.2(\text{kN})$$

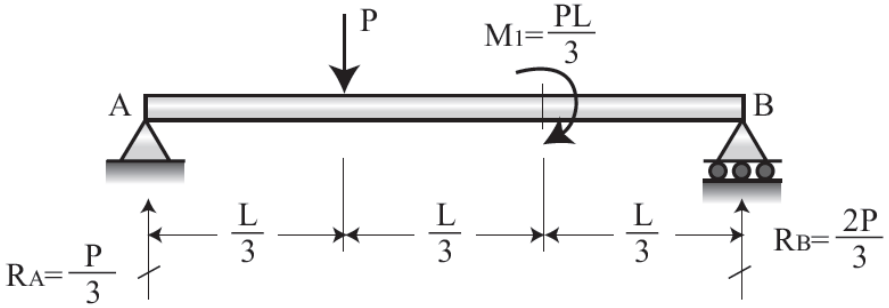

10
0.2
A
3.2
3
3.2
7
0.2(kN)
V_D
-6.8
6.4
(kN-m)
-0.4

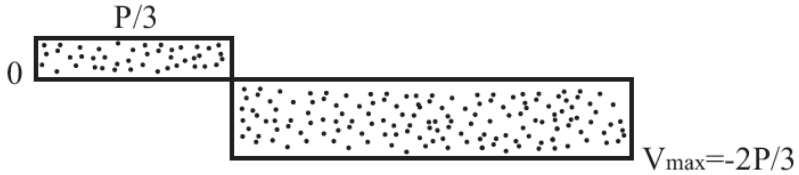
例 6

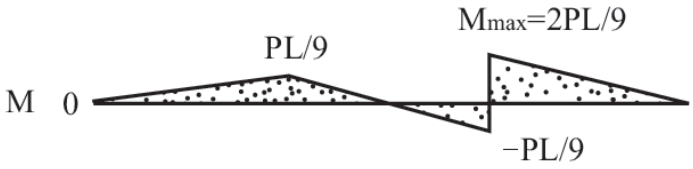
試繪出如圖所示之剪應力和力矩圖(shear and moment diagram)。



【解析】

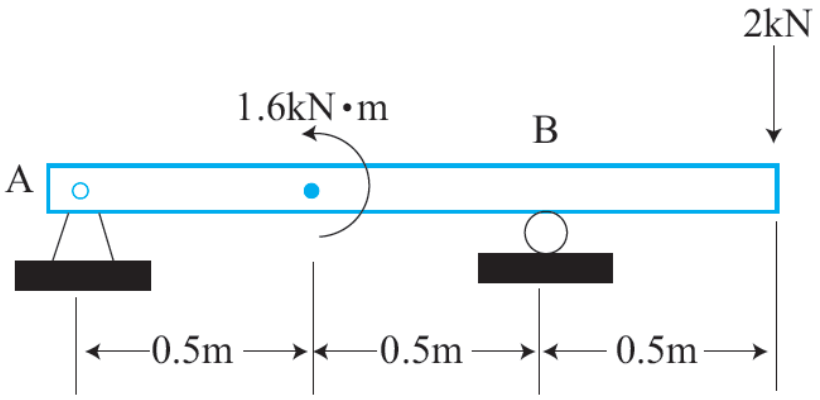






例 7

試繪出如圖所示之剪應力和力矩圖(shear and moment diagram)。



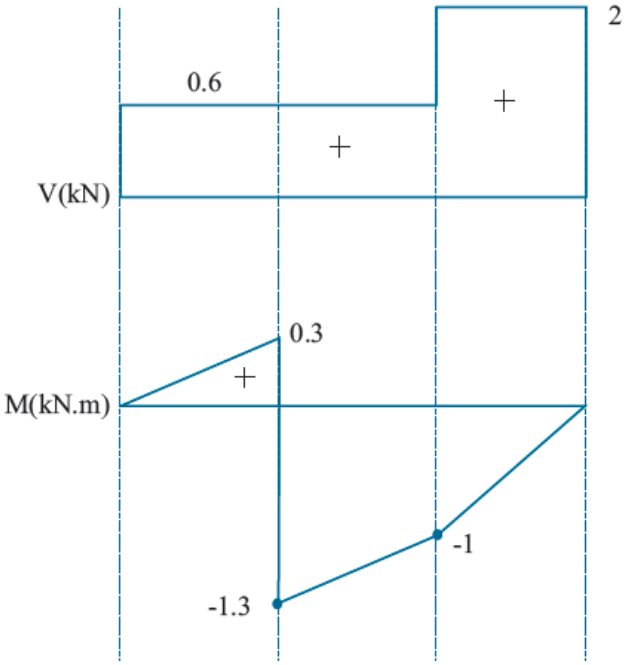
(地特三等)

【解析】

(1) 先求支承反力：$\sum M_A = 0 \Rightarrow R_B = 1.4\text{kN}(\uparrow)$

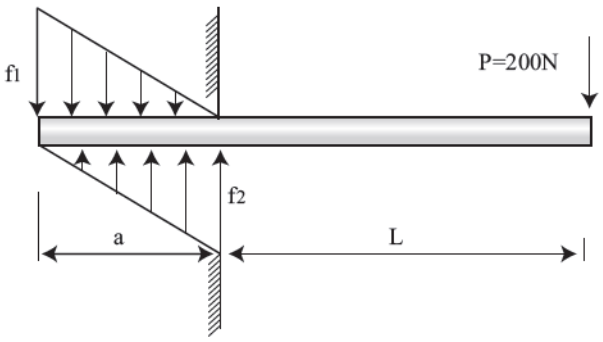
$\sum M_B = 0 \Rightarrow R_A = 0.6\text{kN}(\uparrow)$

(2)剪力圖與彎矩圖：



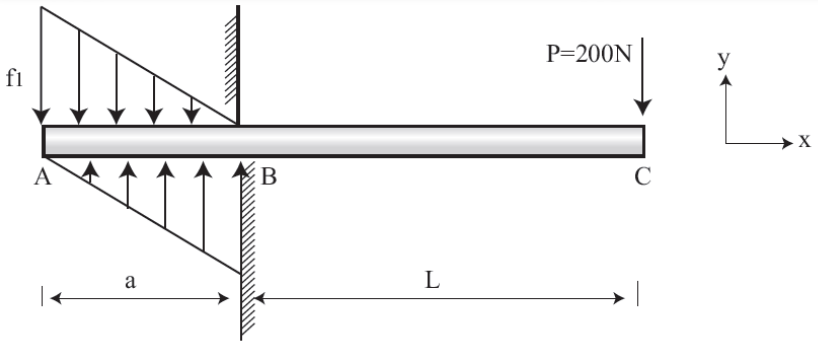
例 8

如圖所示，懸臂樑左端插入直立牆內，牆內長度為a 公尺，牆外長度為L公尺，右端有一集中力P=200N。假設插入牆內端所受的反作用力分布為強度沿樑長度變化，如圖所示。求其最大強度f_1 和f_2。畫出牆外部份樑斷面的剪力圖和彎矩圖。



[機械普考]

【解析】

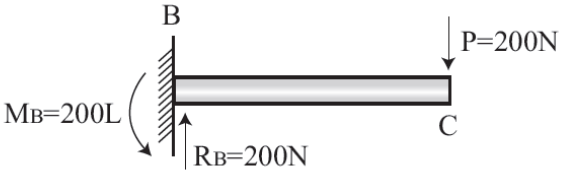


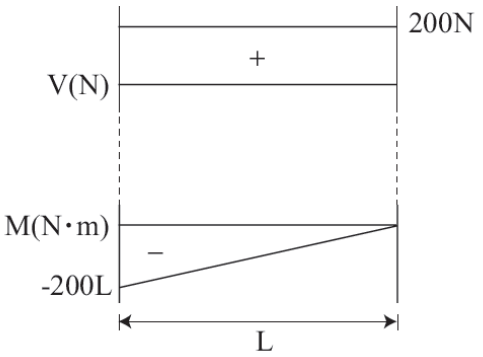
(1) $\Sigma F_y = 0 \Rightarrow f_1 a \times \frac{1}{2} + 200 = f_2 a \times \frac{1}{2}$ ——①

$\Sigma M_B = 0 \Rightarrow f_1 \times a \times \frac{1}{2} \times \frac{2}{3} a = f_2 \times a \times \frac{1}{2} \times \frac{1}{3} a + 200 \times L$ ——②

由①②$\Rightarrow f_2 = (\frac{1200L + 800a}{a^2})$，$f_1 = (\frac{1200L + 400a}{a^2})$

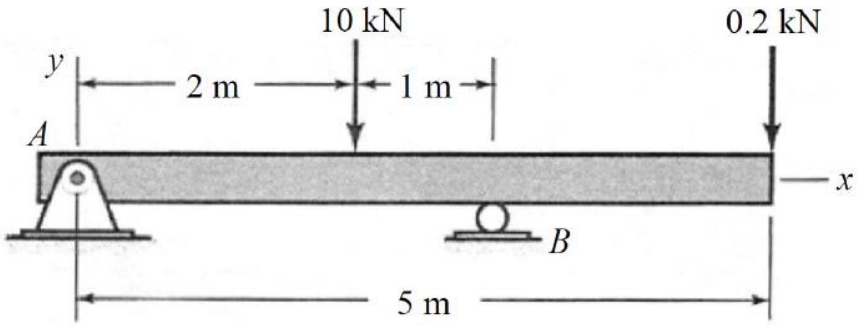
(2) 牆外部樑受力圖及剪力圖與彎矩圖：





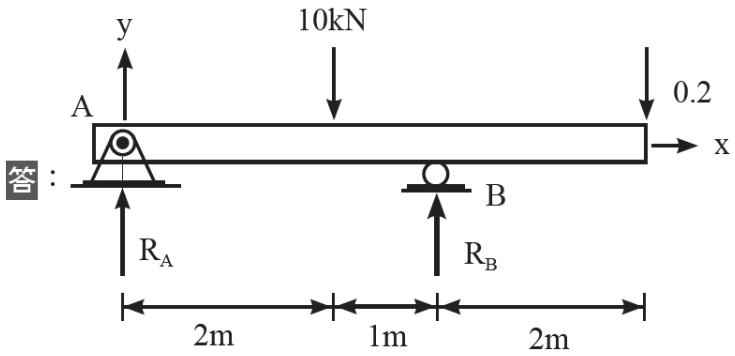
例 9

如圖表示一個等截面的橫樑（prismatic beam）及其受力狀態，A為銷接（pin）支持，B為滾柱（roller）支持。求解A與B處的反作用力，並繪製剪力圖（shear diagram）。



【106 普考】

【解析】



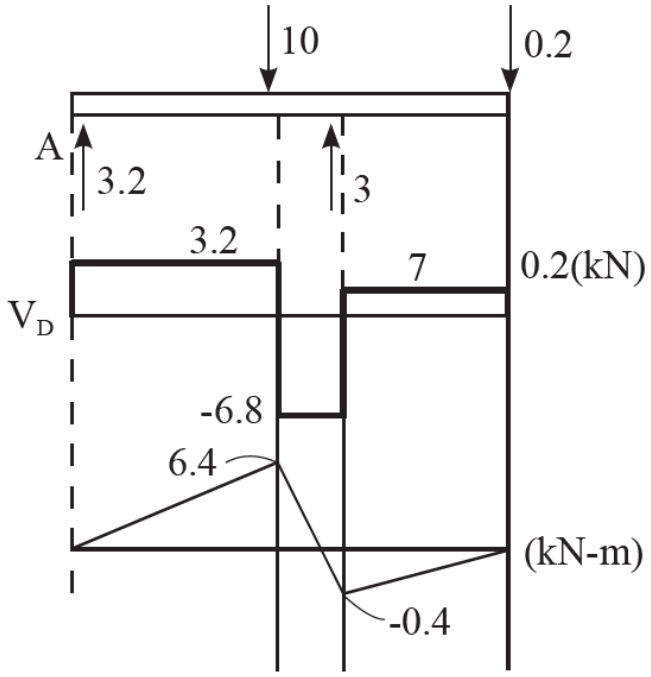
答：

$$\circlearrowleft \sum M_A = 0$$

$$0.2\times 5+10\times 2-R_B\times 3=0 \Rightarrow R_B=7$$

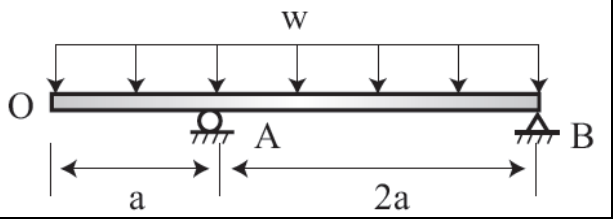
$$+\uparrow \sum F_y = 0$$

$$R_A - 10 - 0.2 + 7 = 0 \Rightarrow R_A = 3.2(\text{kN})$$



例 10

右圖所示的樑受到均佈負載w，請求出此樑所受到的最大剪力及最大彎矩(樑的重量不考慮)。(94年機械普考)



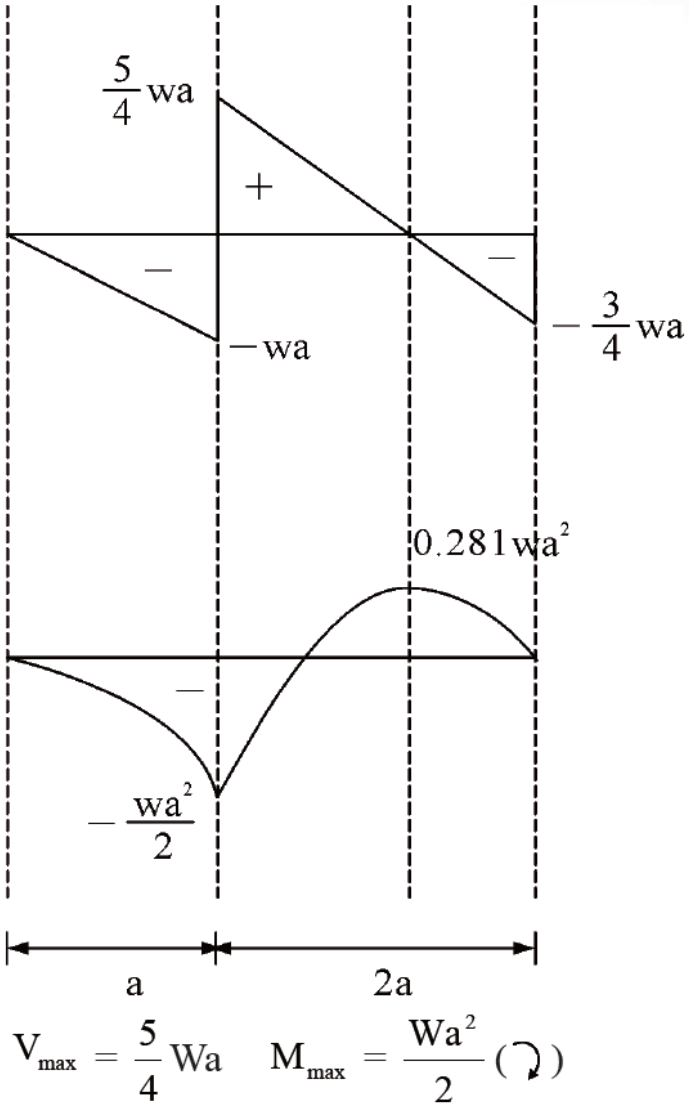
【解析】

(1) 先求支承反力：

$$\sum M_B = 0 \Rightarrow 3Wa \times \frac{3}{2}a = R_A \times 2a \quad \Rightarrow R_A = \frac{9}{4}W \cdot a$$

$$\sum M_A = 0 \Rightarrow R_B = \frac{3}{4}Wa$$

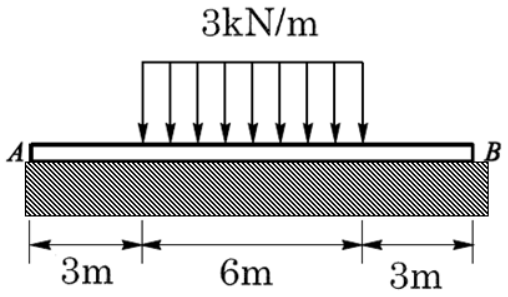
(2) 畫剪力圖及彎矩圖：



$$V_{max} = \frac{5}{4}Wa \qquad M_{max} = \frac{Wa^2}{2}(\circlearrowleft)$$

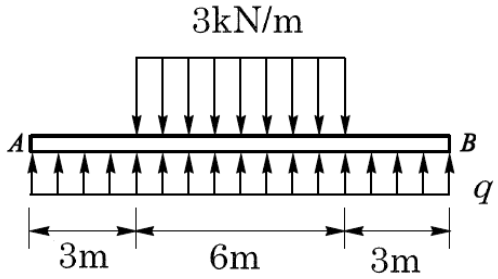
例 11

Assuming the upward reaction of the ground on beam AB to be uniformly distributed, calculate (a) the maximum absolute value of the shear and (b) the maximum absolute value of the bending moment.

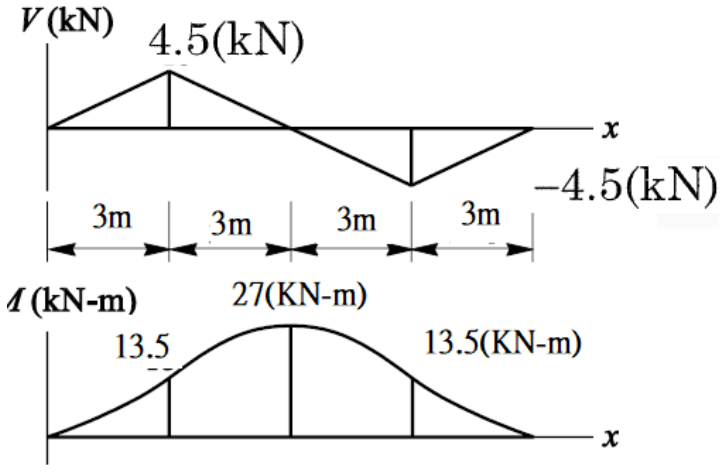


【98 成大工科】

【解析】

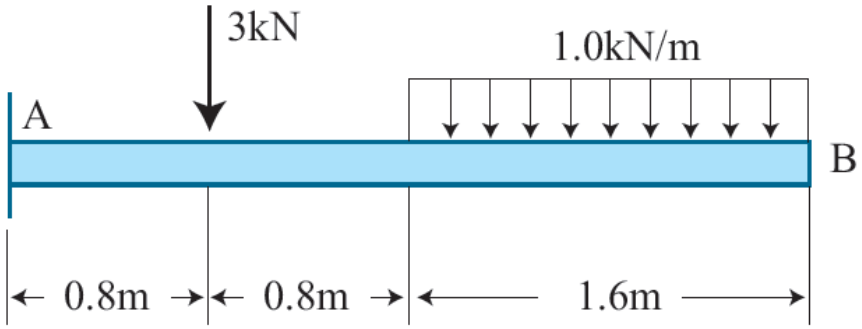


$$\sum F = 0 \Rightarrow 3\times 6 = q\times 12 \Rightarrow q = 1.5\ (\text{kN/m})$$



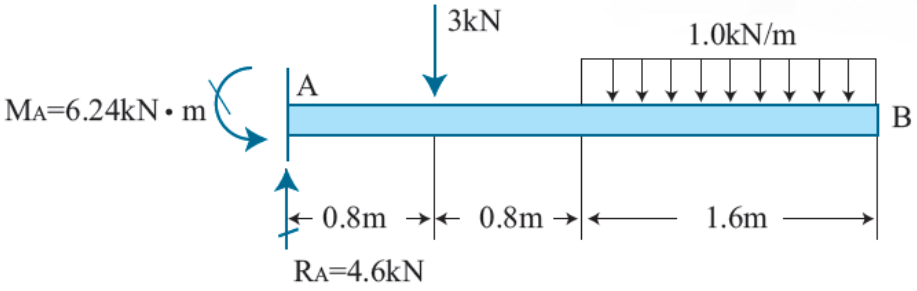
例 12

如圖所示，試繪出樑(Beam)之剪力圖與彎矩圖



【解析】

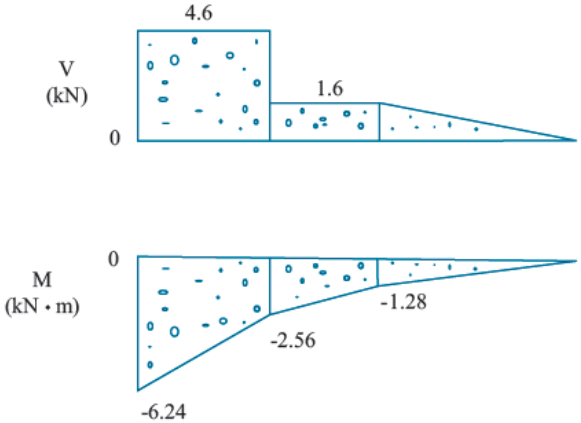
(1)先求支承反力



$$\sum M_A = 0 \Rightarrow M_A = 3\times 0.8 + 1\times 1.6\times (0.8 + 0.8 + \frac{1.6}{2}) = 6.24(KN - m)$$

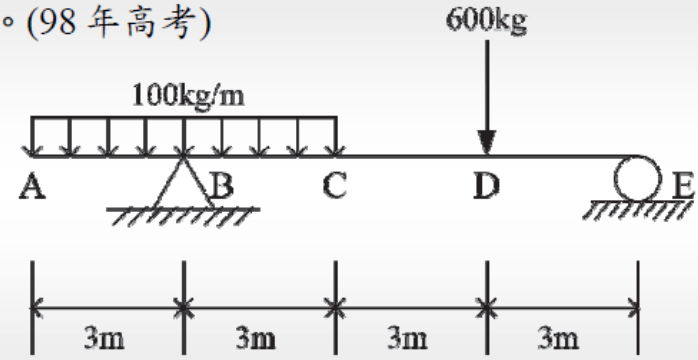
$$\sum F_y = 0 \Rightarrow R_A = 3 + 1\times 1.6 = 4.6(KN)$$

(2)畫剪力圖與彎矩圖：



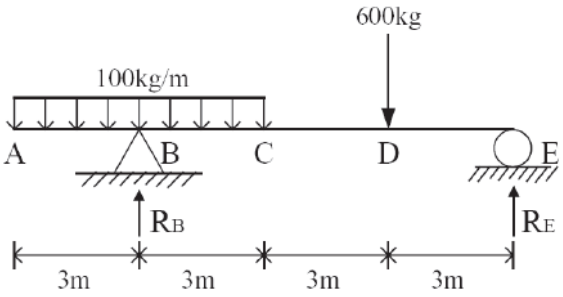
例 13

試繪出下圖結構之剪力圖與彎矩圖。(98 年高考)



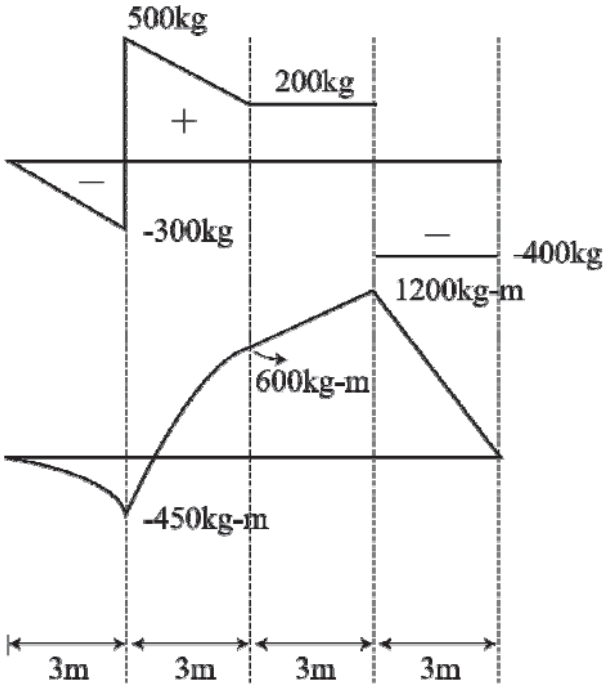
【解析】

解



$\sum M_B = 0 \quad \Rightarrow R_E = 400 \ (kg)$

$\sum F_y = 0 \quad \Rightarrow R_B = 800 \ (kg)$

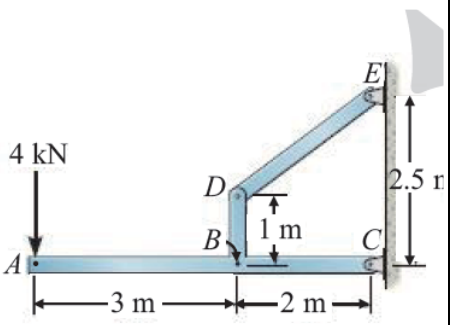


例 14

如圖所示之外伸樑（overhanging beam）ABC具有一凸出臂BD，在樑的端點A承受4kN之負載：

(1)試求在支撐點C、E分別之反力（reactions）。

(2)繪出樑ABC之剪力圖（shear diagram）及彎矩圖（bending moment diagram）。



【103 機械普考】

【解析】

(1) 取整體$\sum M_C = 0$

$4\times(3+2) = R_E \times 2.5 \Rightarrow R_E = 8(kN)$

取ABC樑部分

$\sum F_y = 0 \quad 4 + R_{Cy} = R_{Dy} \Rightarrow R_{Cy} = 2(kN)$

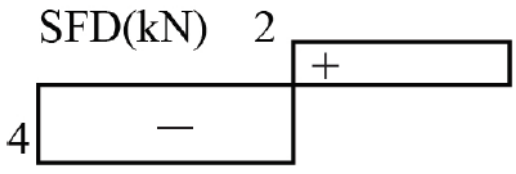
$\sum F_x = 0 \quad R_{Dx} = R_{Cx} \Rightarrow R_{Dx} = R_{Cx} = 8(kN)$

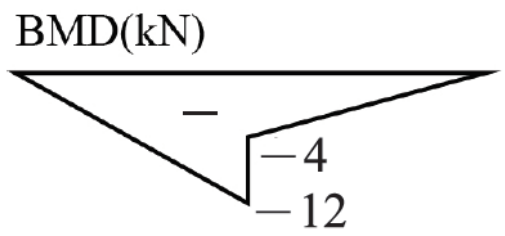
$\sum M_C = 0 \quad 4\times(3+2) = R_{Dx}\times 1 + R_{Dy}\times 2 \Rightarrow R_{Dy} = 6(kN)$

取DE桿部分

$\sum F_x = 0 \quad R_{Ex} = R_{Dx} \Rightarrow R_{Ex} = R_{Dx} = 8(kN)$

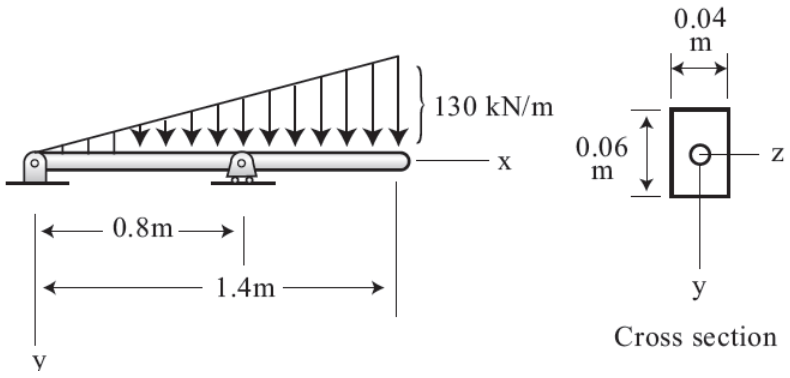
$\sum F_y = 0 \quad R_{Dy} = R_{Ey} \Rightarrow R_{Dy} = R_{Ey} = 6(kN)$





例 15

如圖所示一簡支外伸樑(simple and overhanging beam)及其受力情形，右邊則顯示其斷面。請問在x =0.6m 處，該樑所受到的最大彎曲力矩最大剪力各為何？



【107 台大醫工】

【解析】

(1) 先計算支承反力：

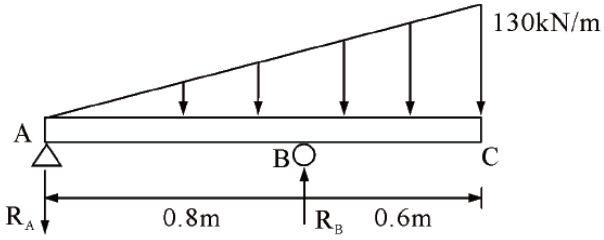
$\Sigma M_A = 0$

$$\Rightarrow R_B = \frac{130\times1.4\times\frac{1}{2}\times1.4\times\frac{2}{3}}{0.8}$$

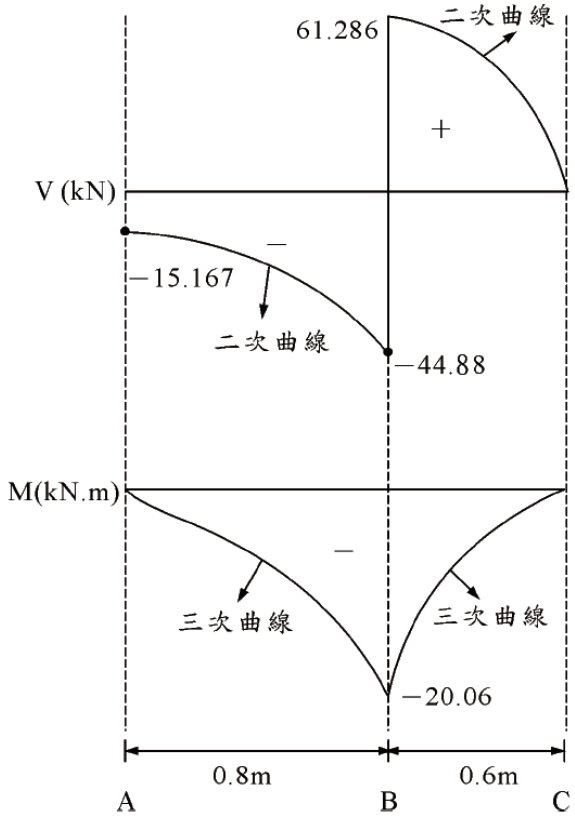
$= 106.167\text{kN}(\uparrow)$

$\Sigma F_y = 0$

$\Rightarrow R_A = 15.167\text{kN}(\downarrow)$

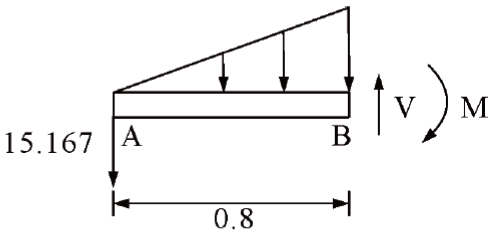


(2) 畫剪力圖及彎矩圖：



取AB段左端之自由體圖

$\Sigma F_y = 0 \Rightarrow$



$V = 15.167 + 130 \times 1.4 \times \frac{1}{2} \times (\frac{0.6}{1.4})^2$

$=31.88kN(\uparrow)$

$\Sigma M_A = 0 \Rightarrow M = 12.44kN \cdot m(\curvearrowright)$

$V_{max} = 3.188kN$ $\qquad \frac{V_{max}}{A} = \frac{3}{2} \times \frac{31.88}{0.04 \times 0.06} = 19925(kpa)$

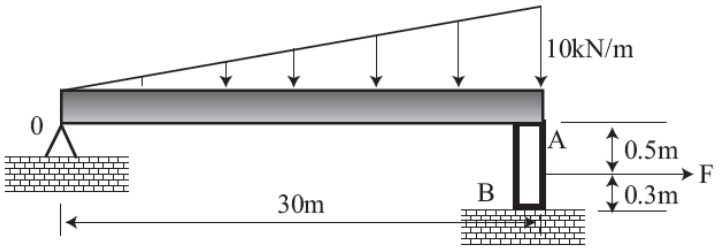
$M_{max} = 12.44kN \cdot m$

例 16

如圖所示之OA簡支樑受線性分布負載（最右端之最大負載強度為10 kN/m），其中樑之左端為鉸支撐，右端之下端以一AB物塊作為簡支撐，其中A端與B端之靜摩擦係數分別為 $\mu_A = 0.6$ 、$\mu_B = 0.4$ ，若不考慮各構件之重量與厚度，重力加速度 $g = 9.81 m/s^2$

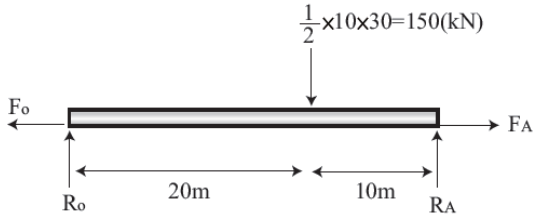
(1) 試繪出OA簡支樑之剪力圖與彎矩圖，並標明剪力為0之位置、彎矩出現最大值之處與其值。

(2) 若將樑下之AB柱推開，試求出推開之最小力F為多少？ (97年機械普考)



【解析】

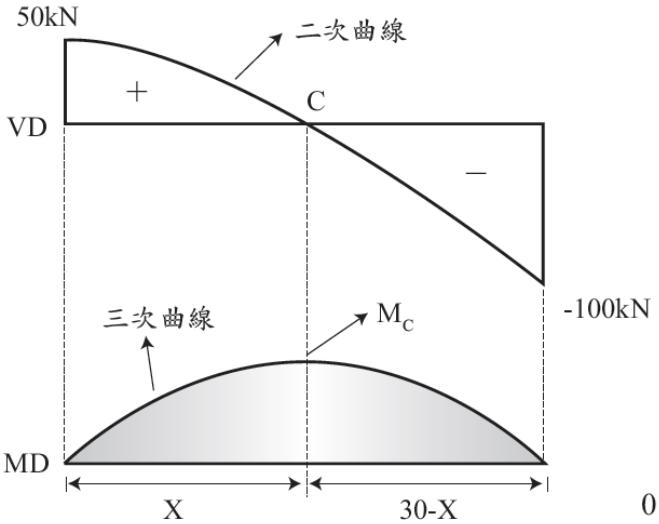
(一) 取OA自由體圖



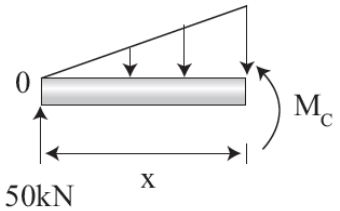
(1) $\Sigma M_o = 0$

$$\Rightarrow R_A = 150 \times 20 \times \frac{1}{30} = 100\,(kN)$$

$$\Sigma F_y = 0 \quad \Rightarrow R_o = 50\,(kN)$$



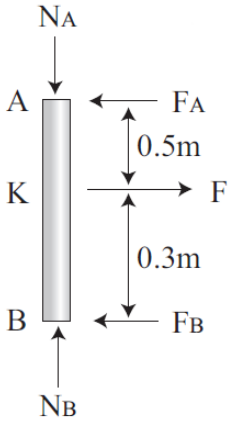
(2) C點為剪力 $= 0$ 之處，取 $\overline{OC}$ 自由體圖



$$\Sigma F_y = 0 \quad \Rightarrow (\frac{x}{30})^2 \times 150 = 50 \quad \Rightarrow x = 17.32\,(m)$$

$$\Sigma M_o = 0 \Rightarrow M_c = (\frac{17.32}{30})^2 \times 150 \times 17.32 \times \frac{2}{3} = 577.3\,(kN\text{-}m)$$

(二)(1) 取AB自由體圖



$\Sigma F_x = 0 \quad \Rightarrow F - F_A - F_B = 0$ ——(1)

$\Sigma F_y = 0 \quad \Rightarrow N_B = 100\,(kN)$——(2)

$\Sigma M_c = 0 \quad \Rightarrow 0.3 = 0.8F_A$ ——(3)

(2) 若支柱僅在A處滑動 $F_B \le \mu_B N_B$

$F_A = 0.6 \times 100 = 60\,(kN)$，代入(1)(2)(3)

$F = 160\,(kN)$，$F_B = 100\,(kN)$，故需考慮另一種滑動

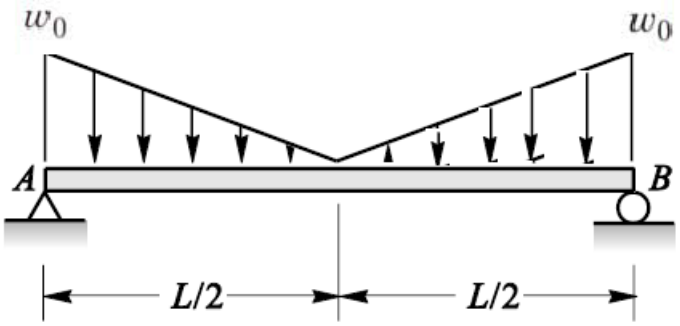
(3) 若支柱僅在B處滑動 $F_A \le \mu_A N_A$

$F_B = \mu_B N_B = 0.4 \times 100 = 40 \quad \sum M_A = 0 \Rightarrow F = 64(KN) \Rightarrow F_A = 24$

故支柱僅在B處滑動之情況先發生

例 17

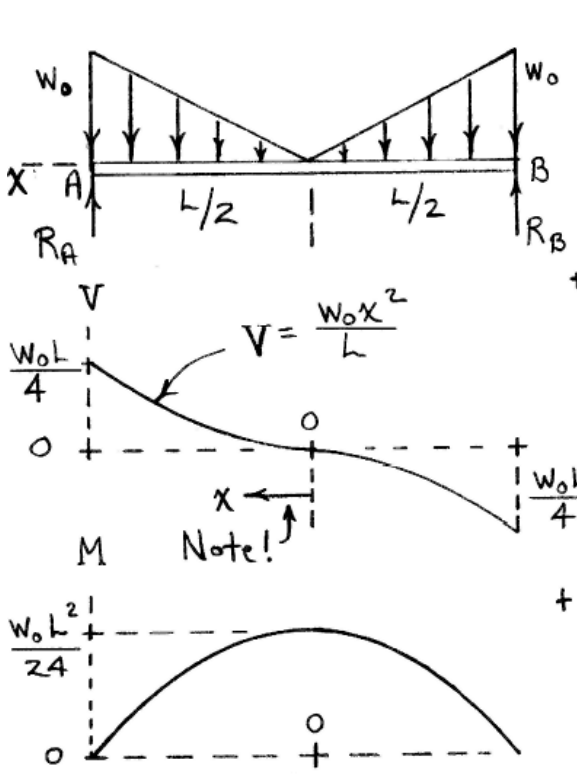
(二). Draw the shear and moment diagrams for the beam loaded as shown. Specify the maximum moment M_{max}. (20 分)



(98 年台大醫工)

【中文翻譯】

【解析】



$$R_A = R_B = \frac{w_0 L}{4}$$

$$x = \frac{L}{2},\ M = 0,\ V = R_A$$

$$+\uparrow \Sigma F = 0:\quad V - \frac{w_0 x^2}{L} = 0$$

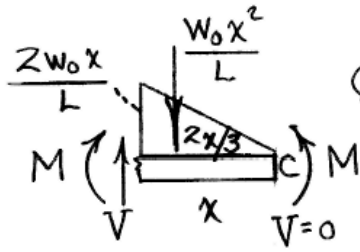
$$V = \frac{w_0 x^2}{L}$$

$$x = L/2:$$

$$\zeta + \Sigma M_C = 0:$$

$$+M_0 - \frac{w_0 L}{4}\left(\frac{L}{2}\right) + \frac{w_0 \left(\frac{L}{2}\right)^2}{L}\left(\frac{2L}{3}\right) = 0$$

$$M_0 = \frac{w_0 L^2}{24}$$



$\circlearrowleft + \sum M_0 = 0$:

$$-M - \frac{w_0 x^2}{L} x + \frac{w_0 x^2}{L} \frac{2x}{3} + \frac{w_0 L^2}{24} = 0$$

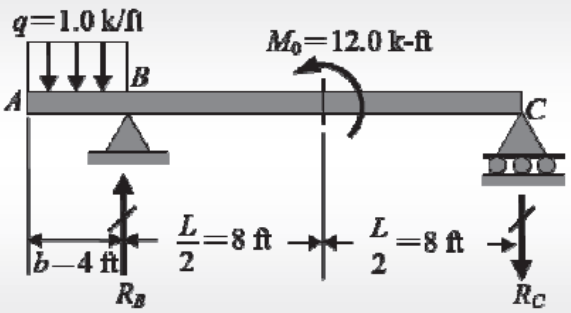
$$M = \frac{w_0}{3L}\left(\frac{L^3}{8} - x^3\right)$$

$M = M_{max}$ @ $x = 0$: $M_{max} = \frac{w_0}{3L}\left(\frac{L^3}{8} - 0\right)$

$$= \underline{\frac{w_0 L^2}{24}}$$

例 18

一梁結構如下，有一均佈外力 q＝1.0k/ft 施於 AB 之間，另有一力矩 M_o＝12.0k-ft 作用於 BC 中點，請分別繪出剪力與彎曲力矩圖。(99 年高考)

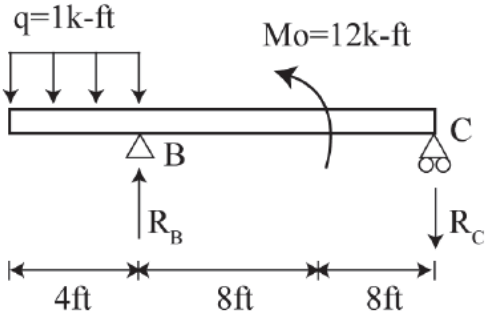


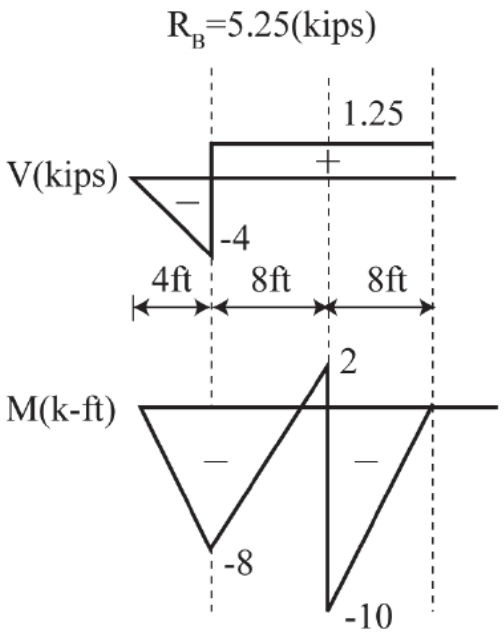
【解析】

解 $\sum M_B = 0 \Rightarrow R_C = \dfrac{1\times 4\times(2)+12}{16} = 1.25(\text{kips})$

R_B=5.25(kips)

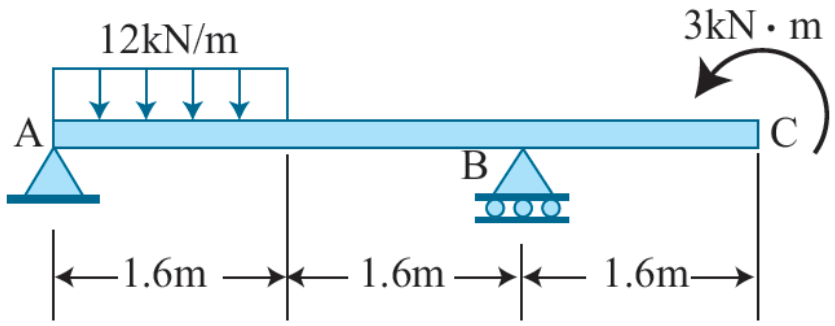
$\varepsilon_{max\,2} = \dfrac{\sigma_2}{E} = -8.08\times 10^{-4}$





例19

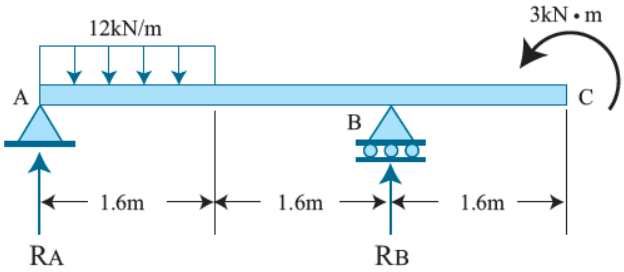
如圖所示，試繪出樑(Beam)之剪力圖與彎矩圖



【101 台大工科類題】

【解析】

(1)先求支承反力



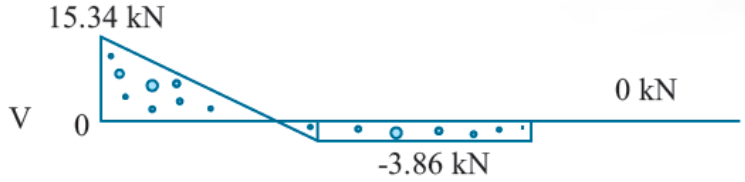
$$\sum M_A = 0 \Rightarrow$$

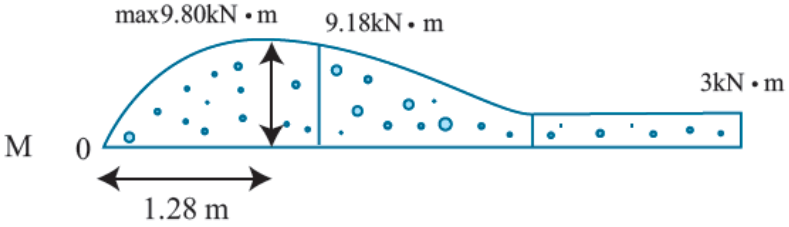
$$12 \times 1.6 \times \frac{1.6}{2} - R_B \times 3.2 - 3 = 0$$

$$\Rightarrow R_B = 3.86(KN)$$

$$\sum F_y = 0 \Rightarrow 12 \times 1.6 = R_A + R_B \Rightarrow R_A = 15.34(KN)$$

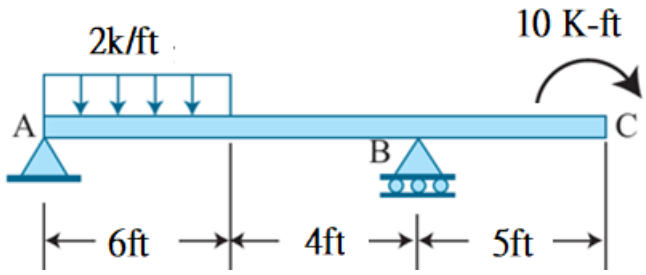
(2)畫剪力圖與彎矩圖：





例20

(15%) Draw the shear-force and bending-moment diagrams for the beam shown in the following figure.



【101 台大工科】

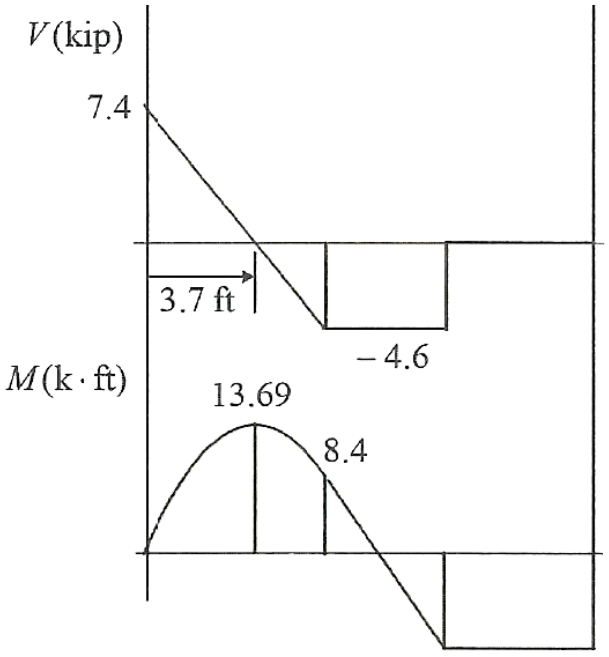
【解析】

$$\Sigma M_A = 0 \qquad 12\times 3+10 = R_B \times 10$$

$$R_B = 4.6\,\text{kip}$$

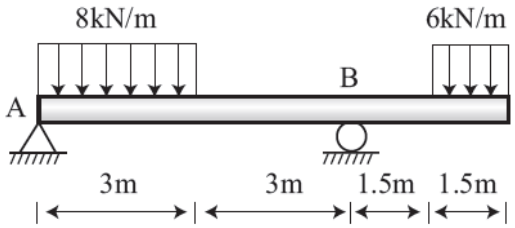
$$\Sigma F_y = 0 \qquad R_A + R_B = 12$$

$$R_A = 7.4\,\text{kip}$$



例 21

如圖所示外伸樑距離A點右側4.5 m處之剖面上剪力V = ？距離A點右側4.5 m處之剖面上，彎曲矩M = ？



【解析】

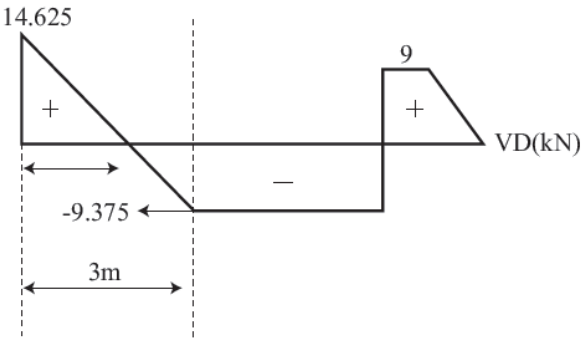
求支承反力

$\sum M_A = 0 \Rightarrow R_B = 18.375\,(\text{kN}) \uparrow$

$\sum F_y = 0 \quad \Rightarrow R_A = 14.625\,(\text{kN}) \uparrow$

承上題

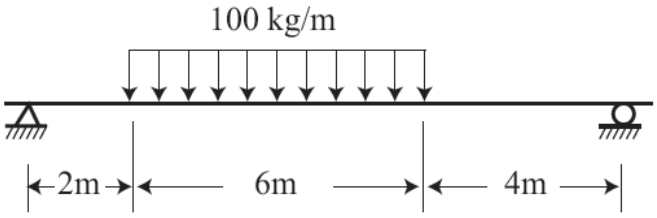
$x : (3-x) = 14.625 : 9.375 \qquad x = 1.828$



$$M = \frac{1}{2}[14.625 \times 1.828 - 9.375 \times (3-1.828)] - 1.5 \times 9.375 = -6.19\,(\text{kN-m})$$

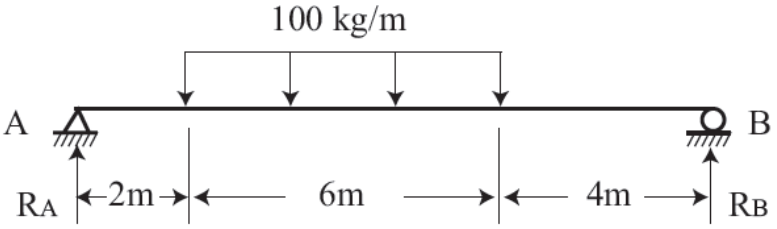
例22

如圖所示，請繪製剪力圖及彎矩圖，並求其危險截面距左端多遠。



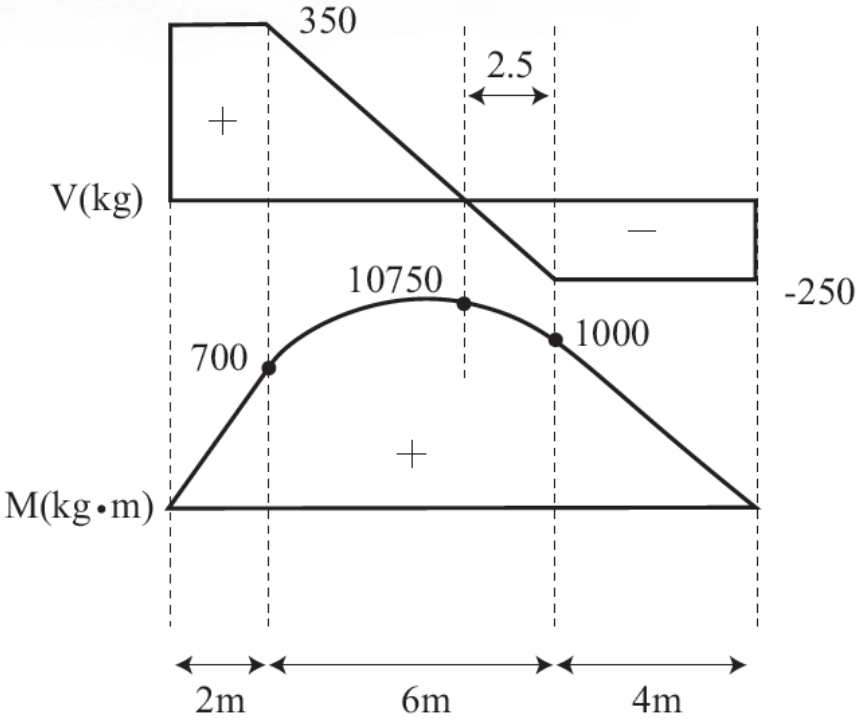
【解析】

(1) 先求支承反力：



$\Sigma M_A = 0 \Rightarrow R_B = 250(kg)$　$\Sigma F_y = 0 \Rightarrow R_A = 350(kg)$

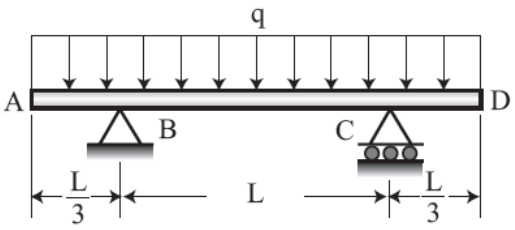
(2) 畫剪力圖及彎矩圖：



故危險截面距左端距離=2+6-2.5=5.5(m)

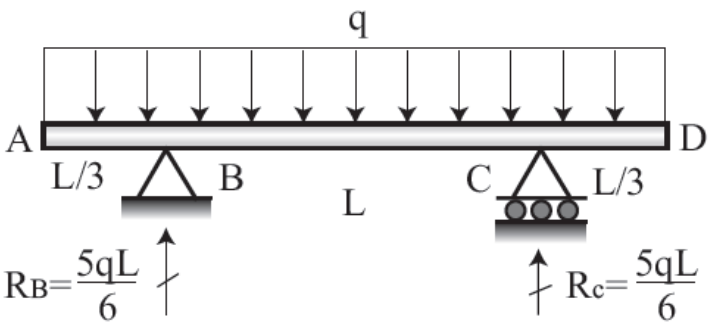
例23

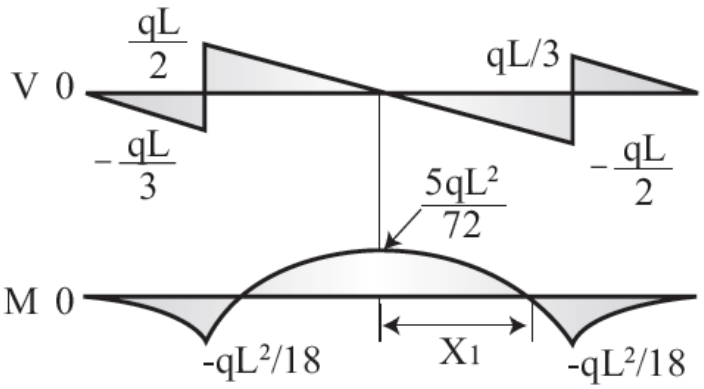
試繪出如圖所示之剪應力和力矩圖(shear and moment diagram)。



【台大機械】

【解析】





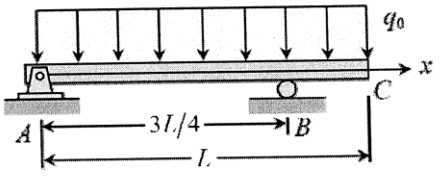
$$x_1 = L\frac{\sqrt{5}}{6} = 0.3727L$$

例24

如圖一所示之簡支撐（simply supported）伸臂梁 ABC 承受均勻分布荷載（distributed loading）q_0，其因次（dimension）為力/單位長度。試求梁 ABC 的：

㈠剪力分布圖（shear force diagram）。（10 分）

㈡彎矩分布圖（bending moment diagram）。（10 分）

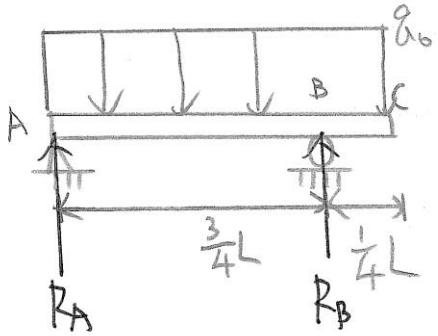


圖一

【106 地特三等】

【解析】

㈠取整体之F.B.D.



$+\uparrow\Sigma F_y=0$

$R_A+R_B=q_0L$

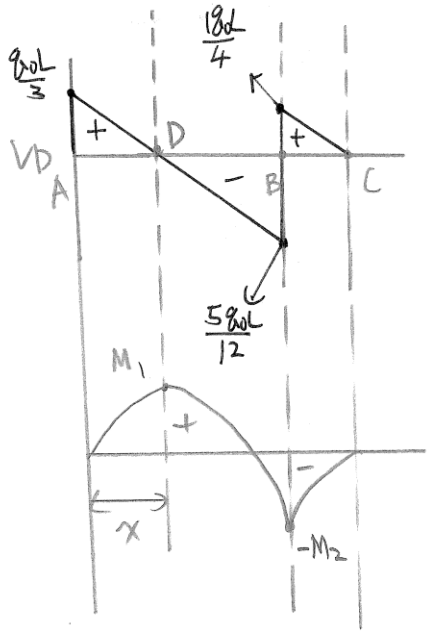
$+\circlearrowleft\Sigma M_A=0$

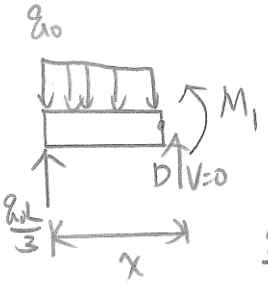
$q_0L\times\frac{L}{2}-R_B\times\frac{3}{4}L=0$

$\Rightarrow R_B=\frac{2}{3}q_0L$

$R_A=\frac{1}{3}q_0L$

(二) 畫剪力圖和彎矩圖





$\frac{q_0L}{3} = q_0x$

$\Rightarrow x = \frac{L}{3}$

㊀ $\Sigma M_D = 0$

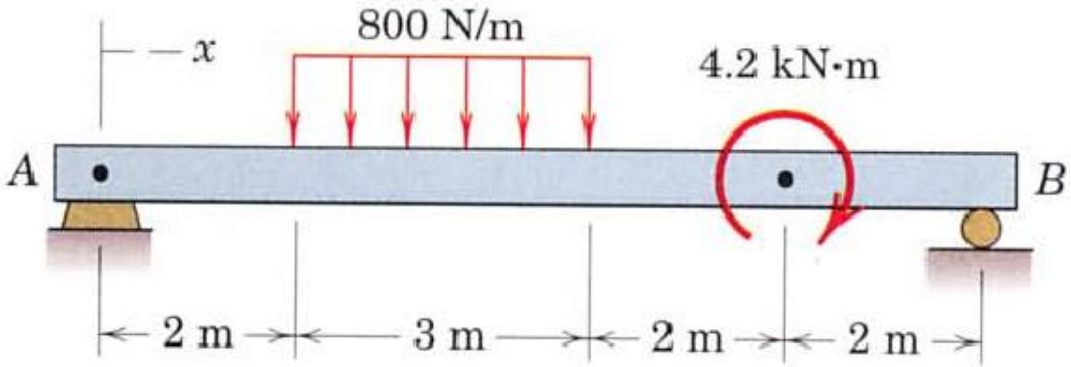
$\frac{q_0L}{3} \times \frac{L}{3} - q_0 \times \frac{L}{3} \times \frac{L}{6} = M_1$

$M_1 = \frac{q_0L^2}{18}$

$-M_2 = -\frac{q_0L}{4} \times \frac{1}{4} \times \frac{1}{2} = \frac{-q_0L}{32}$

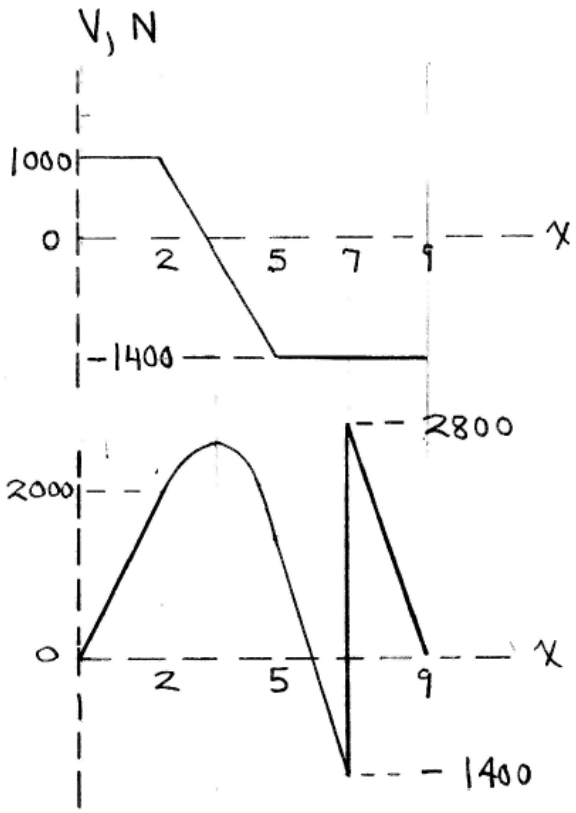
例 25

2. Calculate and plot the shear and moment diagrams for the beam loaded as shown. (20%)



(96 中山光電 丁組靜力學) (96 中山光電 丁組靜力學)

【解析】



$x = 6\ m$

$V = -1400\ N$

$M = 8400 - 1400(6)$

$= 0$

$$\frac{dM}{dx} = 0$$

$$\frac{d}{dx}\left(-400x^2 + 2600x - 1600\right)$$

$$= -800x + 2600 = 0$$

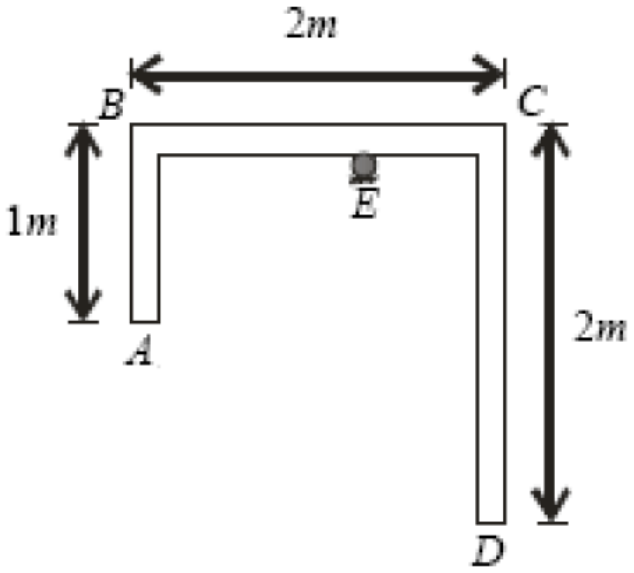
$$x = 3.25 \text{ m}$$

$$M_{x=3.25} = -400(3.25)^2 + 2600(3.25) - 1600 = 2625 \text{ N·m}$$

$$M_{max} = 2800 \text{ N·m}$$

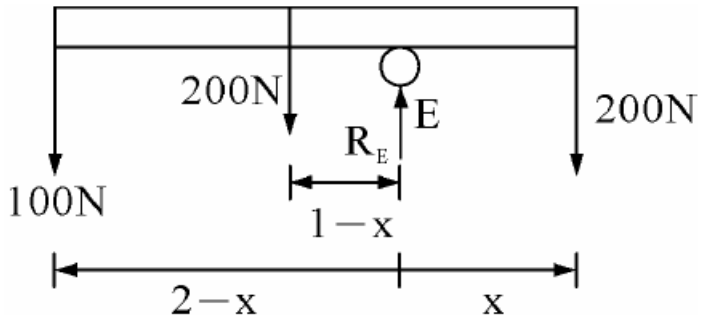
例 26

如圖所示，構件係由 AB、CD 兩段垂直部分與 BC 水平部分所組成，其長度分別如圖示，已知構件每單位長度之重量為 100 N /m，試求支撐點 E 應與 B 點多少距離才能保持平衡？試繪出 BC 段之剪力圖與彎矩圖(必須標示出 B、C、E 各點之大小數值)。(95 年土木普考)



【解析】

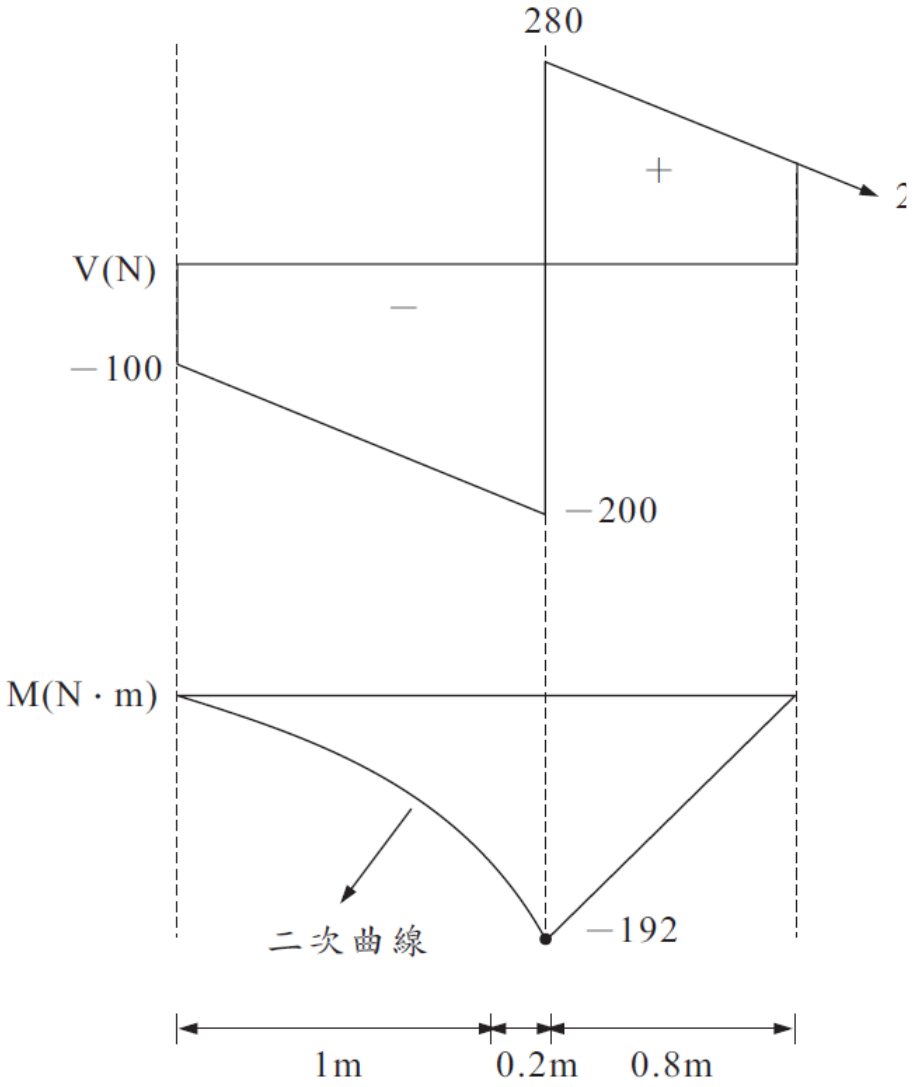
(1) **取 BC 段自由體圖：**



$$\sum M_E = 0 \Rightarrow 200x = 200\times(1-x) + 100\times(2-x) \quad x = 0.8(\text{m})$$

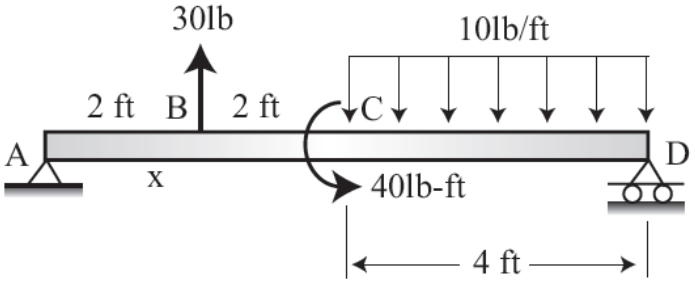
$$\sum F_y = 0 \Rightarrow R_E = 200 + 200 + 100 = 500\text{N}$$

(2) **畫剪力圖及彎矩圖：**



例 27

有一簡支梁承受如圖之外力，請繪出其剪力圖及彎矩圖。



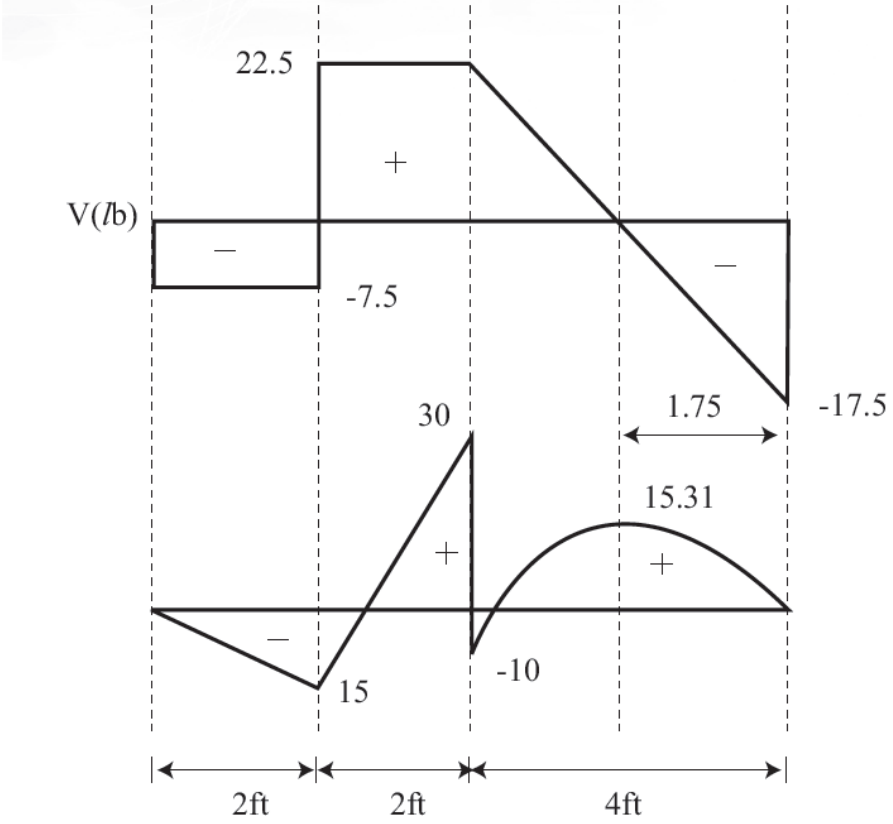
【台大工科類題】

【解析】

(1) 先求支承反力：

$\Sigma M_A = 0 \Rightarrow R_D = 17.5\,\ell b\,(\uparrow)$　　$\Sigma M_D = 0 \Rightarrow R_A = 7.5\,\ell b\,(\downarrow)$

(2) 畫剪力圖及彎矩圖：

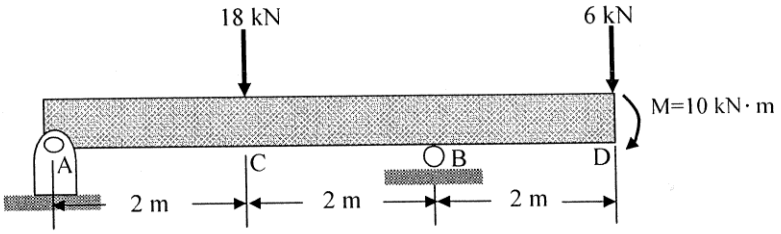


例 28

有一矩形截面樑（忽略其重量）受力與力矩作用如下圖。

㈠繪出此樑的自由體圖並求 A、B 位置之反作用力。（10 分）

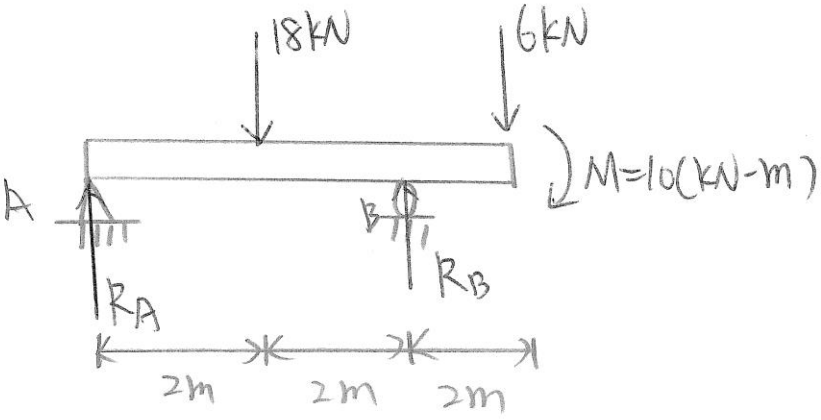
㈡繪出樑之剪力分佈圖及彎矩分佈圖並標明各關鍵位置（A、B、C、D 處）之剪力及彎矩值。（10 分）



【106 關務三等】

【解析】

㈠取樑之F.B.D

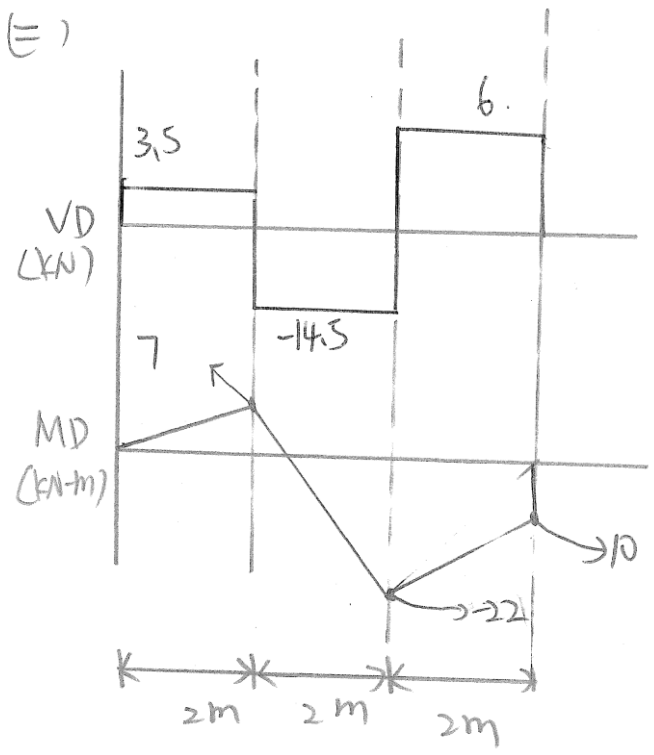


$\curvearrowleft \Sigma M_A = 0$

$18\times2+6\times6+10-R_B\times4=0 \Rightarrow R_B=20.5(kN)$

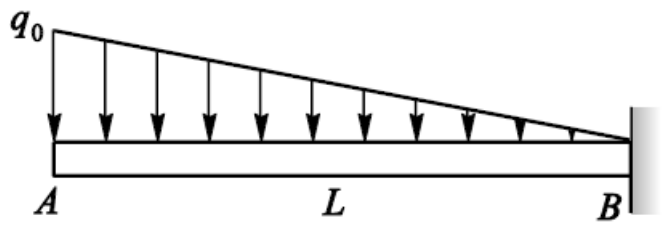
$\uparrow \Sigma F_y=0 \Rightarrow R_A=3.5(kN)$

(三)

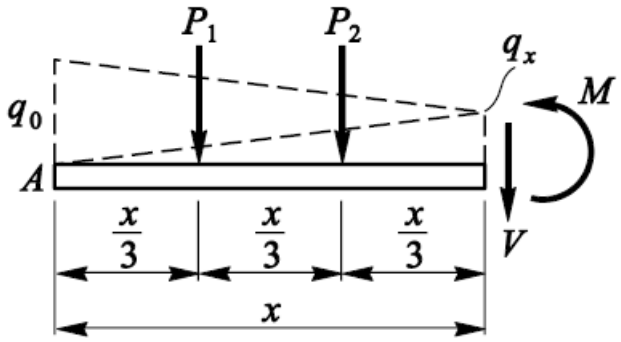


例 29

圖中懸臂梁承受均變負荷，試求梁之剪力及彎矩方程式，並繪剪力圖及彎矩圖。



【解析】

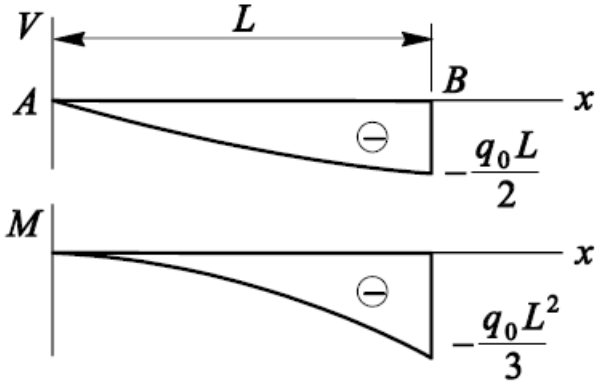


$$q_x = \frac{q_0}{L}(L-x) \quad , \quad P_1 = \frac{1}{2}q_0 x \quad , \quad P_2 = \frac{q_0}{2L}(L-x)x$$

$$\Sigma F_y = 0 \quad , \quad V = -P_1 - P_2 = -\frac{1}{2}q_0 x - \frac{q_0}{2L}(L-x)x = -q_0 x + \frac{q_0}{2L}x^2$$

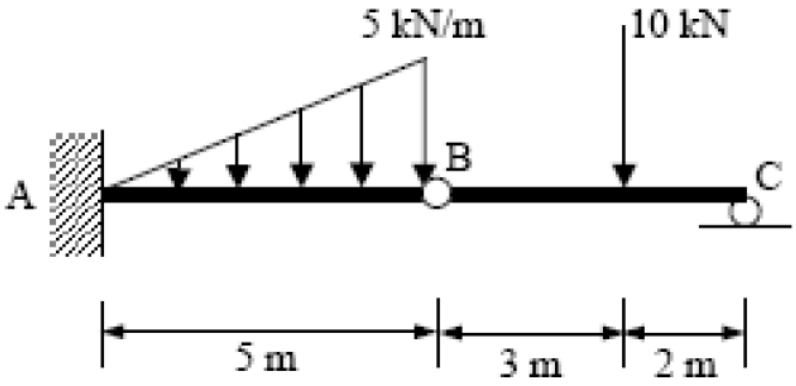
$$\Sigma M = 0 \quad , \quad M = -P_1\left(\frac{2}{3}x\right) - P_2\left(\frac{x}{3}\right) = -\left(\frac{1}{2}q_0 x\right)\left(\frac{2}{3}x\right) - \frac{q_0}{2L}(L-x)x\left(\frac{x}{3}\right)$$

$$= -\frac{1}{2}q_0 x^2 + \frac{q_0}{6L}x^3$$



例 30

圖示樑的 A 為固定端，B 為鉸接，C 為滾支承，請繪此梁的剪力圖和彎矩圖。(96 年土木普考)



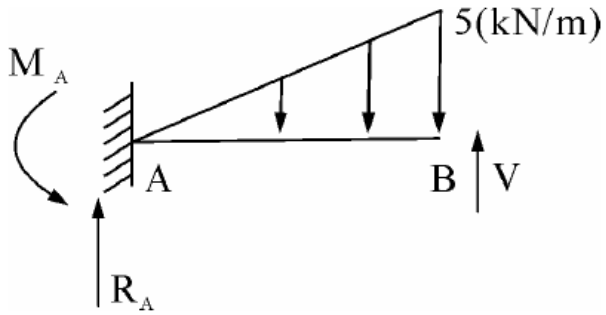
【解析】

(1) 先求支承反力：

$$\sum F_y = 0 \Rightarrow 5\times 5\times \frac{1}{2} + 10 = R_C + R_A \text{——①}$$

$$\sum M_C = 0 \Rightarrow 10\times 2 + 5\times 5\times \frac{1}{2}\times(\frac{5}{3} + 3 + 2) + M_A = R_A \times 10 \text{——②}$$

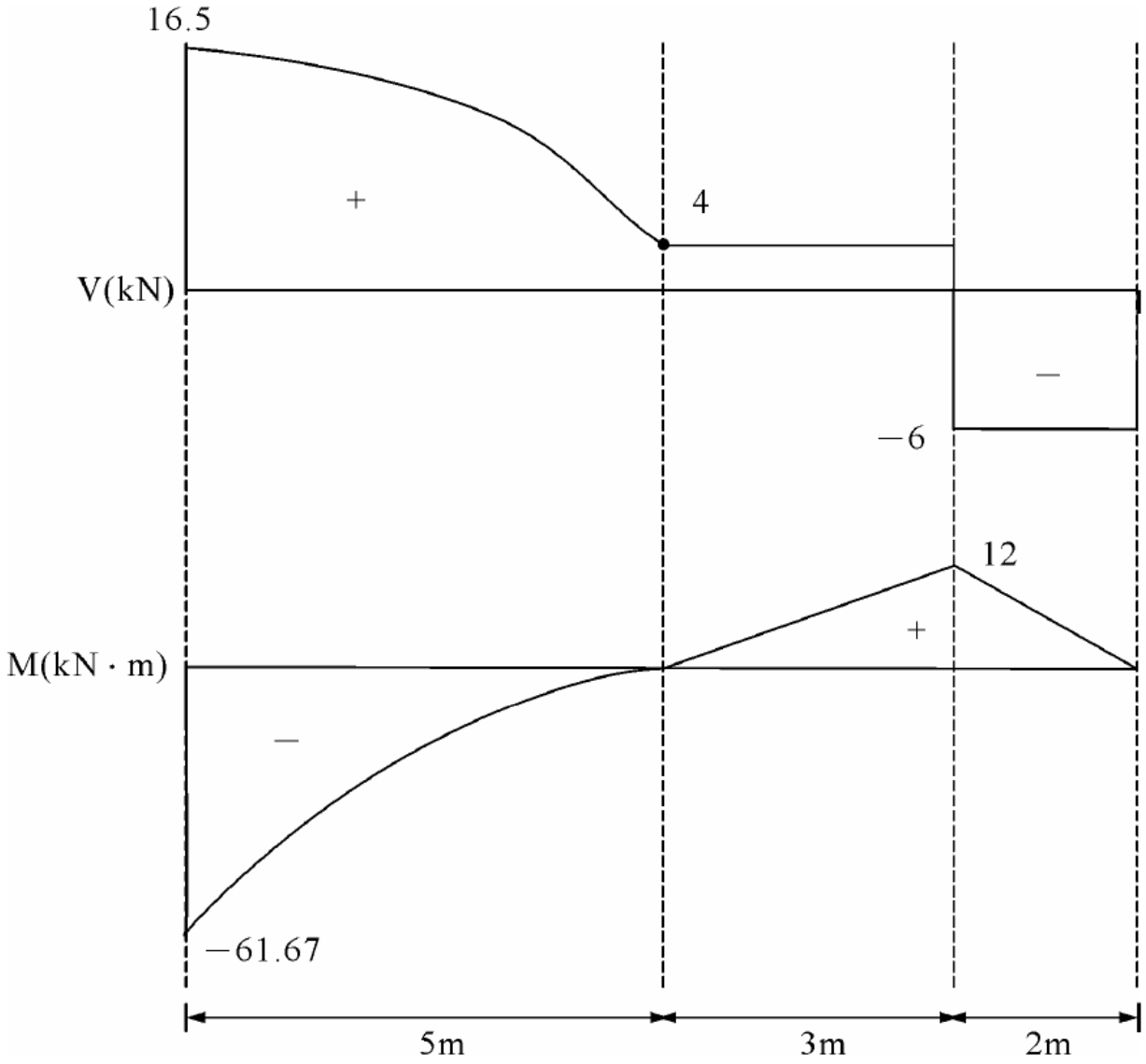
(2) 取 AB 自由體圖：



$$\sum M_B = 0 \Rightarrow M_A + 5\times 5\times \frac{1}{2}\times \frac{5}{3} = 5R_A \text{——③}$$

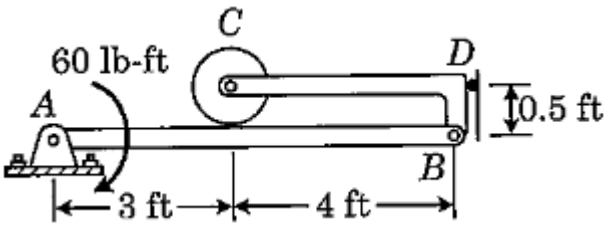
由②③可得 $R_A = 16.5(\text{kN})$，$M_A = 61.67\text{kN}$ 代入①得 $R_C = 6(\text{kN})$

(3) 畫剪力圖及彎矩圖：



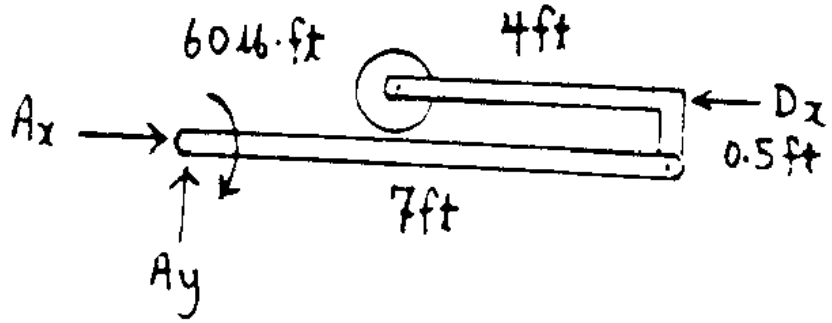
例 31

. Determine the maximum shear force and bending moment in beam AB. Neglect the weights of the frame and roller C. Assume rollers C and D to be smooth. (20%)



(97 年交大應用力學丙組)

(一)取整體自由體圖



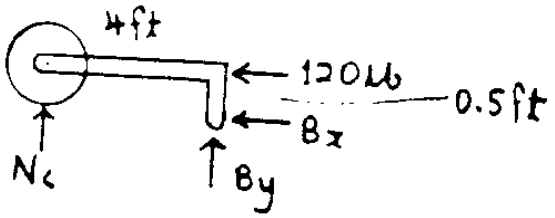
$$(+\Sigma M_A = 0;\quad -60 + D_x(0.5) = 0$$

$$D_x = 120 \text{ lb}$$

$$\overset{+}{\rightarrow}\Sigma F_x = 0;\quad A_x = 120 \text{ lb}$$

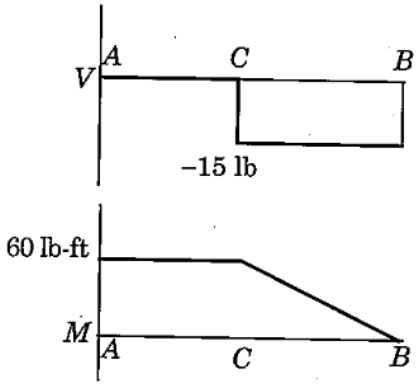
$$+\uparrow \Sigma F_y = 0;\quad A_y = 0$$

(二)取 BCD 自由體圖



$$\circlearrowleft + \Sigma M_B = 0;\ -N_C(4) + 120(0.5) = 0$$

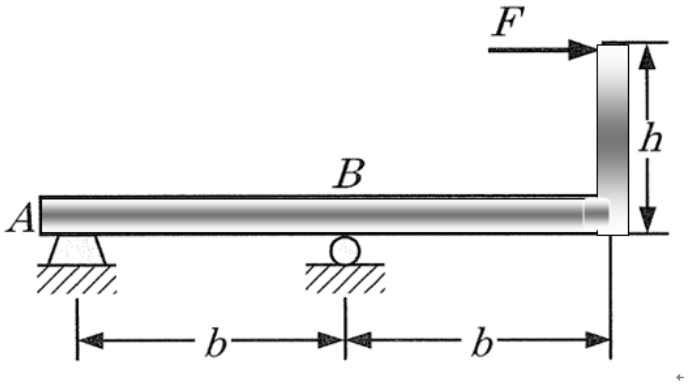
$$N_C = 15.0 \text{ lb}$$



由剪力圖與彎矩圖知：$V_{max} = 15$ (lb)，$M_{max} = 60$ (lb-ft)

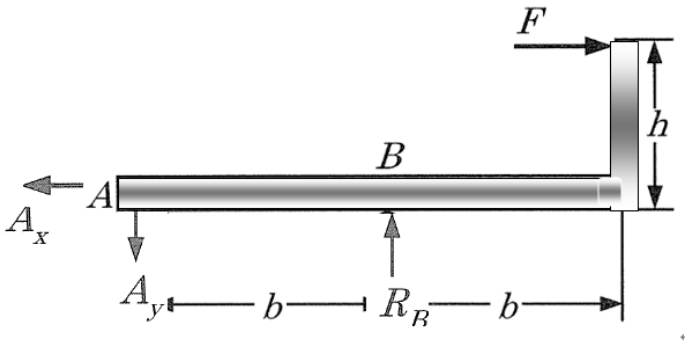
例 32

Draw the shear and moment diagrams for the beam loaded by the force F applied to the strut welded to the beam as shown.



(台大醫工)

(一)取整體自由體圖



$$\sum F_x = 0$$

$$A_x = F(\leftarrow)$$

$$\sum F_y = 0$$

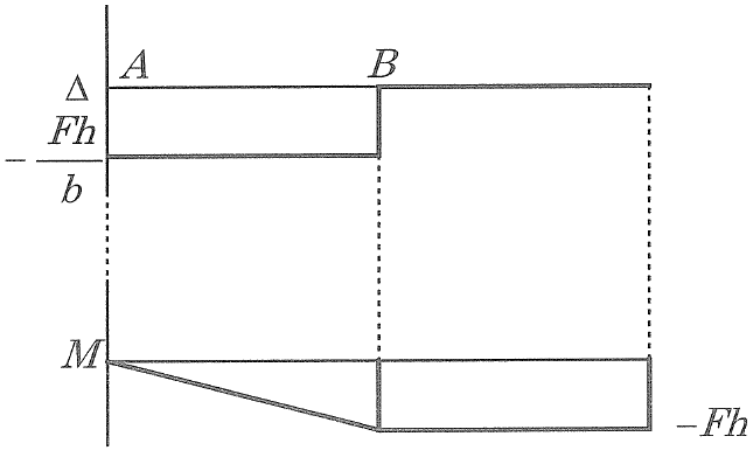
$$A_y = R_B = \frac{Fh}{b}(\downarrow)$$

$$\sum M_A = 0$$

$$Fh - R_B \times b = 0 \Rightarrow R_B = \frac{Fh}{b}(\uparrow)$$

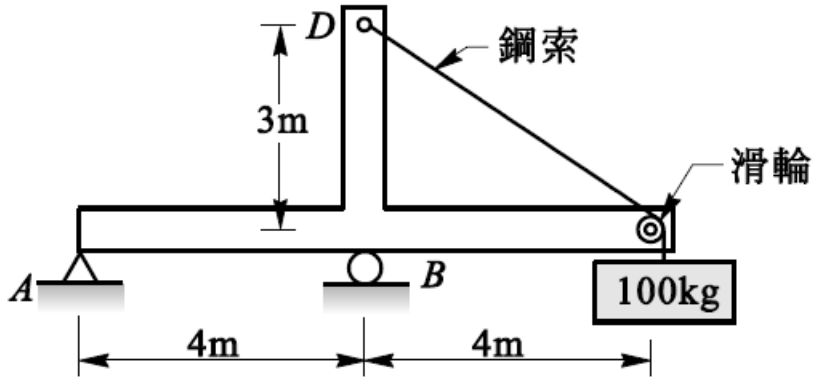
(二)剪力圖及彎矩圖

剪力圖及彎矩圖

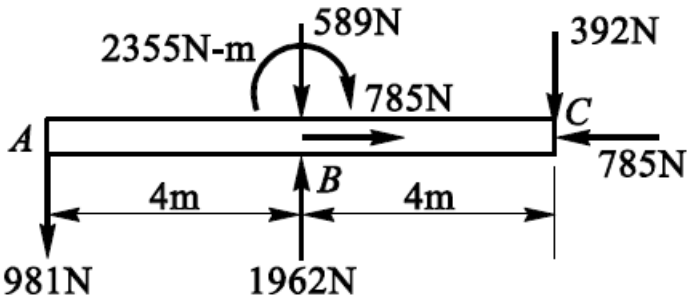


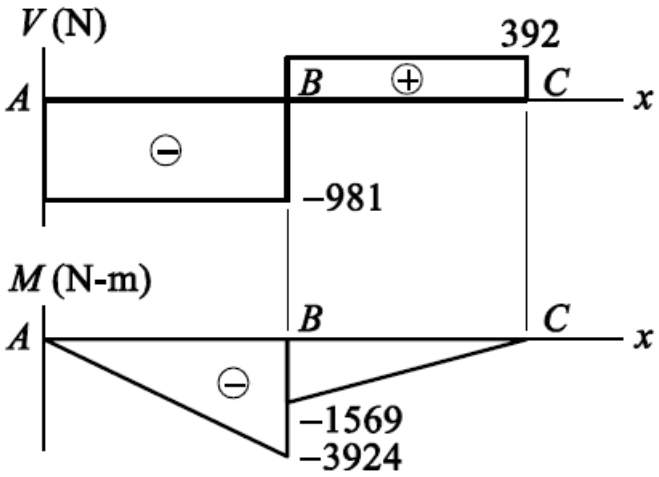
例 33

試繪圖中外伸梁之剪力圖及彎矩圖。



(107 中央機械)



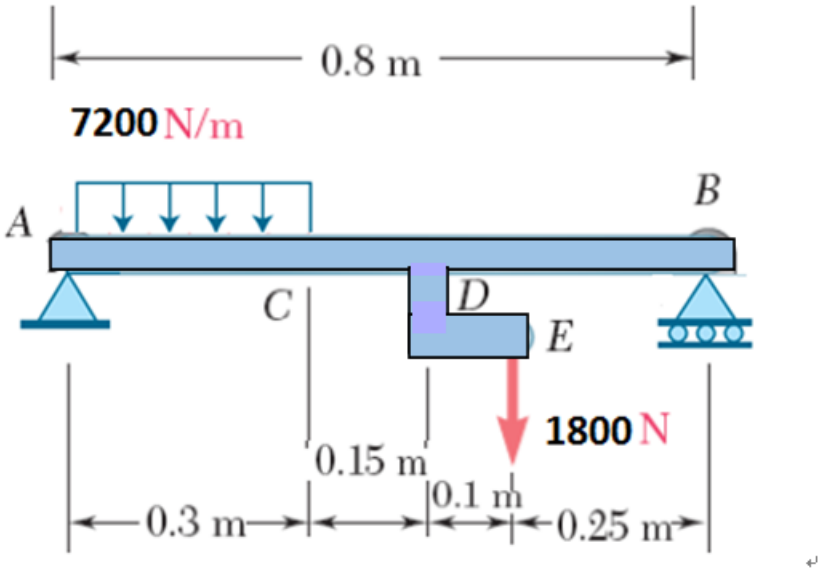


$V_{\max} = 981\ \text{N}$

$M_{\max} = 3924\ \text{N-m}$

例 34

Draw the shear and bending moment diagrams for the beam AB. The distributed load of 40 N/m extends over 0.3 m of the beam, from A to C, and the 400-N load is applied at E.



(102 北科機械類題)

【解析】

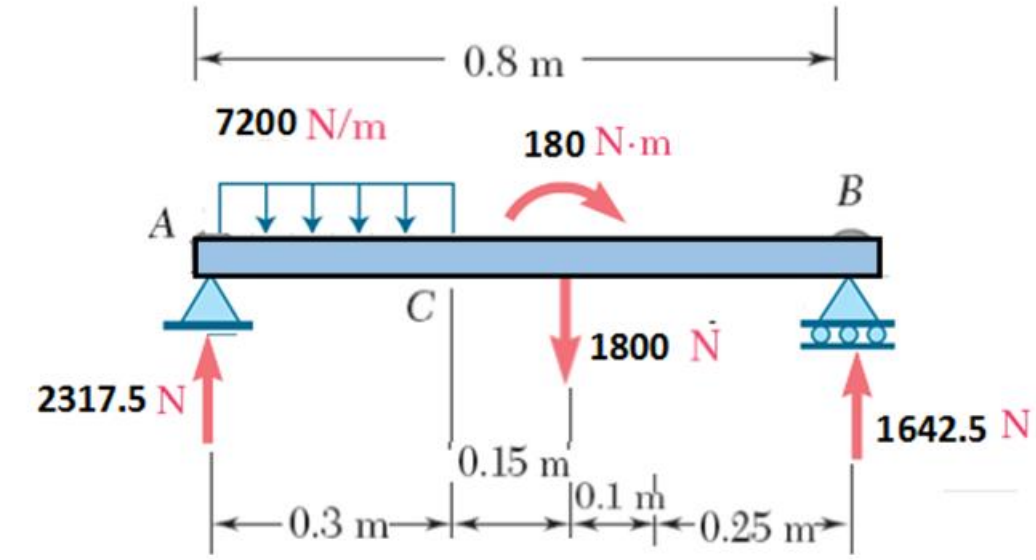
(一) 先求支承反力

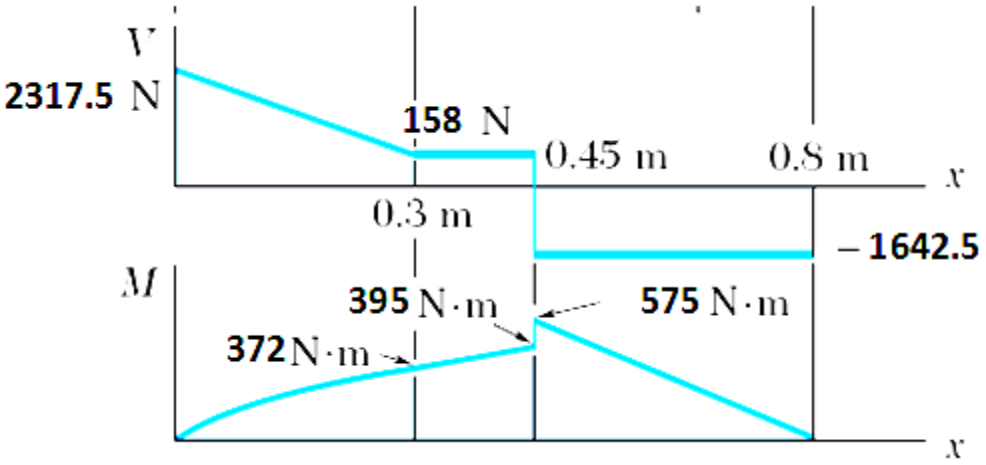
$$\sum M_A = 0 : B_y(0.8\text{m}) - (7200\,\text{N})(0.3\text{m}) \times 0.15 - (1800\,\text{N})(0.55\text{m}) = 0 \Rightarrow B_y = 1642.5(N)$$

$$\sum M_B = 0 : (7200\text{N}) \times 0.3 \times (0.65\text{m}) + (1800\ \text{N})(0.25\text{m}) - A_y(0.8\text{m}) = 0 \Rightarrow A_y = 2317.5(N)$$

(二) 剪力彎矩圖

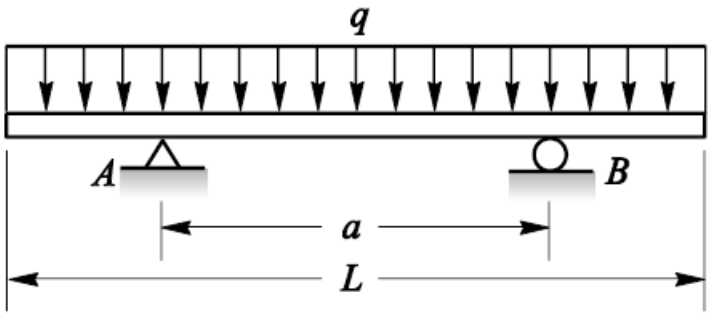
(三)


0.8 m
7200 N/m
180 N·m
A
B
C
1800 N
2317.5 N
1642.5 N
0.15 m
0.3 m
0.1 m
0.25 m


V
2317.5 N
158 N
0.45 m
0.8 m
x
0.3 m
−1642.5
M
395 N·m
575 N·m
372 N·m
x

例 35

A beam of length L is being designed to support a uniform load of intensity q (see figure). If the supports of the beam are placed at the ends, creating a simple beam, the maximum bending moment in the beam is $qL^2/8$. However, if the supports of the beam are moved symmetrically toward the middle of the beam (as pictured), the maximum bending moment is reduced. Determine the distance a between the supports so that the maximum bending moment in the beam has the smallest possible numerical value. Draw the shear-force and bending-moment diagrams for this condition.



【台大機械】

【中文翻譯】

一樑具有兩外伸，支持強度 q 之均勻負載（ 見圖 ）。樑總長 L，支撐之跨距 a。求有樑之最大彎矩之最小值的距離 a。繪出此情況之剪力及彎矩圖。

【解析】

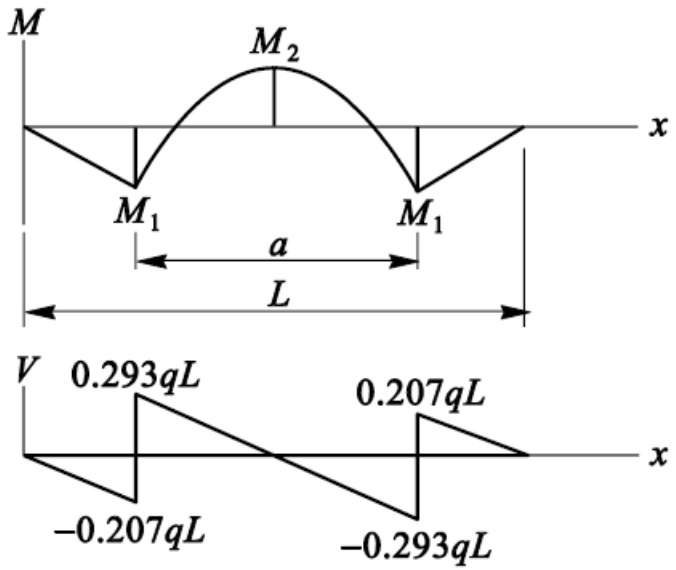
梁彎矩圖之形狀如圖所示

當 $|M_1| = M_2$ 時，梁內之最大彎矩達最小值

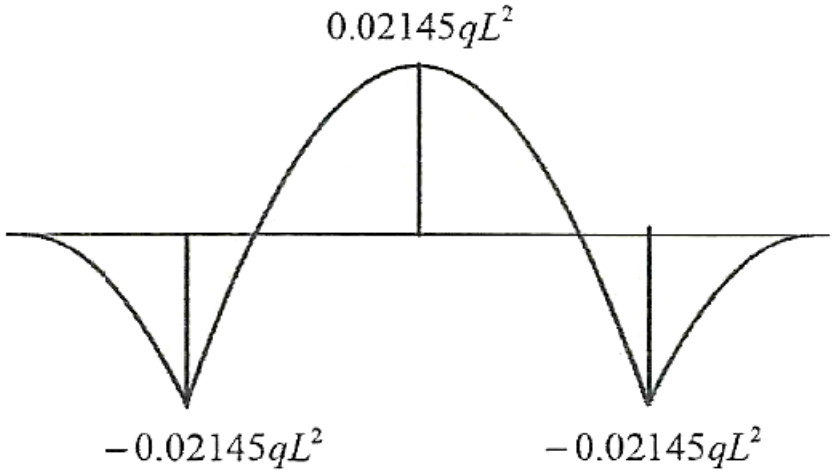
$$|M_1| = \frac{q}{8}(L-a)^2 \quad , \quad M_2 = R_A\left(\frac{a}{2}\right) - \frac{qL^2}{8} = \frac{qL}{8}(2a-L)$$

$$\frac{q}{8}(L-a)^2 = \frac{qL}{8}(2a-L) \text{ ，得 } a = (2-\sqrt{2})L = 0.586L$$

$$|M_1| = M_2 = \frac{q}{8}(L-0.586L) = 0.0214qL^2$$


M
M2
x
M1
M1
a
L

V
0.293qL
0.207qL
x
−0.207qL
−0.293qL


0.02145qL²
−0.02145qL²
−0.02145qL²

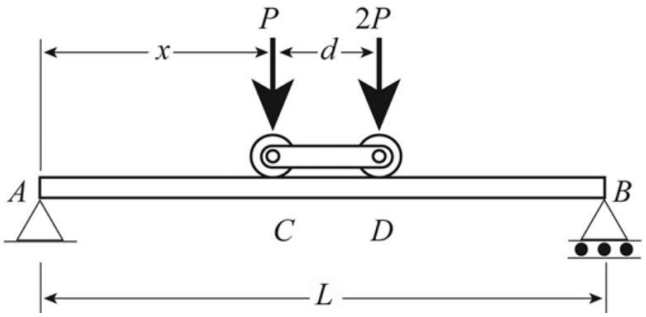
例 36

A simple beam AB supports two connected wheel loads P and $2P$ that are distance d apart (see figure). The wheels may be placed at any distance x from the left-hand support of the beam.

(a) Determine the distance x that will produce the maximum shear force in the beam, and also determine the maximum shear force V_{max}.

(b) Determine the distance x that will produce the maximum bending moment in the beam, and also draw the corresponding bending moment diagram.

(Assume $P = 10$ kN, $d = 2.4$ m, and $L = 12$ m.)



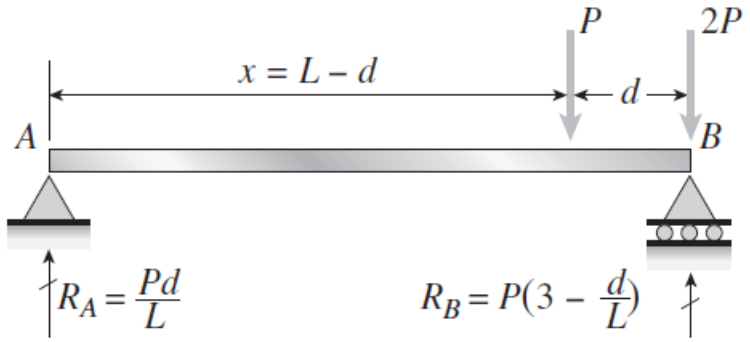
【高應大】【土木高考】【台科機械】

【中文翻譯】

簡樑 AB 承擔兩相連負載 P 及 $2P$，它們分離了距離 d （ 見圖 ）。該負載我們可以置於從樑左端起算任何距離 x 處之樑上。求距離 x 使：(a) 樑具最大剪力，及 (b) 樑具最大彎矩。假設 $P = 10$ kN 、 $d = 2.4$ m 及 $L = 12$ m 。且，求最大剪力 V_{max} 及最大彎矩 M_{max} 。

【解析】

(一)



(二)

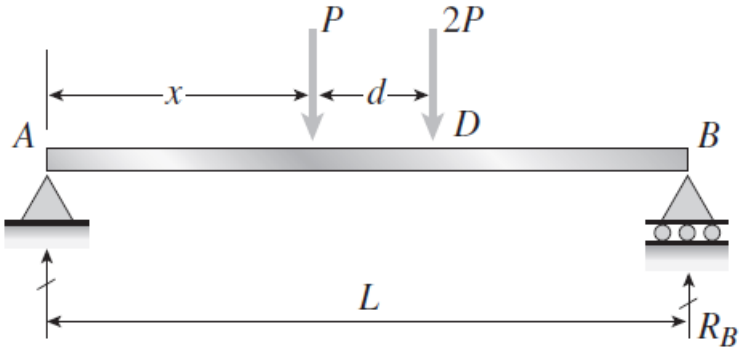
最大剪應力會產生在較大力 $2P$ 靠近 B 點的時候

$$\Sigma M_A = 0 \qquad P(L-d)+2PL = R_B L$$

$$R_B = P\left(3-\frac{d}{L}\right) = (10)\left(3-\frac{2.4}{12}\right) = 28\ \text{kN}$$

$$V_{\max} = 28\ \text{kN}$$

(一) 最大彎矩



最大彎矩會產生在 $2P$ 作用點正下方 D 點。

$$\Sigma M_A = 0 \qquad Px+2P(x+d) = R_B L$$

$$R_B = \frac{P}{L}x + \frac{2P}{L}(x+d) = \frac{P}{L}(2d+3x)$$

$$M_D = R_B(L-x-d)$$

$$= \frac{P}{L}(2d+3x)(L-x-d)$$

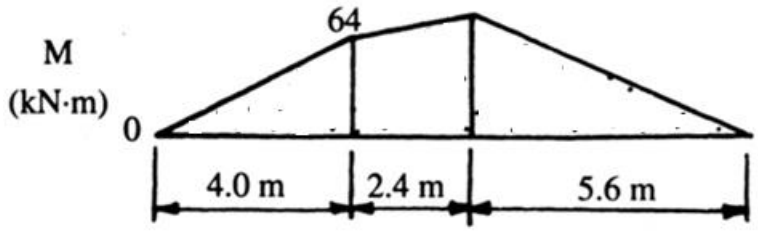
$$= \frac{P}{L}[-3x^2+(3L-5d)x+2d(L-d)]$$

$$\frac{dM_D}{dx} = \frac{P}{L}(-6x+3L-5d) = 0$$

$$x = \frac{L}{6}\left(3-\frac{5d}{L}\right) = 4.0\ \text{m}$$

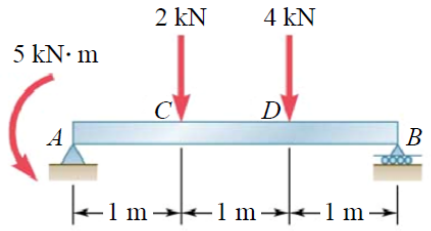
$$M_{\max} = \frac{P}{L}\left[-3\left(\frac{L}{6}\right)^2\left(3-\frac{5d}{L}\right)^2 + (3L-5d) \times \left(\frac{L}{6}\right)\left(3-\frac{5d}{L}\right) + 2d(L-d)\right]$$

$$= \frac{PL}{12}\left(3-\frac{d}{L}\right)^2 = 78.4 \text{ kN} \cdot \text{m} \quad \leftarrow$$



例 37

如圖(1)所示之簡支樑承受所示之負載，試繪出其剪力圖及彎矩圖，並標示其最大之剪力與彎矩。（20 分）

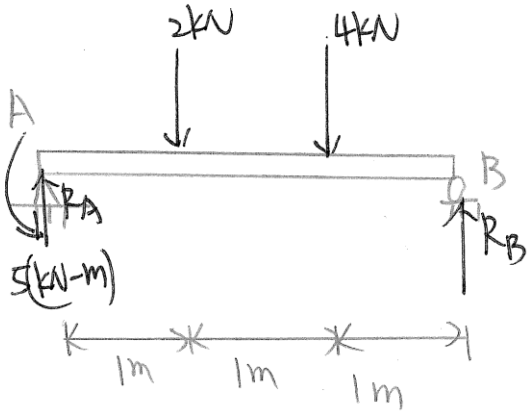


圖(1)

【106 關務四等】

【解析】

(一)



$\curvearrowleft \Sigma M_A = 0$

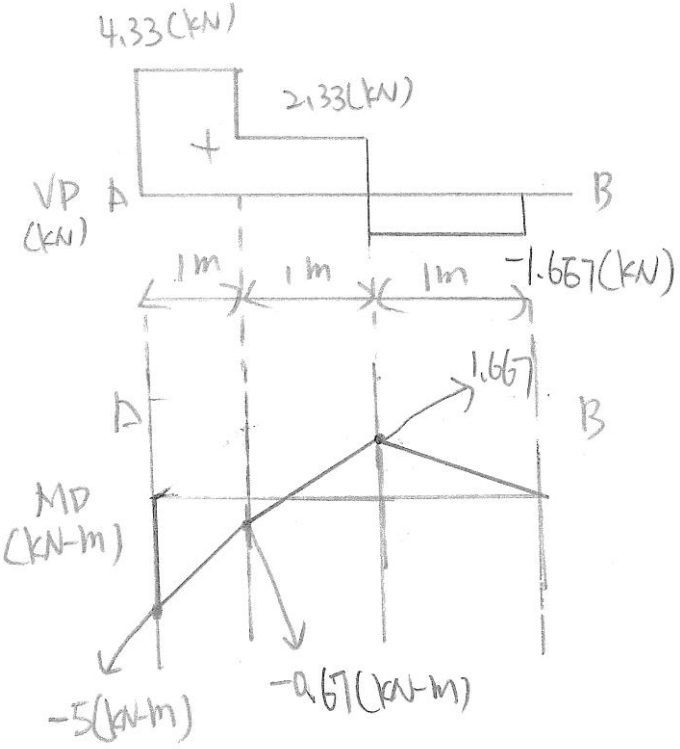
$2\times1+4\times2-R_B\times3-5=0$

$\Rightarrow R_B = 1.667(kN)$

$\uparrow \Sigma F_y = 0$

$R_A + R_B = 6 \Rightarrow R_A = 4.333(kN)$

(二)

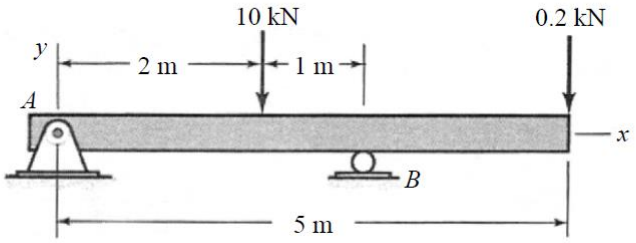


$V_{max} = 4.33(kN)$

$M_{max} = -5(kN\text{-}m)$

例 38

如圖表示一個等截面的橫樑（prismatic beam）及其受力狀態，*A* 為銷接（pin）支持，*B* 為滾柱（roller）支持。求解 *A* 與 *B* 處的反作用力，並繪製剪力圖（shear diagram）。（20 分）

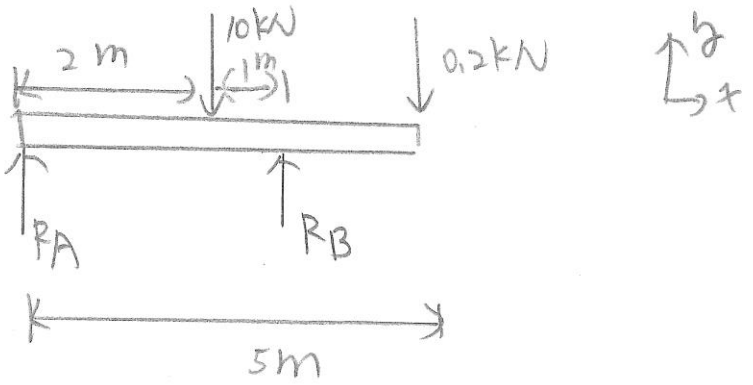


【106 普考】

【解析】

(一)

(一) 取整体之F.B.D



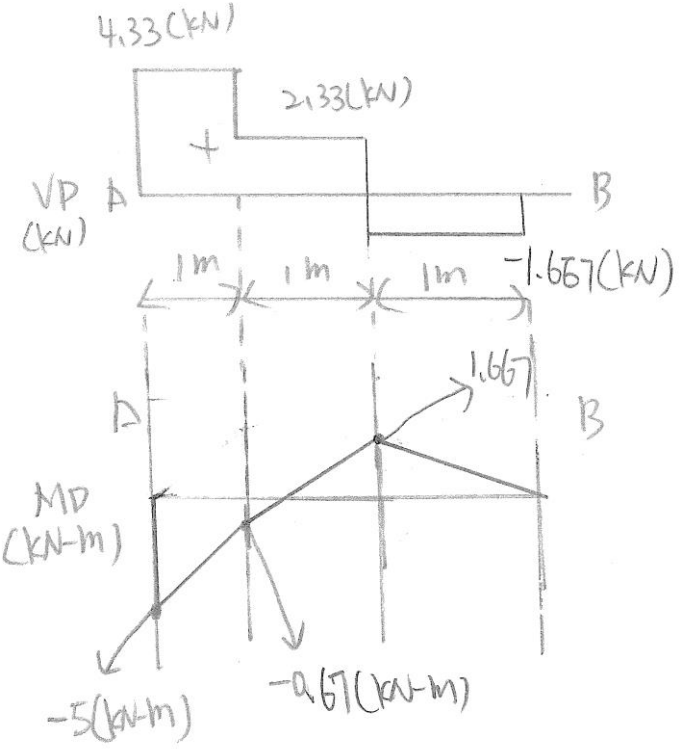
$\curvearrowleft \Sigma M_A = 0$

$R_B \times 3 - 10 \times 2 - 0.2 \times 5 = 0 \Rightarrow R_B = 7\,(kN)$

$\uparrow \Sigma F_y = 0$

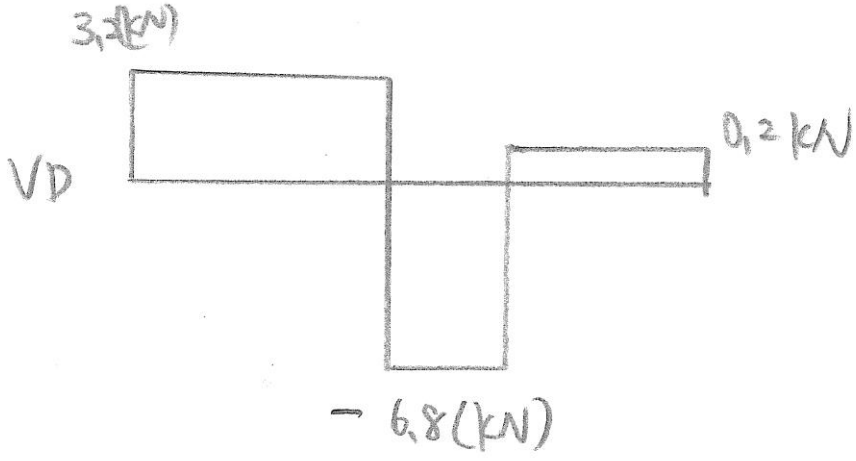
$R_A = 3.2\,(kN)$

(二)



$V_{max} = 4.33(kN)$

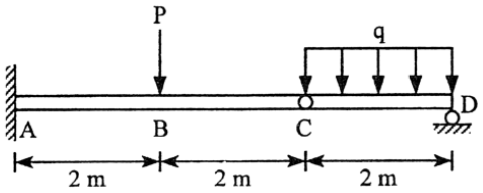
$M_{max} = -5(kN\text{-}m)$





例 39

、如圖所示之連續梁 ABCD 有相同之 *EI* 值，A 爲固定端，C 爲鉸接，D 爲簡支承。如 $P=4\,\text{kN}$ 且 $q=2\,\text{kN/m}$，請計算 A 點之反力、彎矩及 D 點之反力，並繪製此連續梁之剪力圖與彎矩圖。(25 分)



【106 土木普考】

【解析】

依據力的平衡

圖 a$[\Sigma M_C=0]$：$R_{Dy}\times 2-2\times 2\times 1=0$ ，$R_{Dy}=2\,\text{kN}$（↑）

取整體結構

$[\Sigma F_x=0]$：$R_{Ax}=0$

$[\Sigma F_y=0]$：$R_{Ay}+2-4-2\times 2=0$ ，$R_{Ay}=6\,\text{kN}$（↑）

$[\Sigma M_A=0]$：$M_A-4\times 2-2\times 2\times 5+2\times 6=0$ ，$M_A=16\,\text{kN-m}$（↺）

